

GRADE 3 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 4) • Forward 2 (Grade 5)
61 question sets • 1525 questions

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Interpret a product as the total number of objects in equal groups

MAIN	Grade 3 main question	3.OA.A.1
BACKWARD 1	Find the total of an array by adding equal addends	2.OA.C.4
BACKWARD 2	Skip-count by 5s, 10s, and 100s	2.NBT.A.2
FORWARD 1	Interpret a multiplication equation as a comparison	4.OA.A.1
FORWARD 2	Interpret a fraction times a whole number as equal parts of a quan...	5.NF.B.4.a

SET 1 • 3.OA.A.1

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

Which describes 4×6 ?

- A.** 4 groups with 6 objects in each group
- B.** 4 groups with 4 objects in each group
- C.** 6 more than 4
- D.** 4 objects and 6 objects

There are 3 bags with 9 marbles in each bag.

Write a multiplication expression for the total number of marbles. Then find the total.

Expression: _____

Total: _____

True or false?

7×2 can mean 7 groups with 2 objects in each group.

A. True

B. False

Describe a situation that 6×5 could show.

Situation: _____

True or false?

8×3 means $8 + 3$.

A. True

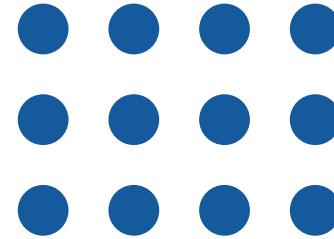
B. False

SET 1 • 3.OA.A.1

BACKWARD 1

Find the total of an array by adding equal addends • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation
with equal addends for the
array.



Equation: _____

True or false?

An array with 2 rows of 5 dots has 10 dots.

- A. True
- B. False

Which addition equation matches an array with 4 rows of 3 dots?

A. $4 + 4 = 8$

B. $4 + 3 = 7$

C. $3 + 4 + 3 = 10$

D. $3 + 3 + 3 + 3 = 12$

How many dots are in an array with 5 rows of 5 dots?

Dots: _____

True or false?

An array with 3 rows of 2 dots has 5 dots.

- A. True
- B. False

SET 1 • 3.OA.A.1

BACKWARD 2

Skip-count by 5s, 10s, and 100s • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.2

Skip-count by 5s.

5, 10, 15, __, __

Next two: _____

True or false?

Skip-counting by 10s from 10 gives 10, 20, 30, 40.

A. True

B. False

Skip-count by 5s.

What number comes after 35?

A. 30

B. 36

C. 45

D. 40

Skip-count by 100s.

100, 200, 300, ____

Next: _____

True or false?

Skip-counting by 5s from 5 gives 5, 10, 20, 25.

A. True

B. False

SET 1 • 3.OA.A.1

FORWARD 1

Interpret a multiplication equation as a comparison • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Which statement matches $32 = 4 \times 8$?

- A.** 32 is 4 times as many as 8.
- B.** 32 is 4 more than 8.
- C.** 8 is 4 times as many as 32.
- D.** 32 is 8 more than 4.

Write a comparison statement for $42 = 6 \times 7$.

Statement: _____

True or false?

$27 = 3 \times 9$ can mean that 27 is 3 times as many as 9.

A. True

B. False

Write an equation for "40 is 5 times as many as 8."

Equation: _____

True or false?

$15 = 3 \times 5$ means that 15 is 3 more than 5.

A. True

B. False

SET 1 • 3.OA.A.1

FORWARD 2

Interpret a fraction times a whole number as equal parts of a quantity • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Which describes $\frac{2}{3} \times 12$?

- A.** Split 12 into 3 equal parts and take 2 of them
- B.** Split 12 into 2 equal parts and take 3 of them
- C.** Add $\frac{2}{3}$ and 12
- D.** Make 12 groups of 3

Find $\frac{3}{4} \times 8$ by splitting 8 into 4 equal parts and taking 3 of them.

Product: _____

True or false?

$\frac{1}{5} \times 20$ is the size of one part when 20 is split into 5 equal parts.

- A. True
- B. False

Multiply.

$$\frac{2}{5} \times 15$$

Product: _____

True or false?

$$\frac{3}{4} \times 12 = 16$$

A. True

B. False

Write a multiplication equation for equal groups and arrays

MAIN	Grade 3 main question	3.OA.A.1
BACKWARD 1	Write an addition equation with equal addends for an array	2.OA.C.4
BACKWARD 2	Add up to four two-digit numbers	2.NBT.B.6
FORWARD 1	Write an equation for a multiplicative comparison word problem	4.OA.A.2
FORWARD 2	Write expressions that record calculations with numbers	5.OA.A.2

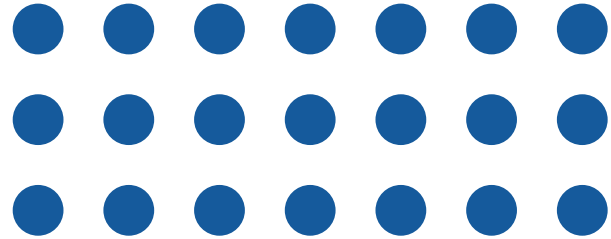
SET 2 • 3.OA.A.1

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

Write a multiplication equation for the array.

Equation: _____



A garden has 5 rows with 8 plants in each row.

Which equation gives the number of plants?

A. $5 \times 8 = 40$

B. $8 - 5 = 3$

C. $5 + 8 = 13$

D. $5 \times 5 = 25$

True or false?

The equation $6 \times 4 = 24$ matches 6 vases with 4 flowers in each vase.

A. True

B. False

Each of 7 spiders has 8 legs.

Write a multiplication equation for the total number of legs.

Equation: _____

True or false?

The equation $4 \times 9 = 36$ matches 4 boxes with 4 pens in each box.

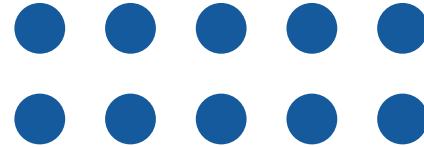
- A. True
- B. False

SET 2 • 3.OA.A.1

BACKWARD 1

Write an addition equation with equal addends for an array • Grade 2 •
Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation
with equal addends for the
array.



Equation: _____

True or false?

$3 + 3 + 3 + 3$ matches an array with 4 rows of 3 dots.

A. True

B. False

An array has 3 rows with 5 dots in each row.

Which equation gives the total?

A. $5 + 5 + 5 = 15$

B. $5 + 5 = 10$

C. $3 + 3 = 6$

D. $3 + 5 = 8$

An array has 4 rows with 4 dots in each row.
Write an addition equation for the total.

Equation: _____

True or false?

An array with 5 rows of 2 dots matches $5 + 2 = 7$.

A. True

B. False

SET 2 • 3.OA.A.1

BACKWARD 2

Add up to four two-digit numbers • Grade 2 • Number & Operations in Base Ten
• 2.NBT.B.6

Add.

$$15 + 15 + 15$$

Sum: _____

True or false?

$$20 + 20 + 20 + 20 = 80$$

A. True

B. False

$$11 + 11 + 11 + 11 = ?$$

A. 22

B. 44

C. 40

D. 33

Add.

$$24 + 24$$

Sum: _____

True or false?

$$13 + 13 + 13 = 36$$

A. True

B. False

SET 2 • 3.OA.A.1

FORWARD 1

Write an equation for a multiplicative comparison word problem • Grade 4 •
Operations & Algebraic Thinking • 4.OA.A.2

A cat weighs 7 pounds. A dog weighs 5 times as much.
Write an equation with a symbol for the dog's weight.
Then solve it.

Equation: _____

Weight: _____

True or false?

Ann has 6 shells. Ben has 3 times as many shells.
Ben has 18 shells.

- A. True
- B. False

A red tower is 9 blocks tall. A blue tower is 4 times as tall.

Which equation gives the blue tower's height, b ?

A. $b = 9 \div 4$

B. $b = 4 + 9$

C. $b = 4 \times 9$

D. $b = 9 - 4$

A plant is 8 cm tall. A tree is 10 times as tall.

How tall is the tree?

Height: _____

True or false?

A rope is 6 feet long. A rope 3 times as long is 9 feet long.

A. True

B. False

SET 2 • 3.OA.A.1

FORWARD 2

Write expressions that record calculations with numbers • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "multiply 4 by the sum of 6 and 3."

Expression: _____

Which expression means "3 times the difference of 10 and 2"?

A. $(3 \times 10) - 2$

B. $10 - 2 \times 3$

C. $3 \times (10 - 2)$

D. $3 \times 10 - 2$

True or false?

$5 \times (7 + 2)$ is 5 times as large as $7 + 2$.

A. True

B. False

There are 4 boxes. Each box has 6 red pens and 2 blue pens.

Write an expression for the total number of pens.

Expression: _____

True or false?

$2 \times (8 + 1)$ is 2 more than $8 + 1$.

A. True

B. False

Interpret a quotient as the number of objects in each share

MAIN	Grade 3 main question	3.OA.A.2
BACKWARD 1	Split a group of up to 20 objects into two equal groups	2.OA.C.3
BACKWARD 2	Split a rectangle into rows and columns of same-size squares and...	2.G.A.2
FORWARD 1	Solve multiplicative comparison word problems by dividing	4.OA.A.2
FORWARD 2	Interpret a fraction as division to find equal shares	5.NF.B.3

SET 3 • 3.OA.A.2

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

28 stickers are shared equally among 4 friends.

What does $28 \div 4$ tell you?

- A.** The number of friends
- B.** The number of stickers each friend gets
- C.** The total number of stickers
- D.** The number of stickers left over

18 apples are shared equally among 3 baskets.

Write a division equation for the number of apples in each basket. Then solve it.

Equation: _____

Apples in each basket: _____

True or false?

In $35 \div 5 = 7$, the 7 can mean the number in each group when 35 objects are shared equally among 5 groups.

A. True

B. False

Describe a situation in which $24 \div 6$ tells how many objects are in each share.

Situation: _____

True or false?

When 20 pencils are shared equally among 4 students, each student gets 16 pencils.

- A. True
- B. False

SET 3 • 3.OA.A.2

BACKWARD 1

Split a group of up to 20 objects into two equal groups • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

Write 16 as the sum of two equal addends.

Sum: _____

True or false?

14 objects can be split into 2 equal groups of 7.

A. True

B. False

Which number of objects can be split into 2 equal groups?

A. 12

B. 9

C. 7

D. 15

12 counters are split into 2 equal groups.

How many counters are in each group?

Counters: _____

True or false?

11 objects can be split into 2 equal groups with none left over.

A. True

B. False

SET 3 • 3.OA.A.2

BACKWARD 2

Split a rectangle into rows and columns of same-size squares and count them •
Grade 2 • Geometry • 2.G.A.2

A rectangle is split into 3 rows of 4 same-size squares.
How many squares are there?

Squares: _____

True or false?

A rectangle split into 2 rows of 5 same-size squares has 10 squares.

- A.** True
- B.** False

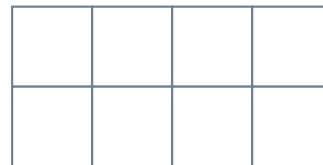
How many same-size squares are in the rectangle?

A. 8

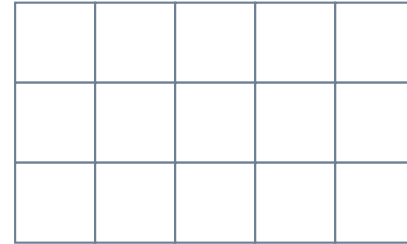
B. 6

C. 10

D. 4



How many same-size squares
are in each row of the
rectangle?



Squares in each row: _____

True or false?

A rectangle split into 4 rows of 4 same-size squares has 8 squares.

- A. True
- B. False

SET 3 • 3.OA.A.2

FORWARD 1

Solve multiplicative comparison word problems by dividing • Grade 4 •
Operations & Algebraic Thinking • 4.OA.A.2

A tree is 24 feet tall. That is 4 times as tall as a bush.
How tall is the bush?

Height: _____

True or false?

Mia has 45 beads. That is 5 times as many beads as Jo has. Jo has 9 beads.

A. True

B. False

A truck weighs 6 tons. That is 3 times as much as a van weighs.

How much does the van weigh?

A. 18 tons

B. 2 tons

C. 3 tons

D. 9 tons

A book costs \$36. That is 9 times as much as a pen costs.

How much does the pen cost?

Cost: _____

True or false?

A barn is 40 feet tall. That is 8 times as tall as a fence. The fence is 32 feet tall.

A. True

B. False

SET 3 • 3.OA.A.2

FORWARD 2

Interpret a fraction as division to find equal shares • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Pizza: _____

True or false?

When 2 sandwiches are shared equally by 5 people, each person gets $\frac{2}{5}$ of a sandwich.

A. True

B. False

6 people share 5 pounds of rice equally.

How much rice does each person get?

A. $\frac{5}{6}$ pound

B. $\frac{6}{5}$ pounds

C. 1 pound

D. $\frac{1}{6}$ pound

Write $7 \div 3$ as a fraction or a mixed number.

Answer: _____

True or false?

$$3 \div 8 = \frac{8}{3}$$

A. True

B. False

Interpret a quotient as the number of equal shares

MAIN	Grade 3 main question	3.OA.A.2
BACKWARD 1	Skip-count by 5s and 10s	2.NBT.A.2
BACKWARD 2	Rows and columns of arrays up to 5 by 5	2.OA.C.4
FORWARD 1	Interpret remainders when dividing into groups	4.OA.A.3
FORWARD 2	Divide a whole number by a unit fraction to find the number of eq...	5.NF.B.7.c

SET 4 • 3.OA.A.2

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

30 pencils are put in boxes, with 5 pencils in each box.

What does $30 \div 5$ tell you?

- A.** The number of pencils left over
- B.** The number of boxes
- C.** The number of pencils in each box
- D.** The total number of pencils

24 eggs are packed 6 to a carton.

Write a division equation for the number of cartons.
Then solve it.

Equation: _____

Cartons: _____

True or false?

In $40 \div 8 = 5$, the 5 can mean the number of groups when 40 objects are put into groups of 8.

A. True

B. False

Describe a situation in which $21 \div 3$ tells how many groups there are.

Situation: _____

True or false?

A coach puts 36 balls into bags of 4. There are 32 bags.

A. True

B. False

SET 4 • 3.OA.A.2

BACKWARD 1

Skip-count by 5s and 10s • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.2

Skip-count by 10s from 0 to 60.

How many 10s did you count?

Tens: _____

True or false?

Counting by 5s from 0 to 25 takes 5 counts.

A. True

B. False

Starting at 0, how many 5s do you count to reach 40?

A. 35

B. 8

C. 40

D. 5

Skip-count by 10s.

40, 50, 60, __, __

Next two: _____

True or false?

Counting by 10s from 0 to 90 takes 10 counts.

A. True

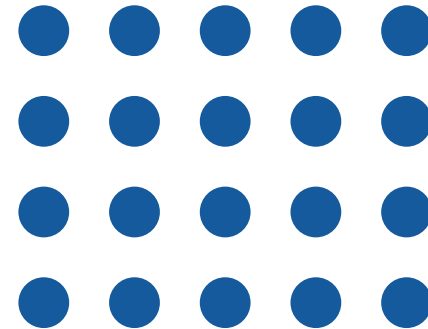
B. False

SET 4 • 3.OA.A.2

BACKWARD 2

Rows and columns of arrays up to 5 by 5 • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many rows does the array have? How many dots are in each row?



Rows: _____

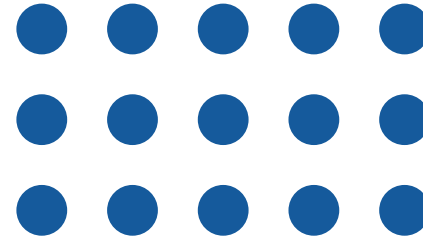
Dots in each row: _____

True or false?

The array has 3 rows.

A. True

B. False



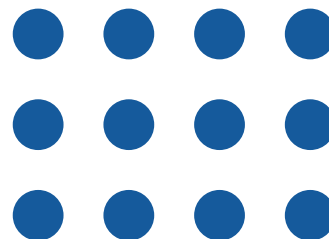
How many dots are in each row of the array?

A. 12

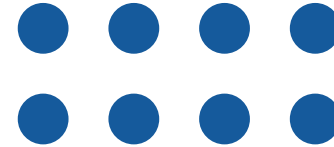
B. 4

C. 3

D. 7



Write an addition equation
with equal addends for the
array.



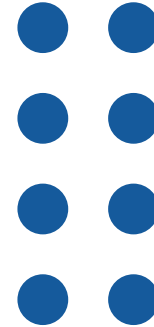
Equation: _____

True or false?

The array has 5 rows.

A. True

B. False



SET 4 • 3.OA.A.2

FORWARD 1

Interpret remainders when dividing into groups • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A teacher puts 26 students in groups of 4.

How many full groups are there, and how many students are left over?

Full groups: _____

Left over: _____

29 people ride in boats that hold 6 people each.
How many boats are needed so everyone can ride?

Boats: _____

A baker puts 50 rolls in bags of 8.

How many bags can she fill completely?

A. 6

B. 7

C. 8

D. 2

True or false?

43 cards put in piles of 5 make 8 full piles with 3 cards left over.

A. True

B. False

True or false?

38 markers put in boxes of 6 fill 7 boxes.

A. True

B. False

SET 4 • 3.OA.A.2

FORWARD 2

Divide a whole number by a unit fraction to find the number of equal shares •
Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{3}$ -cup servings are in 4 cups?

Servings: _____

True or false?

$$6 \div \frac{1}{2} = 12$$

A. True

B. False

A 5-foot board is cut into pieces that are each $\frac{1}{4}$ foot long.

How many pieces are there?

A. $\frac{1}{20}$

B. $\frac{5}{4}$

C. 20

D. 9

Divide.

$$3 \div \frac{1}{5}$$

Quotient: _____

True or false?

$$2 \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

Solve multiplication word problems within 100

MAIN	Grade 3 main question	3.OA.A.3
BACKWARD 1	One-step addition word problems within 100	2.OA.A.1
BACKWARD 2	Add lengths in word problems within 100	2.MD.B.5
FORWARD 1	Multiply a two-digit number by a one-digit number to solve word...	4.NBT.B.5
FORWARD 2	Multiply multi-digit whole numbers to solve word problems	5.NBT.B.5

SET 5 • 3.OA.A.3

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

A bakery puts 6 muffins in each box. It fills 7 boxes.
How many muffins are in the boxes?

Muffins: _____

A wall has 8 rows of tiles with 9 tiles in each row.
How many tiles are on the wall?

Tiles: _____

Each ribbon is 4 feet long. Mia has 6 ribbons.

What is the total length of the ribbons?

- A.** 28 feet
- B.** 24 feet
- C.** 2 feet
- D.** 10 feet

True or false?

A parking lot has 5 rows with 9 cars in each row.
There are 54 cars.

A. True

B. False

Tickets cost \$7 each. A family buys 8 tickets.

Write an equation with a symbol for the total cost. Then solve it.

Equation: _____

Total cost: _____

SET 5 • 3.OA.A.3

BACKWARD 1

One-step addition word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A box has 24 crayons. Another box has 24 crayons.
How many crayons are there in all?

Crayons: _____

True or false?

Jo has 35 cards and gets 15 more. She has 50 cards.

A. True

B. False

A bus has 27 children and 18 adults.

How many people are on the bus?

A. 45

B. 35

C. 9

D. 55

A farmer picks 46 apples in the morning and 38 apples in the afternoon.

How many apples does he pick?

Apples: _____

True or false?

A class has 16 girls and 17 boys. There are 23 students.

A. True

B. False

SET 5 • 3.OA.A.3

BACKWARD 2

Add lengths in word problems within 100 • Grade 2 • Measurement & Data •
2.MD.B.5

A red ribbon is 25 cm long. A blue ribbon is 30 cm long.
What is their total length?

Length: _____

True or false?

Two 20-foot ropes laid end to end are 40 feet long.

A. True

B. False

One path is 48 m long and another is 36 m long.
What is their total length?

- A.** 74 m
- B.** 12 m
- C.** 94 m
- D.** 84 m

A board is 47 inches long. Ben glues a 25-inch board to its end.

How long is it now?

Length: _____

True or false?

A 60-cm stick and a 35-cm stick laid end to end are 85 cm long.

- A. True
- B. False

SET 5 • 3.OA.A.3

FORWARD 1

Multiply a two-digit number by a one-digit number to solve word problems •
Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

A school buys 6 boxes of 24 pencils.

How many pencils does it buy?

Pencils: _____

True or false?

A theater has 9 rows of 32 seats. It has 288 seats.

A. True

B. False

A bike trip covers 45 miles each day for 4 days.

How long is the trip?

- A.** 180 miles
- B.** 49 miles
- C.** 90 miles
- D.** 160 miles

Tickets cost \$18 each. A club buys 7 tickets.

What is the total cost?

Cost: _____

True or false?

A farmer plants 5 rows of 76 trees. He plants 350 trees.

A. True

B. False

SET 5 • 3.OA.A.3

FORWARD 2

Multiply multi-digit whole numbers to solve word problems • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

A stadium has 48 sections with 125 seats in each section.

How many seats are there?

Seats: _____

True or false?

A truck carries 36 crates with 144 apples in each crate. It carries 5,184 apples.

A. True

B. False

A printer prints 215 pages each hour for 24 hours.

How many pages does it print?

- A.** 4,300
- B.** 5,060
- C.** 5,160
- D.** 1,290

A school orders 32 boxes with 250 sheets of paper in each box.

How many sheets is that?

Sheets: _____

True or false?

A farm packs 64 cartons of 18 eggs. That is 1,052 eggs.

A. True

B. False

Solve division word problems within 100

MAIN	Grade 3 main question	3.OA.A.3
BACKWARD 1	One-step subtraction word problems within 100	2.OA.A.1
BACKWARD 2	Split circles and rectangles into 2, 3, or 4 equal shares	2.G.A.3
FORWARD 1	Divide up to three-digit numbers by one-digit numbers to solve w...	4.NBT.B.6
FORWARD 2	Divide by two-digit divisors to solve word problems	5.NBT.B.6

SET 6 • 3.OA.A.3

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

56 students are split equally into 7 teams.

How many students are on each team?

Students: _____

A gardener plants 63 seeds in rows of 9.
How many rows does she plant?

Rows: _____

A 48-inch ribbon is cut into 6 equal pieces.

How long is each piece?

- A.** 42 inches
- B.** 8 inches
- C.** 6 inches
- D.** 54 inches

True or false?

81 chairs are set up in 9 equal rows. There are 9 chairs in each row.

A. True

B. False

A coach has 54 balls. She puts the same number of balls in each of 6 bags.

Write an equation with a symbol for the number of balls in each bag. Then solve it.

Equation: _____

Balls in each bag: _____

SET 6 • 3.OA.A.3

BACKWARD 1

One-step subtraction word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A shelf has 64 books. 28 books are checked out.
How many books are left?

Books: _____

True or false?

Mo had 50 stickers and gave away 18. He has 32 stickers left.

- A.** True
- B.** False

A tank has 90 fish. 45 fish are moved to another tank.
How many fish are left?

- A.** 35
- B.** 135
- C.** 55
- D.** 45

A rope is 75 feet long. 30 feet are cut off.

How much rope is left?

Rope left: _____

True or false?

A box had 40 pears. 15 were eaten, so 35 are left.

A. True

B. False

SET 6 • 3.OA.A.3

BACKWARD 2

Split circles and rectangles into 2, 3, or 4 equal shares • Grade 2 • Geometry •
2.G.A.3

A rectangle is split into 4 equal shares.

What is each share called?

Name: _____

True or false?

A circle cut into 3 equal shares is cut into thirds.

A. True

B. False

A sandwich is cut into 2 equal shares.

What is each share called?

- A.** A third
- B.** A whole
- C.** A fourth
- D.** A half

How many thirds make a whole?

Thirds: _____

True or false?

A rectangle cut into 4 shares of different sizes is cut into fourths.

A. True

B. False

SET 6 • 3.OA.A.3

FORWARD 1

Divide up to three-digit numbers by one-digit numbers to solve word problems

• Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

A farmer packs 96 eggs into cartons of 6.
How many cartons does he fill?

Cartons: _____

True or false?

A 135-page book is read in 5 equal parts. Each part is 27 pages.

A. True

B. False

A school splits 252 students into 4 equal groups.
How many students are in each group?

- A.** 248
- B.** 61
- C.** 63
- D.** 1,008

A rope 128 feet long is cut into 8 equal pieces.
How long is each piece?

Length: _____

True or false?

7 friends share 196 marbles equally. Each friend gets 29 marbles.

A. True

B. False

SET 6 • 3.OA.A.3

FORWARD 2

Divide by two-digit divisors to solve word problems • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

A school has 672 students in 24 classes of the same size.

How many students are in each class?

Students: _____

True or false?

A 1,350-mile trip is driven in 15 days, the same distance each day. Each day is 90 miles.

A. True

B. False

A factory packs 2,016 cans in boxes of 48.

How many boxes does it fill?

- A.** 43
- B.** 42
- C.** 2,064
- D.** 41

An orchard has 1,155 trees in 35 equal rows.

How many trees are in each row?

Trees: _____

True or false?

3,600 seats are split into 25 equal sections. Each section has 154 seats.

A. True

B. False

Represent a multiplication or division word problem with a drawing and an equation

MAIN	Grade 3 main question	3.OA.A.3
BACKWARD 1	Use an equation with a symbol for the unknown in a one-step wo...	2.OA.A.1
BACKWARD 2	Draw arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Use a drawing and an equation to solve a multiplicative comparis...	4.OA.A.2
FORWARD 2	Use a tape diagram to find a fraction of a whole number in context	5.NF.B.6

SET 7 • 3.OA.A.3

MAIN

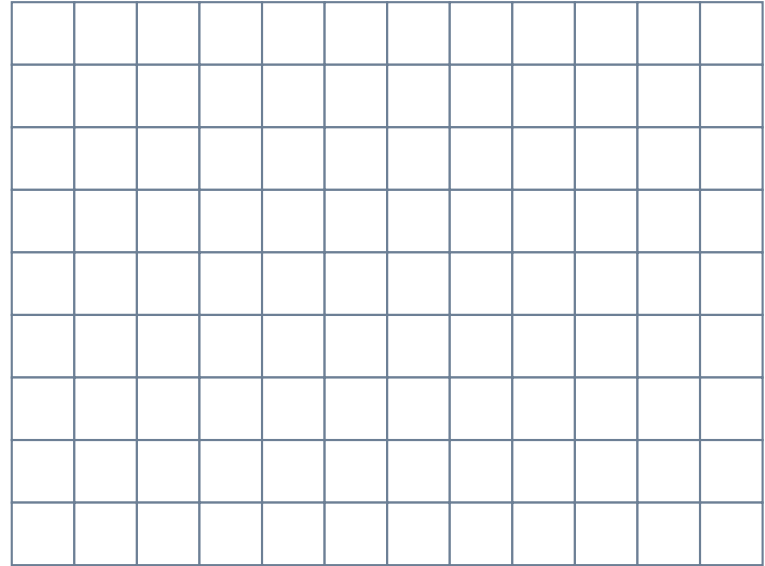
Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 4 plates with 5 grapes on each plate.

Draw a picture of the problem. Then write an equation and find the total.

Equation: _____

Total: _____

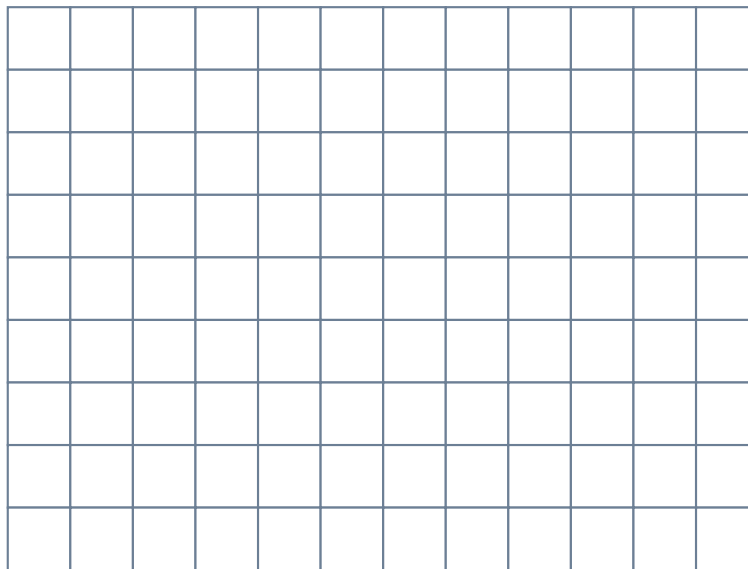


18 crackers are shared equally
among 3 children.

Draw a picture of the problem.
Then write a division equation
and find how many crackers each
child gets.

Equation: _____

Each child: _____



A drawing shows 3 rows with 6 stars in each row.
Which equation matches the drawing?

A. $3 + 6 = ?$

B. $6 \div 3 = ?$

C. $3 \times 6 = ?$

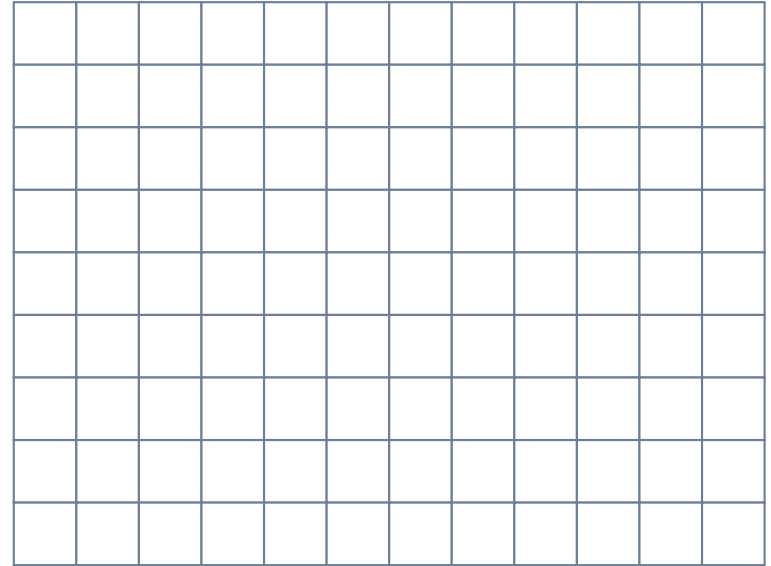
D. $6 - 3 = ?$

A 20-foot board is cut into pieces that are each 4 feet long.

Draw a diagram. Then write an equation with a symbol for the unknown and solve it.

Equation: _____

Pieces: _____



True or false?

A drawing of 5 groups with 3 dots in each group matches the equation $5 + 3 = 8$.

A. True

B. False

SET 7 • 3.OA.A.3

BACKWARD 1

Use an equation with a symbol for the unknown in a one-step word problem •
Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lena has 23 shells. She finds some more and now has 41.

Write an equation with a symbol for the unknown. Then solve it.

Equation: _____

Shells found: _____

True or false?

The equation $52 - b = 30$ matches "52 birds were on a wire. Some flew away, and 30 were left."

- A. True
- B. False

A box had some pens. 14 more were put in, and now there are 50.

Which equation matches?

A. $p + 14 = 50$

B. $p - 14 = 50$

C. $14 - p = 50$

D. $50 + 14 = p$

A rope is 60 inches long. Kim cuts off 25 inches.

Write an equation with a symbol for the length left.
Then solve it.

Equation: _____

Length left: _____

True or false?

The equation $18 + 9 = n$ matches "There were 18 cats. 9 cats left. How many are still here?"

A. True

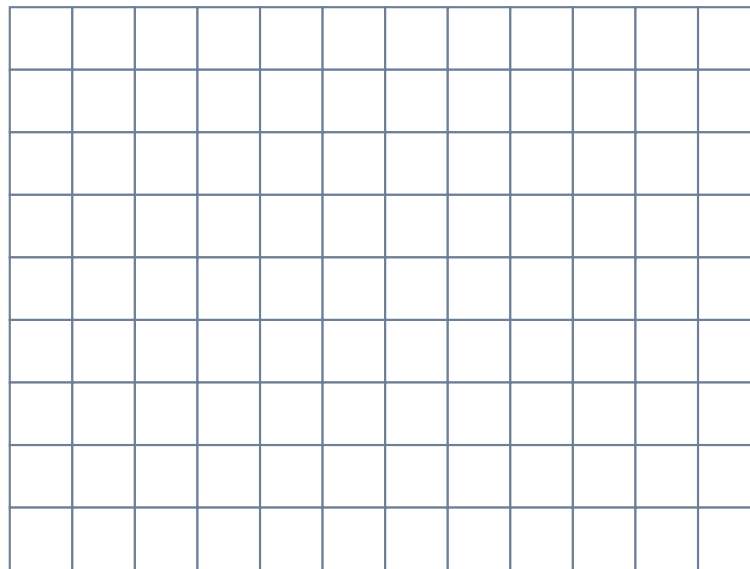
B. False

SET 7 • 3.OA.A.3

BACKWARD 2

Draw arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Draw an array with 3 rows of 4 dots.



True or false?

An array with 4 rows of 2 dots shows
 $2 + 2 + 2 + 2$.

- A. True
- B. False

How many dots do you draw for an array with 5 rows of 3 dots?

A. 53

B. 8

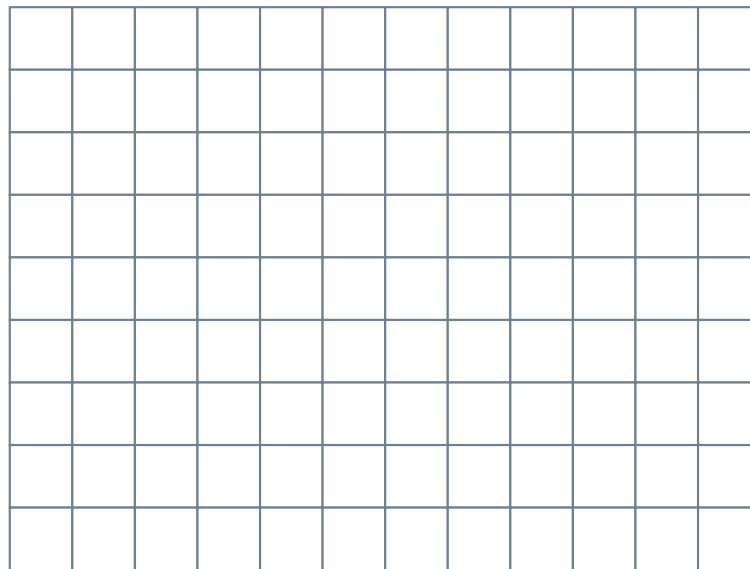
C. 15

D. 12

Draw an array with 2 rows of 5 dots.

Then write an addition equation for it.

Equation: _____



True or false?

An array with 3 rows of 3 dots has 6 dots.

- A.** True
- B.** False

SET 7 • 3.OA.A.3

FORWARD 1

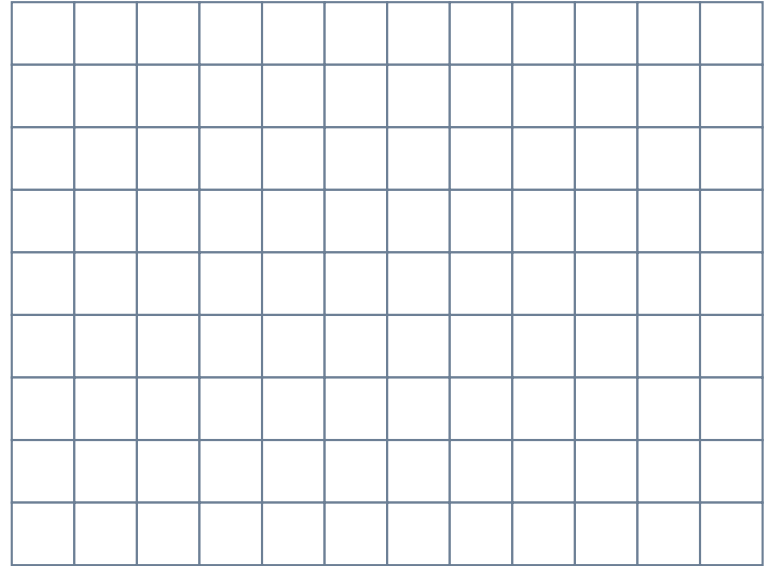
Use a drawing and an equation to solve a multiplicative comparison problem •
Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A blue scarf is 3 feet long. A red scarf is 4 times as long.

Draw a tape diagram. Then write an equation and find the red scarf's length.

Equation: _____

Length: _____



True or false?

The equation $36 = 4 \times n$ matches "36 is 4 times as many as n."

- A. True
- B. False

A school has 8 teachers and 5 times as many parent helpers.

Which equation finds the number of parent helpers, p ?

A. $p = 8 - 5$

B. $p = 5 + 8$

C. $p = 5 \times 8$

D. $p = 8 \div 5$

Ty swam 30 laps. That is 6 times as many laps as Al swam.

Write an equation with a symbol for Al's laps. Then solve it.

Equation: _____

Laps: _____

True or false?

"21 is 3 times as many as 7" can be written as
 $21 = 3 + 7$.

A. True

B. False

SET 7 • 3.OA.A.3

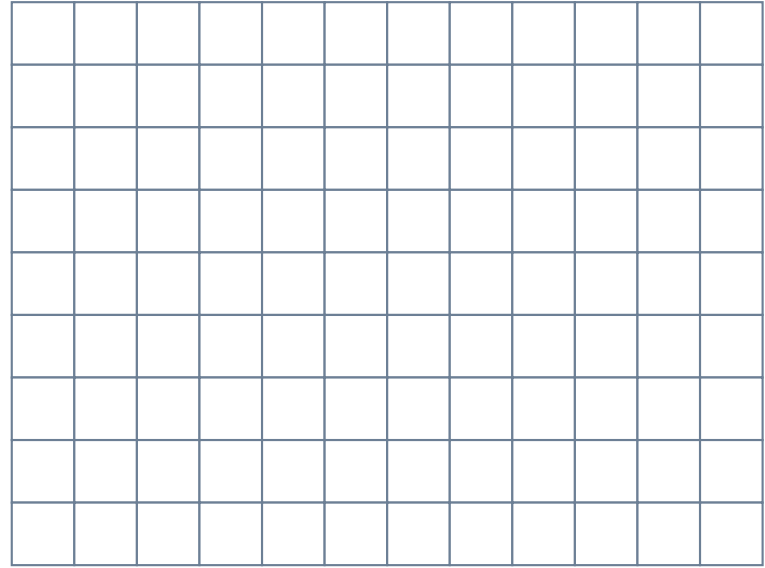
FORWARD 2

Use a tape diagram to find a fraction of a whole number in context • Grade 5 •
Number & Operations–Fractions • 5.NF.B.6

A path is 6 miles long. Kai walked $\frac{2}{3}$ of it.

Draw a tape diagram. Then find how far Kai walked.

Distance: _____



True or false?

$\frac{3}{4}$ of a 12-pound bag of rice is 9 pounds.

A. True

B. False

A recipe uses 2 cups of milk. Leo makes $\frac{1}{2}$ of the recipe.

How much milk does he use?

A. 1 cup

B. $2\frac{1}{2}$ cups

C. 4 cups

D. $\frac{1}{4}$ cup

A rope is 10 feet long. $\frac{3}{5}$ of it is red.

How many feet of the rope are red?

Feet: _____

True or false?

$\frac{2}{5}$ of 20 students is 5 students.

A. True

B. False

Find the unknown whole number in a multiplication equation

MAIN	Grade 3 main question	3.OA.A.4
BACKWARD 1	Find the unknown number in an addition or subtraction equation w...	1.OA.D.8
BACKWARD 2	Skip-count to find a number in a counting sequence	2.NBT.A.2
FORWARD 1	Find factor pairs and decide whether a number is a factor	4.OA.B.4
FORWARD 2	Find an unknown factor with a two-digit factor	5.NBT.B.6

SET 8 • 3.OA.A.4

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$7 \times ? = 56$$

Unknown: _____

Find the unknown number.

$$? \times 4 = 36$$

Unknown: _____

Which number makes the equation true?

$$6 \times n = 42$$

A. 36

B. 7

C. 48

D. 6

True or false?

In $4 \times \square = 32$, the unknown number is 9.

A. True

B. False

Find the unknown number.

$$63 = 9 \times ?$$

Unknown: _____

SET 8 • 3.OA.A.4

BACKWARD 1

Find the unknown number in an addition or subtraction equation within 20 •

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

$$9 + ? = 15$$

Unknown: _____

True or false?

In $13 - ? = 6$, the unknown number is 7.

A. True

B. False

$$? + 7 = 16$$

A. 10

B. 9

C. 23

D. 8

Find the unknown number.

$$14 = ? + 5$$

Unknown: _____

True or false?

In $? - 4 = 9$, the unknown number is 5.

A. True

B. False

SET 8 • 3.OA.A.4

BACKWARD 2

Skip-count to find a number in a counting sequence • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 5s from 5.

How many numbers do you say to reach 35?

Numbers: _____

True or false?

Skip-counting by 10s from 10, the 4th number is 40.

A. True

B. False

Skip-count by 5s from 5.

What is the 6th number?

A. 11

B. 35

C. 25

D. 30

Skip-count by 100s from 100.

What is the 5th number?

Number: _____

True or false?

Skip-counting by 10s from 10, the 7th number is 80.

A. True

B. False

SET 8 • 3.OA.A.4

FORWARD 1

Find factor pairs and decide whether a number is a factor • Grade 4 •
Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 20.

Factor pairs: _____

True or false?

7 is a factor of 63.

A. True

B. False

Which is a factor pair of 36?

A. 5×7

B. 4×9

C. 3×11

D. 6×5

Is 6 a factor of 44? Explain.

Answer: _____

Explanation: _____

True or false?

8 is a factor of 30.

A. True

B. False

SET 8 • 3.OA.A.4

FORWARD 2

Find an unknown factor with a two-digit factor • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Find the unknown number.

$$24 \times ? = 912$$

Unknown: _____

True or false?

If $15 \times n = 645$, then $n = 43$.

A. True

B. False

$$? \times 32 = 1,248$$

- A.** 41
- B.** 1,280
- C.** 38
- D.** 39

Find the unknown number.

$$2,100 = 75 \times ?$$

Unknown: _____

True or false?

If $18 \times n = 864$, then $n = 46$.

A. True

B. False

Find the unknown whole number in a division equation

MAIN	Grade 3 main question	3.OA.A.4
BACKWARD 1	Find the unknown number in a subtraction or addition equation w...	1.OA.D.8
BACKWARD 2	Add and subtract within 20 mentally	2.OA.B.2
FORWARD 1	Find the unknown number in a division equation with a larger divi...	4.NBT.B.6
FORWARD 2	Find the unknown number in a division equation with a two-digit d...	5.NBT.B.6

SET 9 • 3.OA.A.4

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$45 \div ? = 9$$

Unknown: _____

Find the unknown number.

$$? \div 6 = 7$$

Unknown: _____

Which number makes the equation true?

$$72 \div n = 8$$

A. 8

B. 64

C. 80

D. 9

True or false?

In $? \div 4 = 8$, the unknown number is 32.

A. True

B. False

True or false?

In $49 \div \square = 7$, the unknown number is 42.

A. True

B. False

SET 9 • 3.OA.A.4

BACKWARD 1

Find the unknown number in a subtraction or addition equation within 20 •

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

$$17 - ? = 9$$

Unknown: _____

True or false?

In $? + 6 = 14$, the unknown number is 8.

A. True

B. False

$$12 = ? - 3$$

A. 15

B. 14

C. 4

D. 9

Find the unknown number.

$$? - 7 = 7$$

Unknown: _____

True or false?

In $16 - ? = 10$, the unknown number is 4.

A. True

B. False

SET 9 • 3.OA.A.4

BACKWARD 2

Add and subtract within 20 mentally • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Subtract.

$$15 - 8$$

Difference: _____

True or false?

$$9 + 8 = 17$$

A. True

B. False

$$13 - 6 = ?$$

A. 6

B. 7

C. 19

D. 8

Add.

$$7 + 6$$

Sum: _____

True or false?

$$16 - 9 = 8$$

A. True

B. False

SET 9 • 3.OA.A.4

FORWARD 1

Find the unknown number in a division equation with a larger dividend • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the unknown number.

$$? \div 4 = 125$$

Unknown: _____

True or false?

In $312 \div n = 6$, $n = 52$.

A. True

B. False

$$848 \div ? = 8$$

- A.** 16
- B.** 856
- C.** 840
- D.** 106

Find the unknown number.

$$? \div 7 = 36$$

Unknown: _____

True or false?

In $450 \div n = 5$, $n = 9$.

A. True

B. False

SET 9 • 3.OA.A.4

FORWARD 2

Find the unknown number in a division equation with a two-digit divisor • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Find the unknown number.

$$? \div 25 = 36$$

Unknown: _____

True or false?

In $1,092 \div n = 42$, $n = 26$.

A. True

B. False

$$1,512 \div ? = 36$$

- A.** 1,476
- B.** 54
- C.** 42
- D.** 48

Find the unknown number.

$$? \div 16 = 45$$

Unknown: _____

True or false?

In $2,000 \div n = 80$, $n = 20$.

A. True

B. False

Use the commutative property of multiplication

MAIN	Grade 3 main question	3.OA.B.5
BACKWARD 1	Add in either order	1.OA.B.3
BACKWARD 2	Find the total of an array by rows or by columns	2.OA.C.4
FORWARD 1	Use the commutative property with multi-digit products	4.NBT.B.5
FORWARD 2	Multiply a fraction and a whole number in either order	5.NF.B.4.a

SET 10 • 3.OA.B.5

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

$$7 \times 8 = 56.$$

Without multiplying, what is 8×7 ? Explain.

Product: _____

Explanation: _____

Which is equal to 9×3 ?

A. 3×9

B. 93

C. 9×9

D. $9 + 3$

True or false?

$$5 \times 9 = 9 \times 5$$

A. True

B. False

An array has 4 rows of 7 dots. It is turned on its side, so it now has 7 rows.

How many dots are in each row now? How many dots are there in all?

Dots in each row: _____

Total: _____

True or false?

$$8 \times 4 = 4 + 8$$

A. True

B. False

SET 10 • 3.OA.B.5

BACKWARD 1

Add in either order • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3

$$9 + 5 = 14.$$

What is $5 + 9$?

Sum: _____

True or false?

$$7 + 6 = 6 + 7$$

A. True

B. False

Which is equal to $4 + 9$?

A. 49

B. $4 + 4$

C. $9 + 4$

D. $9 - 4$

Fill in the blank.

$$6 + 8 = 8 + \underline{\quad}$$

Blank: _____

True or false?

$$5 + 2 = 5 - 2$$

A. True

B. False

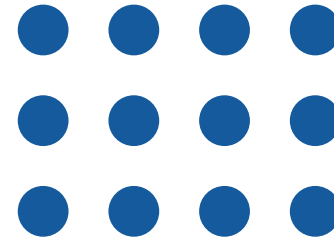
SET 10 • 3.OA.B.5

BACKWARD 2

Find the total of an array by rows or by columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Add the rows. Then add the columns.

What total do you get each way?



By rows: _____

By columns: _____

True or false?

An array with 2 rows of 5 dots has the same number of dots as an array with 5 rows of 2 dots.

- A. True
- B. False

An array has 3 rows of 5 dots.

Which addition adds its columns?

A. $5 + 5 + 5$

B. $5 + 3 + 5$

C. $3 + 5$

D. $3 + 3 + 3 + 3 + 3$

How many dots are in an array with 4 rows of 5 dots?

Dots: _____

True or false?

An array with 4 rows of 3 dots has more dots than an array with 3 rows of 4 dots.

- A.** True
- B.** False

SET 10 • 3.OA.B.5

FORWARD 1

Use the commutative property with multi-digit products • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

True or false?

$$24 \times 15 = 15 \times 24$$

A. True

B. False

$$36 \times 7 = 252.$$

What is 7×36 ?

Product: _____

Which is equal to 48×25 ?

A. 25×48

B. 48×52

C. $48 + 25$

D. 84×25

Is 19×6 equal to 6×19 ? Find the product.

Equal? _____

Product: _____

True or false?

$$125 \times 8 = 8 \times 152$$

A. True

B. False

SET 10 • 3.OA.B.5

FORWARD 2

Multiply a fraction and a whole number in either order • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

True or false?

$$\frac{2}{3} \times 6 = 6 \times \frac{2}{3}$$

A. True

B. False

Multiply.

$$8 \times \frac{3}{4}$$

Product: _____

Which is equal to $\frac{5}{6} \times 12$?

A. $12 \times \frac{6}{5}$

B. $\frac{5}{6} + 12$

C. $12 \times \frac{5}{6}$

D. $\frac{6}{5} \times 12$

$$\frac{3}{5} \times 10 = 6.$$

What is $10 \times \frac{3}{5}$?

Product: _____

True or false?

$$\frac{1}{2} \times 14 = 14 \times 2$$

A. True

B. False

Use the associative property of multiplication

MAIN	Grade 3 main question	3.OA.B.5
BACKWARD 1	Add three numbers in any grouping	1.OA.B.3
BACKWARD 2	Add within 20 mentally	2.OA.B.2
FORWARD 1	Use grouping and multiples of 10 to multiply	4.NBT.B.5
FORWARD 2	Multiply by powers of 10 written with exponents	5.NBT.A.2

SET 11 • 3.OA.B.5

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

Find the product by first multiplying 2×5 .

$$2 \times 5 \times 7$$

Product: _____

Which is equal to $(4 \times 2) \times 3$?

A. $4 + (2 \times 3)$

B. $4 \times 2 + 3$

C. $4 \times (2 \times 3)$

D. $(4 + 2) \times 3$

True or false?

$$(3 \times 2) \times 4 = 3 \times (2 \times 4)$$

A. True

B. False

Show two different ways to group $5 \times 2 \times 6$. Then find the product.

Way 1: _____

Way 2: _____

Product: _____

True or false?

$$(2 \times 4) \times 5 = 2 + (4 \times 5)$$

A. True

B. False

SET 11 • 3.OA.B.5

BACKWARD 1

Add three numbers in any grouping • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3

Add.

$$7 + 3 + 5$$

Sum: _____

True or false?

$$(4 + 6) + 3 = 4 + (6 + 3)$$

A. True

B. False

Which sum is equal to $8 + 5 + 2$?

A. $10 + 5$

B. $8 + 7 + 2$

C. $10 + 7$

D. $13 + 5$

Add by making a ten first.

$$6 + 9 + 4$$

Sum: _____

True or false?

$$(5 + 5) + 4 = 10 + 9$$

A. True

B. False

SET 11 • 3.OA.B.5

BACKWARD 2

Add within 20 mentally • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Add.

$$8 + 9$$

Sum: _____

True or false?

$$6 + 7 = 13$$

A. True

B. False

$$9 + 9 = ?$$

A. 81

B. 17

C. 19

D. 18

Add.

$$10 + 7$$

Sum: _____

True or false?

$$5 + 8 = 12$$

A. True

B. False

SET 11 • 3.OA.B.5

FORWARD 1

Use grouping and multiples of 10 to multiply • Grade 4 • Number & Operations
in Base Ten • 4.NBT.B.5

Find $4 \times 5 \times 13$ by first multiplying 4×5 .

Product: _____

True or false?

$$6 \times 25 \times 4 = 6 \times 100$$

A. True

B. False

$$2 \times 37 \times 5 = 37 \times ?$$

- A.** 2
- B.** 74
- C.** 10
- D.** 7

Multiply.

$$3 \times 20 \times 5$$

Product: _____

True or false?

$$25 \times 3 \times 4 = 75 \times 12$$

A. True

B. False

SET 11 • 3.OA.B.5

FORWARD 2

Multiply by powers of 10 written with exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$4 \times 10^2$$

Product: _____

True or false?

$$7 \times 10 \times 10 \times 10 = 7 \times 10^3$$

A. True

B. False

Which is equal to 36×100 ?

A. $36 + 100$

B. 36×10^3

C. 36×10^2

D. 36×10

Find $25 \times 4 \times 10^2$.

Product: _____

True or false?

$$3 \times 10^2 = 3,000$$

A. True

B. False

Use the distributive property to multiply

MAIN	Grade 3 main question	3.OA.B.5
BACKWARD 1	Add two-digit numbers within 100	2.NBT.B.5
BACKWARD 2	Split an array into two parts and add the parts	2.OA.C.4
FORWARD 1	Multiply a two-digit number by a one-digit number with the distrib...	4.NBT.B.5
FORWARD 2	Evaluate expressions with parentheses that use the distributive pr...	5.OA.A.1

SET 12 • 3.OA.B.5

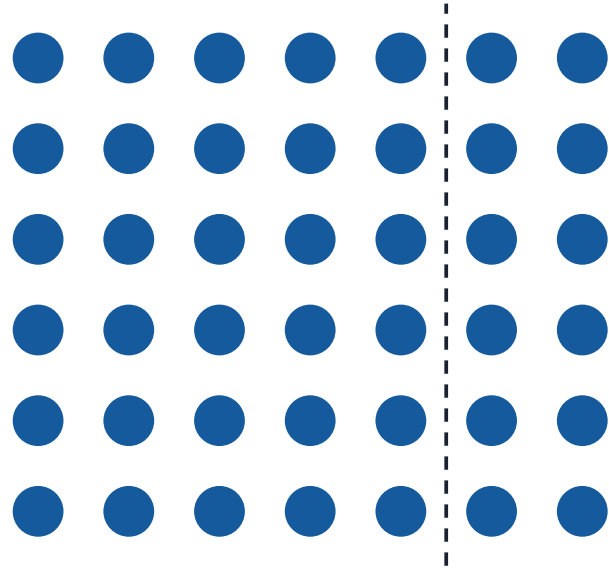
MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

The dashed line splits the array of 6 rows of 7 into 6 rows of 5 and 6 rows of 2.

Use $6 \times 7 = 6 \times 5 + 6 \times 2$ to find 6×7 .

Product: _____



Which is equal to 4×9 ?

A. $4 \times 5 + 4$

B. $4 \times 5 \times 4$

C. $4 + 9$

D. $4 \times 5 + 4 \times 4$

True or false?

$$9 \times 6 = 9 \times 3 + 9 \times 3$$

A. True

B. False

Use the distributive property to find 8×6 . Show the parts you add.

Product: _____

True or false?

$$7 \times 8 = 7 \times 4 + 4$$

A. True

B. False

SET 12 • 3.OA.B.5

BACKWARD 1

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$40 + 16$$

Sum: _____

True or false?

$$35 + 21 = 56$$

A. True

B. False

$$30 + 18 = ?$$

A. 38

B. 12

C. 48

D. 58

Add.

$$26 + 26$$

Sum: _____

True or false?

$$45 + 27 = 62$$

A. True

B. False

SET 12 • 3.OA.B.5

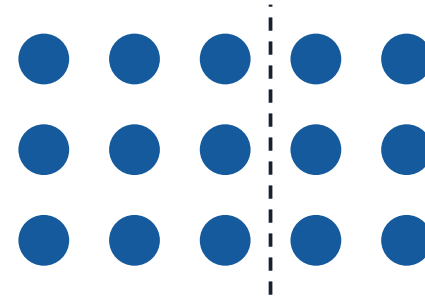
BACKWARD 2

Split an array into two parts and add the parts • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

The dashed line splits the array.

How many dots are on each side?

How many dots are there in all?



Left: _____

Right: _____

Total: _____

True or false?

An array of 4 rows of 5 dots can be split into 4 rows of 3 dots and 4 rows of 2 dots.

A. True

B. False

An array of 2 rows of 5 dots is split into 2 rows of 4 dots and 2 rows of 1 dot.

How many dots are there in all?

A. 10

B. 8

C. 12

D. 2

An array of 5 rows of 4 dots is split into two arrays of 5 rows of 2 dots.

How many dots are in each part?

Dots in each part: _____

True or false?

Splitting an array into two parts changes the total number of dots.

- A.** True
- B.** False

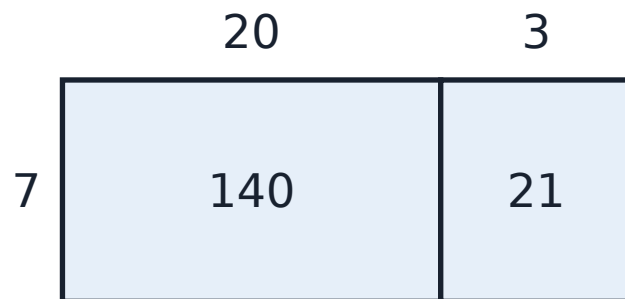
SET 12 • 3.OA.B.5

FORWARD 1

Multiply a two-digit number by a one-digit number with the distributive property • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Use the area model to find
 7×23 .

Product: _____



True or false?

$$6 \times 34 = 6 \times 30 + 6 \times 4$$

A. True

B. False

Which is equal to 8×45 ?

A. $8 \times 4 + 8 \times 5$

B. $8 \times 40 + 8 \times 5$

C. $8 \times 40 + 5$

D. $8 + 40 + 5$

Use the distributive property to find 5×68 . Show your work.

Product: _____

True or false?

$$9 \times 52 = 450 + 2$$

A. True

B. False

SET 12 • 3.OA.B.5

FORWARD 2

Evaluate expressions with parentheses that use the distributive property •

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (10 + 3)$$

Value: _____

True or false?

$$4 \times (25 + 5) = 4 \times 25 + 4 \times 5$$

A. True

B. False

Evaluate.

$$(7 \times 20) + (7 \times 4)$$

A. 168

B. 140

C. 31

D. 161

Evaluate.

$$3 \times (40 - 2)$$

Value: _____

True or false?

$$5 \times (8 + 2) = 42$$

A. True

B. False

Understand division as an unknown-factor problem

MAIN	Grade 3 main question	3.OA.B.6
BACKWARD 1	Understand subtraction as an unknown-addend problem	1.OA.B.4
BACKWARD 2	Write even numbers as the sum of two equal addends	2.OA.C.3
FORWARD 1	Use multiplication to find and check quotients	4.NBT.B.6
FORWARD 2	Divide a whole number by a unit fraction and check with multipli...	5.NF.B.7.b

SET 13 • 3.OA.B.6

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $36 \div 9$ by finding the unknown factor.

$$9 \times ? = 36$$

Quotient: _____

Which multiplication fact helps you find $42 \div 6$?

A. $42 - 6 = 36$

B. $6 + 7 = 13$

C. $6 \times 7 = 42$

D. $6 \times 6 = 36$

Write the unknown-factor equation for $54 \div 9$. Then find the quotient.

Equation: _____

Quotient: _____

True or false?

$56 \div 7 = 8$ because $7 \times 8 = 56$.

A. True

B. False

True or false?

$27 \div 3 = 8$ because $3 \times 8 = 27$.

A. True

B. False

SET 13 • 3.OA.B.6

BACKWARD 1

Understand subtraction as an unknown-addend problem • Grade 1 •
Operations & Algebraic Thinking • 1.OA.B.4

Find $15 - 9$ by thinking $9 + ? = 15$.

Difference: _____

True or false?

$13 - 5 = 8$ because $5 + 8 = 13$.

A. True

B. False

Which addition fact helps you find $17 - 8$?

A. $8 + 9 = 17$

B. $8 + 8 = 16$

C. $17 + 8 = 25$

D. $9 - 8 = 1$

Write the unknown-addend equation for $12 - 7$. Then find the difference.

Equation: _____

Difference: _____

True or false?

$14 - 6 = 9$ because $6 + 9 = 14$.

A. True

B. False

SET 13 • 3.OA.B.6

BACKWARD 2

Write even numbers as the sum of two equal addends • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

Write 18 as the sum of two equal addends.

Sum: _____

True or false?

$10 = 5 + 5$, so 10 is even.

A. True

B. False

Which number can be written as the sum of two equal whole-number addends?

A. 17

B. 14

C. 13

D. 9

What number plus itself makes 12?

Number: _____

True or false?

15 can be written as the sum of two equal whole-number addends.

A. True

B. False

SET 13 • 3.OA.B.6

FORWARD 1

Use multiplication to find and check quotients • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find $84 \div 4$ by thinking $4 \times ? = 84$.

Quotient: _____

True or false?

$96 \div 3 = 32$ because $3 \times 32 = 96$.

A. True

B. False

Which equation checks $135 \div 5 = 27$?

A. $135 \times 5 = 675$

B. $27 - 5 = 22$

C. $5 \times 27 = 135$

D. $5 + 27 = 32$

Write the unknown-factor equation for $224 \div 7$. Then find the quotient.

Equation: _____

Quotient: _____

True or false?

$152 \div 8 = 18$ because $8 \times 18 = 152$.

A. True

B. False

SET 13 • 3.OA.B.6

FORWARD 2

Divide a whole number by a unit fraction and check with multiplication • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Find $3 \div \frac{1}{4}$ by thinking $? \times \frac{1}{4} = 3$.

Quotient: _____

True or false?

$5 \div \frac{1}{2} = 10$ because $10 \times \frac{1}{2} = 5$.

A. True

B. False

Which equation checks $4 \div \frac{1}{3} = 12$?

A. $4 \times \frac{1}{3} = 12$

B. $12 \div 4 = 3$

C. $\frac{1}{3} + 12 = 4$

D. $12 \times \frac{1}{3} = 4$

Divide. Then write the multiplication equation that checks your answer.

$$2 \div \frac{1}{6}$$

Quotient: _____

Check: _____

True or false?

$6 \div \frac{1}{3} = 2$ because $2 \times \frac{1}{3} = 6$.

A. True

B. False

Multiply fluently within 100

MAIN	Grade 3 main question	3.OA.C.7
BACKWARD 1	Add equal addends to find a total	2.OA.C.4
BACKWARD 2	Add doubles within 20	2.OA.B.2
FORWARD 1	Multiply a two-digit number by a one-digit number	4.NBT.B.5
FORWARD 2	Multiply multi-digit whole numbers with the standard algorithm	5.NBT.B.5

SET 14 • 3.OA.C.7

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

Multiply.

$$9 \times 8$$

Product: _____

$$6 \times 8 = ?$$

A. 42

B. 48

C. 54

D. 14

True or false?

$$7 \times 7 = 49$$

A. True

B. False

True or false?

$$8 \times 4 = 36$$

A. True

B. False

SET 14 • 3.OA.C.7

BACKWARD 1

Add equal addends to find a total • Grade 2 • Operations & Algebraic Thinking
• 2.OA.C.4

Add the equal addends.

$$4 + 4 + 4 + 4 + 4$$

Sum: _____

True or false?

$$3 + 3 + 3 + 3 = 12$$

A. True

B. False

$$5 + 5 + 5 = ?$$

A. 555

B. 15

C. 10

D. 20

An array has 4 rows of 3 dots.

How many dots are there?

Dots: _____

True or false?

$$2 + 2 + 2 + 2 + 2 = 12$$

A. True

B. False

SET 14 • 3.OA.C.7

BACKWARD 2

Add doubles within 20 • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Add.

$$7 + 7$$

Sum: _____

True or false?

$$9 + 9 = 18$$

A. True

B. False

$$8 + 8 = ?$$

A. 15

B. 88

C. 18

D. 16

Add.

$$6 + 6$$

Sum: _____

True or false?

$$5 + 5 = 11$$

A. True

B. False

SET 14 • 3.OA.C.7

FORWARD 1

Multiply a two-digit number by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$7 \times 46$$

Product: _____

True or false?

$$8 \times 29 = 232$$

A. True

B. False

$$6 \times 57 = ?$$

A. 342

B. 63

C. 302

D. 352

Multiply.

$$9 \times 84$$

Product: _____

True or false?

$$4 \times 73 = 282$$

A. True

B. False

SET 14 • 3.OA.C.7

FORWARD 2

Multiply multi-digit whole numbers with the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$346 \times 27$$

Product: _____

Multiply.

$$58 \times 64$$

Product: _____

$$214 \times 35 = ?$$

- A.** 1,712
- B.** 7,390
- C.** 7,590
- D.** 7,490

True or false?

$$125 \times 48 = 6,000$$

A. True

B. False

True or false?

$$63 \times 47 = 2,861$$

A. True

B. False

Divide fluently within 100

MAIN

Grade 3 main question

3.OA.C.7

BACKWARD 1

Subtract within 100

2.NBT.B.5

BACKWARD 2

Split up to 20 objects into two equal groups

2.OA.C.3

FORWARD 1

Divide by a one-digit number, with remainders

4.NBT.B.6

FORWARD 2

Divide by a two-digit divisor

5.NBT.B.6

SET 15 • 3.OA.C.7

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$63 \div 7$$

Quotient: _____

Divide.

$$48 \div 8$$

Quotient: _____

$$54 \div 6 = ?$$

A. 48

B. 8

C. 9

D. 60

True or false?

$$72 \div 9 = 8$$

A. True

B. False

True or false?

$$42 \div 6 = 8$$

A. True

B. False

SET 15 • 3.OA.C.7

BACKWARD 1

Subtract within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$63 - 27$$

Difference: _____

True or false?

$$80 - 35 = 45$$

A. True

B. False

$$72 - 18 = ?$$

A. 56

B. 54

C. 90

D. 64

Subtract.

$$91 - 46$$

Difference: _____

True or false?

$$50 - 22 = 38$$

A. True

B. False

SET 15 • 3.OA.C.7

BACKWARD 2

Split up to 20 objects into two equal groups • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

16 counters are split into 2 equal groups.

How many counters are in each group?

Counters: _____

True or false?

20 objects make 2 equal groups of 10.

A. True

B. False

14 objects are split into 2 equal groups.

How many objects are in each group?

A. 28

B. 6

C. 12

D. 7

12 objects are split into 2 equal groups.

How many objects are in each group?

Objects: _____

True or false?

18 objects make 2 equal groups of 8.

A. True

B. False

SET 15 • 3.OA.C.7

FORWARD 1

Divide by a one-digit number, with remainders • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide. Write the quotient and the remainder.

$$58 \div 7$$

Quotient: _____

Remainder: _____

True or false?

$$76 \div 9 = 8 \text{ R } 4$$

A. True

B. False

Divide.

$$93 \div 4$$

A. 23

B. 23 R 1

C. 24 R 1

D. 22 R 5

Divide.

$$87 \div 3$$

Quotient: _____

True or false?

$$65 \div 8 = 7 \text{ R } 9$$

A. True

B. False

SET 15 • 3.OA.C.7

FORWARD 2

Divide by a two-digit divisor • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$864 \div 24$$

Quotient: _____

True or false?

$$1,104 \div 46 = 24$$

A. True

B. False

Divide.

$$975 \div 39$$

A. 25

B. 26

C. 24

D. 936

Divide. Write the quotient and the remainder.

$$700 \div 32$$

Quotient: _____

Remainder: _____

True or false?

$$1,512 \div 27 = 54$$

A. True

B. False

Use the relationship between multiplication and division

MAIN	Grade 3 main question	3.OA.C.7
BACKWARD 1	Use the relationship between addition and subtraction within 20	1.OA.C.6
BACKWARD 2	Subtract within 20 mentally	2.OA.B.2
FORWARD 1	Use multiplication to check division, including remainders	4.NBT.B.6
FORWARD 2	Divide a unit fraction by a whole number and check with multipli...	5.NF.B.7.a

SET 16 • 3.OA.C.7

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Write the four facts in the fact family for 6, 9, and 54.

Facts: _____

$$7 \times 8 = 56.$$

Which division fact is in the same fact family?

A. $56 \div 7 = 8$

B. $56 \times 7 = 8$

C. $7 \div 56 = 8$

D. $8 \div 56 = 7$

True or false?

Because $4 \times 9 = 36$, you know that $36 \div 4 = 9$.

A. True

B. False

Use multiplication to check: is $63 \div 9 = 7$ correct? Write the multiplication fact you used.

Correct? _____

Fact: _____

True or false?

Because $5 \times 8 = 40$, you know that $40 \div 8 = 4$.

A. True

B. False

SET 16 • 3.OA.C.7

BACKWARD 1

Use the relationship between addition and subtraction within 20 • Grade 1 •
Operations & Algebraic Thinking • 1.OA.C.6

Write the four facts in the fact family for 7, 8, and 15.

Facts: _____

True or false?

Because $9 + 4 = 13$, you know that $13 - 4 = 9$.

A. True

B. False

$$6 + 5 = 11.$$

Which subtraction fact is in the same fact family?

A. $6 - 5 = 1$

B. $11 - 5 = 6$

C. $11 - 4 = 7$

D. $5 - 6 = 11$

Is $16 - 7 = 9$ correct? Write the addition fact you used to check.

Correct? _____

Fact: _____

True or false?

Because $8 + 6 = 14$, you know that $14 - 6 = 7$.

A. True

B. False

SET 16 • 3.OA.C.7

BACKWARD 2

Subtract within 20 mentally • Grade 2 • Operations & Algebraic Thinking •
2.OA.B.2

Subtract.

$$17 - 9$$

Difference: _____

True or false?

$$14 - 7 = 7$$

A. True

B. False

$$12 - 5 = ?$$

A. 7

B. 8

C. 17

D. 6

Subtract.

$$13 - 8$$

Difference: _____

True or false?

$$16 - 8 = 9$$

A. True

B. False

SET 16 • 3.OA.C.7

FORWARD 1

Use multiplication to check division, including remainders • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Check: is $47 \div 5 = 9 \text{ R } 2$ correct? Write the multiplication and addition you used.

Correct? _____

Check: _____

True or false?

$34 \div 6 = 5 \text{ R } 4$ because $6 \times 5 + 4 = 34$.

A. True

B. False

Which equation shows that $75 \div 8 = 9 \text{ R } 3$?

A. $8 + 9 + 3 = 75$

B. $8 \times 9 = 75$

C. $9 \times 3 + 8 = 75$

D. $8 \times 9 + 3 = 75$

Use multiplication to check: is $96 \div 4 = 24$ correct?
Write the multiplication you used.

Correct? _____

Check: _____

True or false?

$59 \div 7 = 8 \text{ R } 2$ because $7 \times 8 + 2 = 59$.

A. True

B. False

SET 16 • 3.OA.C.7

FORWARD 2

Divide a unit fraction by a whole number and check with multiplication • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a

Find $\frac{1}{3} \div 4$. Then write the multiplication that checks it.

Quotient: _____

Check: _____

True or false?

$$\frac{1}{2} \div 3 = \frac{1}{6} \text{ because } \frac{1}{6} \times 3 = \frac{1}{2}.$$

A. True

B. False

Which equation checks $\frac{1}{4} \div 2 = \frac{1}{8}$?

A. $\frac{1}{4} \times 2 = \frac{1}{8}$

B. $\frac{1}{8} \times 2 = \frac{1}{4}$

C. $2 \times 4 = 8$

D. $\frac{1}{8} \times \frac{1}{4} = 2$

Divide.

$$\frac{1}{5} \div 2$$

Quotient: _____

True or false?

$$\frac{1}{6} \div 2 = \frac{1}{3}$$

A. True

B. False

Solve two-step word problems using the four operations

MAIN	Grade 3 main question	3.OA.D.8
BACKWARD 1	One-step take-from word problems within 100	2.OA.A.1
BACKWARD 2	Find the total of equal rows by repeated addition	2.OA.C.4
FORWARD 1	Solve multistep word problems with whole numbers	4.OA.A.3
FORWARD 2	Write expressions that record multistep calculations	5.OA.A.2

SET 17 • 3.OA.D.8

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Mia has 4 bags with 6 marbles in each bag. She gives away 9 marbles.

How many marbles does she have left?

Marbles: _____

A class has 27 students, and 3 more students join. The class is split into 5 equal teams.

How many students are on each team?

Students: _____

Pens cost \$3 each. Leo buys 7 pens and pays with a \$50 bill.

How much change does he get?

A. \$21

B. \$53

C. \$47

D. \$29

True or false?

A farmer has 32 eggs. He packs them 8 to a carton and sells 2 cartons. He has 2 cartons left.

A. True

B. False

Ana read 8 pages on Monday and 4 times as many pages on Tuesday.

How many pages did she read on the two days?

Pages: _____

SET 17 • 3.OA.D.8

BACKWARD 1

One-step take-from word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

There were 72 people on a train. 26 got off.
How many people are still on the train?

People: _____

True or false?

Kim had 90 cents and spent 45 cents. She has 45 cents left.

A. True

B. False

A store had 60 hats and sold 34.

How many hats are left?

A. 36

B. 26

C. 34

D. 94

A tree had 83 apples. 47 fell off.

How many apples are still on the tree?

Apples: _____

True or false?

A box had 55 nails. 18 were used, so 47 are left.

A. True

B. False

SET 17 • 3.OA.D.8

BACKWARD 2

Find the total of equal rows by repeated addition • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

There are 4 rows of 5 chairs.

Use repeated addition to find the number of chairs.

Chairs: _____

True or false?

3 rows of 4 tiles is $4 + 4 + 4 = 12$ tiles.

A. True

B. False

There are 5 rows of 2 trees.

Which addition gives the total?

A. $2 + 2 + 2 + 2 + 2$

B. $5 + 2$

C. $2 + 5 + 2$

D. $5 + 5$

There are 3 rows of 3 cups.

How many cups are there?

Cups: _____

True or false?

2 rows of 4 desks is $4 + 2 = 6$ desks.

A. True

B. False

SET 17 • 3.OA.D.8

FORWARD 1

Solve multistep word problems with whole numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A store gets 6 boxes of 48 cans. It sells 175 cans.
How many cans are left?

Cans: _____

True or false?

Tickets cost \$12 for adults and \$7 for children. 3 adults and 4 children pay \$64 in all.

- A. True
- B. False

A school has 4 classes of 26 students and 3 classes of 24 students.

How many students are in these classes?

A. 72

B. 160

C. 176

D. 104

A library has 1,250 books. It buys 8 boxes of 35 books.
How many books does it have now?

Books: _____

True or false?

Leo saves \$15 a week for 6 weeks and then spends \$38. He has \$62 left.

- A.** True
- B.** False

SET 17 • 3.OA.D.8

FORWARD 2

Write expressions that record multistep calculations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for: buy 4 packs of 12 pencils, then give away 9 pencils.

Expression: _____

27 students are joined by 3 more, and then they are split into 5 equal teams.

Which expression matches?

A. $(27 + 3) \div 5$

B. $27 + 3 \div 5$

C. $27 + (3 \div 5)$

D. $5 \div (27 + 3)$

True or false?

The expression $50 - 3 \times 7$ gives the change from \$50 after buying 7 pens that cost \$3 each.

- A. True
- B. False

Evaluate.

$$(18 + 6) \div 4 \times 3$$

Value: _____

True or false?

The expression $8 + 4 \times 8$ has a value of 96.

- A. True
- B. False

Represent two-step word problems with an equation with a letter for the unknown

MAIN	Grade 3 main question	3.OA.D.8
BACKWARD 1	Write an equation with a symbol for the unknown in a one-step wo...	2.OA.A.1
BACKWARD 2	Subtract two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Write and solve equations with a letter for multistep problems	4.OA.A.3
FORWARD 2	Write and interpret numerical expressions without evaluating th...	5.OA.A.2

SET 18 • 3.OA.D.8

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

A baker makes 5 trays of 8 rolls. She sells 23 rolls.

Write an equation with a letter for the number of rolls left. Then solve it.

Equation: _____

Rolls left: _____

Sam has \$40. He buys 3 books that cost \$9 each.

Which equation gives the money he has left, m ?

A. $m = 40 - 3 + 9$

B. $m = 3 \times 9 - 40$

C. $m = 40 + 3 \times 9$

D. $m = 40 - 3 \times 9$

There are 36 cupcakes. 4 are eaten, and the rest are shared equally by 8 friends.

Write an equation with a letter for each friend's share. Then solve it.

Equation: _____

Each friend: _____

True or false?

"Jo had 15 stickers and then got 6 packs of 5 stickers" can be written as $s = 15 + 6 \times 5$, and $s = 45$.

A. True

B. False

True or false?

"7 boxes of 6 crayons, then 10 crayons are lost" can be written as $c = 7 + 6 - 10$.

A. True

B. False

SET 18 • 3.OA.D.8

BACKWARD 1

Write an equation with a symbol for the unknown in a one-step word problem •
Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Ty had some marbles. He got 25 more and now has 61.
Write an equation with a symbol for the unknown.

Equation: _____

True or false?

"44 ducks were on a pond, and 19 flew away" can be written as $44 - 19 = d$.

A. True

B. False

There are 28 red cars and some blue cars. There are 50 cars in all.

Which equation matches?

A. $b - 50 = 28$

B. $50 + 28 = b$

C. $28 - b = 50$

D. $28 + b = 50$

A pole is 70 cm long. Lu cuts off some, and 42 cm is left.
Write an equation with a symbol for the unknown.

Equation: _____

True or false?

"There were 36 books, and 12 more came" can be written as $36 - 12 = b$.

A. True

B. False

SET 18 • 3.OA.D.8

BACKWARD 2

Subtract two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$70 - 34$$

Difference: _____

True or false?

$$58 - 29 = 29$$

A. True

B. False

$$84 - 47 = ?$$

A. 37

B. 131

C. 43

D. 47

Subtract.

$$65 - 38$$

Difference: _____

True or false?

$$93 - 55 = 48$$

A. True

B. False

SET 18 • 3.OA.D.8

FORWARD 1

Write and solve equations with a letter for multistep problems • Grade 4 •
Operations & Algebraic Thinking • 4.OA.A.3

A club buys 6 packs of 15 cups and uses 58 cups.

Write an equation with a letter for the cups left. Then solve it.

Equation: _____

Cups left: _____

A farmer has 120 eggs. She sells 7 cartons of 12 eggs.
Which equation gives the eggs left, e ?

A. $e = 120 - 7 \times 12$

B. $e = 120 - 7 + 12$

C. $e = 7 \times 12 - 120$

D. $e = 120 \div 7 \times 12$

True or false?

After a \$16 gift, the rest of the cost of a \$96 bike is paid in 8 equal parts. This can be written as $p = (96 - 16) \div 8$, and $p = 10$.

- A.** True
- B.** False

4 vans carry 12 students each, and 3 cars carry 5 students each.

Write an equation with a letter for the number of students. Then solve it.

Equation: _____

Students: _____

True or false?

"A hall has 250 seats, and 9 rows of 24 seats are filled" can be written as $s = 250 - 9 + 24$ for the empty seats.

A. True

B. False

SET 18 • 3.OA.D.8

FORWARD 2

Write and interpret numerical expressions without evaluating them • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "3 times the difference of 40 and 12."

Expression: _____

True or false?

$(24 + 16) \div 8$ is one-eighth of $24 + 16$.

A. True

B. False

Without calculating, how does $5 \times (92 - 17)$ compare with $92 - 17$?

- A.** It is 5 times as large.
- B.** It is 5 less.
- C.** It is 5 more.
- D.** It is the same.

Write an expression for "add 15 and 9, then divide by 6."

Expression: _____

True or false?

$7 \times (18 + 3)$ is 7 more than $18 + 3$.

A. True

B. False

Check whether answers are reasonable using mental computation and rounding

MAIN	Grade 3 main question	3.OA.D.8
BACKWARD 1	Add or subtract 10 or 100 mentally	2.NBT.B.8
BACKWARD 2	Compare three-digit numbers	2.NBT.A.4
FORWARD 1	Use rounding to check answers to multistep problems	4.OA.A.3
FORWARD 2	Round decimals to estimate sums, differences, and products	5.NBT.A.4

SET 19 • 3.OA.D.8

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Round each number to the nearest ten to estimate
 $48 + 31$.

Then tell whether a sum of 79 is reasonable.

Estimate: _____

Reasonable? _____

Leo says $412 - 189 = 323$.

Round each number to the nearest hundred to estimate.
Is his answer reasonable?

Estimate: _____

Reasonable? _____

A store sold 296 shirts on Monday and 404 shirts on Tuesday.

Rounding each number to the nearest hundred, which is the estimate of the total?

A. 600

B. 900

C. 800

D. 700

True or false?

Rounding each number to the nearest ten,
 $62 + 27$ is about 90.

- A. True
- B. False

True or false?

Rounding each number to the nearest hundred,
 $589 - 312$ is about 200.

- A. True
- B. False

SET 19 • 3.OA.D.8

BACKWARD 1

Add or subtract 10 or 100 mentally • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8

What is 100 more than 453?

Answer: _____

True or false?

10 less than 280 is 270.

A. True

B. False

What is 100 less than 615?

A. 614

B. 715

C. 515

D. 605

What is 10 more than 395?

Answer: _____

True or false?

100 more than 860 is 870.

A. True

B. False

SET 19 • 3.OA.D.8

BACKWARD 2

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Write $>$, $=$, or $<$.

482 428

Symbol: _____

True or false?

$$609 < 690$$

A. True

B. False

Which number is greatest?

A. 713

B. 731

C. 371

D. 317

Write $>$, $=$, or $<$.

350 305

Symbol: _____

True or false?

$$298 > 302$$

A. True

B. False

SET 19 • 3.OA.D.8

FORWARD 1

Use rounding to check answers to multistep problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Ty says $6 \times 48 - 95 = 193$.

Round 48 and 95 to the nearest ten. Estimate, then tell whether his answer is reasonable.

Estimate: _____

Reasonable? _____

True or false?

Rounding each number to the nearest hundred,
 $1,212 + 389$ is about 1,600.

A. True

B. False

Rounding each number to the nearest ten, which is the estimate of $4 \times 39 + 52$?

- A.** 160
- B.** 210
- C.** 190
- D.** 200

Round each number to the nearest thousand to estimate $6,814 - 2,390$.

Estimate: _____

True or false?

Rounding 21 to the nearest ten, 8×21 is about 240.

A. True

B. False

SET 19 • 3.OA.D.8

FORWARD 2

Round decimals to estimate sums, differences, and products • Grade 5 •
Number & Operations in Base Ten • 5.NBT.A.4

Round each number to the nearest whole number. Then estimate.

$$3.8 + 6.4$$

Estimate: _____

True or false?

Rounding each number to the nearest whole number, $9.7 - 2.2$ is about 8.

A. True

B. False

Round each number to the nearest whole number.

Which is the estimate of 5.6×3.1 ?

A. 16

B. 15

C. 20

D. 18

Round each number to the nearest whole number. Then estimate.

$$7.4 \times 2.8$$

Estimate: _____

True or false?

Rounding each number to the nearest whole number, $12.4 + 3.7$ is about 15.

A. True

B. False

Identify arithmetic patterns in the addition and multiplication tables

MAIN	Grade 3 main question	3.OA.D.9
BACKWARD 1	Odd and even numbers up to 20	2.OA.C.3
BACKWARD 2	Skip-count by 5s, 10s, and 100s and see patterns	2.NBT.A.2
FORWARD 1	Generate a pattern from a rule and identify its features	4.OA.C.5
FORWARD 2	Generate two numerical patterns and compare their terms	5.OA.B.3

SET 20 • 3.OA.D.9

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

True or false?

Every product in the 4s row of the multiplication table is even.

- A. True
- B. False

The 5s row of the multiplication table is 5, 10, 15, 20, 25, ...

What pattern do you see in the ones digits?

Pattern: _____

In the 9s row of the multiplication table (9, 18, 27, 36, ...), what happens to the digits from one product to the next?

- A.** The tens digit goes up by 1 and the ones digit goes down by 1.
- B.** Both digits go up by 1.
- C.** The digits stay the same.
- D.** Both digits go down by 1.

In the addition table, look at the sums $1 + 9$, $2 + 8$, $3 + 7$, and $4 + 6$.

What do you notice?

Pattern: _____

True or false?

Every product in the 3s row of the multiplication table is odd.

- A. True
- B. False

SET 20 • 3.OA.D.9

BACKWARD 1

Odd and even numbers up to 20 • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.3

Is 17 odd or even?

Answer: _____

True or false?

14 is even.

A. True

B. False

Which number is odd?

A. 16

B. 19

C. 12

D. 20

Write 14 as the sum of two equal addends.

Sum: _____

True or false?

An odd number of objects can be put into two equal groups with none left over.

- A.** True
- B.** False

SET 20 • 3.OA.D.9

BACKWARD 2

Skip-count by 5s, 10s, and 100s and see patterns • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 10s from 230.

230, 240, 250, __, __

Next two: _____

True or false?

When you skip-count by 5s from 5, every number ends in 5 or 0.

A. True

B. False

What comes next?

400, 500, 600, ____

A. 601

B. 610

C. 800

D. 700

Skip-count by 5s from 65.

65, 70, 75, __, __

Next two: _____

True or false?

When you skip-count by 10s from 7, every number ends in 0.

A. True

B. False

SET 20 • 3.OA.D.9

FORWARD 1

Generate a pattern from a rule and identify its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 6," starting at 3.

Write the first five terms.

Terms: _____

True or false?

The rule "add 4," starting at 1, makes only odd numbers.

A. True

B. False

The rule is "multiply by 2," starting at 5.

What is the 4th term?

A. 20

B. 80

C. 13

D. 40

The rule is "add 5," starting at 2: 2, 7, 12, 17, ...

What do you notice about the ones digits?

Pattern: _____

True or false?

The rule "add 3," starting at 4, makes only even numbers.

A. True

B. False

SET 20 • 3.OA.D.9

FORWARD 2

Generate two numerical patterns and compare their terms • Grade 5 •
Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 6.

Write the first four terms of each pattern.

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. Each term of B is 2 times the matching term of A.

A. True

B. False

Pattern A: 0, 4, 8, 12. Pattern B: 0, 12, 24, 36.

How does each B term compare with the matching A term?

- A.** It is the same.
- B.** It is 4 times as large.
- C.** It is 3 times as large.
- D.** It is 8 more.

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8.

What is the B term when the A term is 10?

B term: _____

True or false?

Pattern A: 0, 2, 4, 6. Pattern B: 0, 6, 12, 18. Each B term is 4 more than the matching A term.

A. True

B. False

Explain arithmetic patterns using properties of operations

MAIN

Grade 3 main question

3.OA.D.9

BACKWARD 1

Explain why a number is even or odd

2.OA.C.3

BACKWARD 2

Explain that addends can be added in any order

1.OA.B.3

FORWARD 1

Explain features of a number pattern

4.OA.C.5

FORWARD 2

Explain the relationship between two numerical patterns

5.OA.B.3

SET 21 • 3.OA.D.9

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

Explain why 4 times any whole number is always even.

Explanation: _____

Why is every product in the 6s row of the multiplication table even?

- A.** The 6s row has 10 products.
- B.** 6 is greater than 5.
- C.** $6 = 2 \times 3$, so each product is 2 times a number.
- D.** Every number is even.

True or false?

The sum of two even numbers is always even.

A. True

B. False

Is the product of two odd numbers odd or even? Give an example.

Answer: _____

Example: _____

True or false?

An odd number times 2 is always odd.

A. True

B. False

SET 21 • 3.OA.D.9

BACKWARD 1

Explain why a number is even or odd • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

Explain why 12 is even.

Explanation: _____

True or false?

An even number can be written as the sum of two equal addends.

A. True

B. False

Which equation shows that 10 is even?

A. $10 = 10 + 0$

B. $10 = 7 + 3$

C. $10 = 5 + 5$

D. $10 = 9 + 1$

Is 15 even or odd? Explain.

Answer: _____

Explanation: _____

True or false?

7 can be written as the sum of two equal whole-number addends.

- A. True
- B. False

SET 21 • 3.OA.D.9

BACKWARD 2

Explain that addends can be added in any order • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3

Explain why $6 + 9 = 9 + 6$.

Explanation: _____

True or false?

$$5 + 8 = 8 + 5$$

A. True

B. False

Which equation shows adding in a different order?

A. $4 + 7 = 47$

B. $4 + 7 = 7 + 4$

C. $4 + 7 = 4 - 7$

D. $4 + 7 = 7 - 4$

$$9 + 3 = 12.$$

What is $3 + 9$?

Sum: _____

True or false?

$$8 + 2 = 2 + 6$$

A. True

B. False

SET 21 • 3.OA.D.9

FORWARD 1

Explain features of a number pattern • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 4," starting at 2: 2, 6, 10, 14, ...

Why is every term even?

Explanation: _____

True or false?

The rule "add 5," starting at 1, makes terms that go odd, even, odd, even.

A. True

B. False

The rule is "add 10," starting at 3.

Why does every term end in 3?

- A.** Adding 10 does not change the ones digit.
- B.** The terms get larger.
- C.** 3 is odd.
- D.** 10 is even.

The rule is "multiply by 3," starting at 1: 1, 3, 9, 27, ...

Are the terms odd or even? Explain.

Answer: _____

Explanation: _____

True or false?

The rule "add 2," starting at 5, makes only even numbers.

A. True

B. False

SET 21 • 3.OA.D.9

FORWARD 2

Explain the relationship between two numerical patterns • Grade 5 •
Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

Explain why each B term is 3 times the matching A term.

Explanation: _____

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 15. Each B term is 3 times the matching A term.

A. True

B. False

Pattern A starts at 1 and adds 1. Pattern B starts at 1 and adds 2.

Why is B NOT always 2 times A?

- A.** A has more terms.
- B.** Both rules add.
- C.** B adds less than A.
- D.** They both start at 1, not at 0.

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12.

What is the A term when the B term is 36?

A term: _____

True or false?

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9. Each B term is 6 more than the matching A term.

A. True

B. False

Round whole numbers to the nearest 10

MAIN	Grade 3 main question	3.NBT.A.1
BACKWARD 1	Hundreds, tens, and ones in three-digit numbers	2.NBT.A.1
BACKWARD 2	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Round multi-digit whole numbers to any place	4.NBT.A.3
FORWARD 2	Round decimals to any place	5.NBT.A.4

SET 22 • 3.NBT.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.1

Round 47 to the nearest ten.

Rounded: _____

True or false?

118 rounded to the nearest ten is 110.

A. True

B. False

Round 685 to the nearest ten.

A. 600

B. 680

C. 690

D. 700

True or false?

234 rounded to the nearest ten is 230.

A. True

B. False

Use the number line.

Is 76 closer to 70 or to 80? Round 76 to the nearest ten.



Closer to: _____

Rounded: _____

SET 22 • 3.NBT.A.1

BACKWARD 1

Hundreds, tens, and ones in three-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1

What digit is in the tens place of 563?

Digit: _____

True or false?

In 708, the 0 means 0 tens.

A. True

B. False

Which number has 3 hundreds, 8 tens, and 2 ones?

- A.** 328
- B.** 832
- C.** 3,082
- D.** 382

What is the value of the 4 in 245?

Value: _____

True or false?

In 961, the 6 is worth 6.

A. True

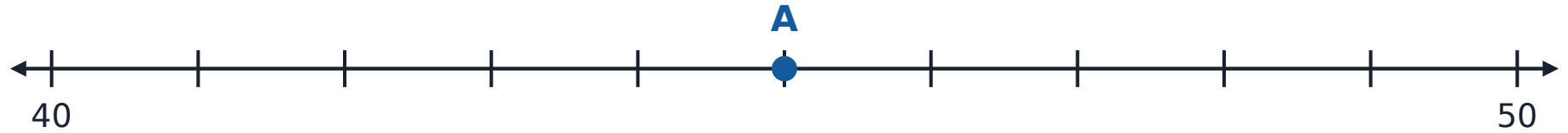
B. False

SET 22 • 3.NBT.A.1

BACKWARD 2

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line, 68 is between 60 and 70.

A. True

B. False

Which number is closest to 30 on a number line?

A. 32

B. 37

C. 39

D. 36

What number is at point B?



B = _____

True or false?

On a number line, 54 is closer to 60 than to 50.

A. True

B. False

SET 22 • 3.NBT.A.1

FORWARD 1

Round multi-digit whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 3,467 to the nearest ten.

Rounded: _____

True or false?

12,845 rounded to the nearest hundred is 12,800.

- A. True
- B. False

Round 58,349 to the nearest thousand.

- A.** 58,000
- B.** 60,000
- C.** 59,000
- D.** 58,300

Round 7,996 to the nearest ten.

Rounded: _____

True or false?

41,705 rounded to the nearest ten thousand is 50,000.

A. True

B. False

SET 22 • 3.NBT.A.1

FORWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 4.67 to the nearest tenth.

Rounded: _____

True or false?

2.349 rounded to the nearest hundredth is 2.35.

A. True

B. False

Round 8.52 to the nearest whole number.

A. 8.6

B. 9

C. 8

D. 8.5

Round 0.938 to the nearest hundredth.

Rounded: _____

True or false?

6.45 rounded to the nearest tenth is 6.4.

A. True

B. False

Round whole numbers to the nearest 100

MAIN	Grade 3 main question	3.NBT.A.1
BACKWARD 1	Read and write numbers to 1000 in expanded form	2.NBT.A.3
BACKWARD 2	Three-digit numbers on a number line	2.MD.B.6
FORWARD 1	Round multi-digit whole numbers to the hundreds, thousands, an...	4.NBT.A.3
FORWARD 2	Round decimals to the nearest whole number, tenth, or hundredth	5.NBT.A.4

SET 23 • 3.NBT.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.1

Round 382 to the nearest hundred.

Rounded: _____

True or false?

749 rounded to the nearest hundred is 700.

A. True

B. False

Round 650 to the nearest hundred.

A. 700

B. 610

C. 650

D. 600

True or false?

216 rounded to the nearest hundred is 300.

A. True

B. False

A school has 538 students.

Round the number of students to the nearest hundred.

Rounded: _____

SET 23 • 3.NBT.A.1

BACKWARD 1

Read and write numbers to 1000 in expanded form • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3

Write 627 in expanded form.

Expanded form: _____

True or false?

Four hundred five is written 405.

A. True

B. False

Which number is $300 + 70 + 4$?

- A.** 734
- B.** 374
- C.** 3,074
- D.** 347

Write the number eight hundred sixty with digits.

Number: _____

True or false?

$$500 + 9 = 590$$

A. True

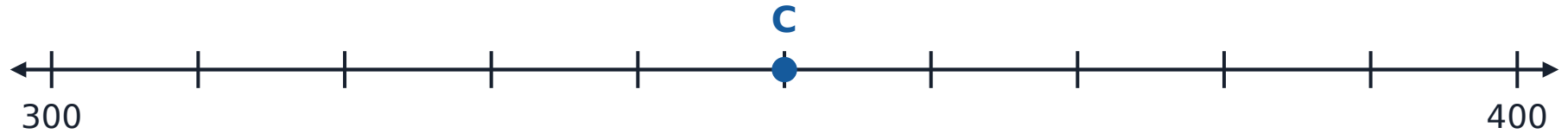
B. False

SET 23 • 3.NBT.A.1

BACKWARD 2

Three-digit numbers on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

What number is at point C?



C = _____

True or false?

On a number line, 720 is between 700 and 800.

A. True

B. False

Which number is closer to 600 than to 500?

- A.** 510
- B.** 520
- C.** 540
- D.** 580

What number is halfway between 200 and 300 on a number line?

Number: _____

True or false?

On a number line, 430 is closer to 500 than to 400.

A. True

B. False

SET 23 • 3.NBT.A.1

FORWARD 1

Round multi-digit whole numbers to the hundreds, thousands, and beyond •

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 6,251 to the nearest hundred.

Rounded: _____

True or false?

94,512 rounded to the nearest thousand is 95,000.

A. True

B. False

Round 238,670 to the nearest hundred thousand.

- A.** 240,000
- B.** 300,000
- C.** 200,000
- D.** 238,700

Round 9,950 to the nearest hundred.

Rounded: _____

True or false?

3,049 rounded to the nearest hundred is 3,100.

A. True

B. False

SET 23 • 3.NBT.A.1

FORWARD 2

Round decimals to the nearest whole number, tenth, or hundredth • Grade 5 •
Number & Operations in Base Ten • 5.NBT.A.4

Round 13.48 to the nearest whole number.

Rounded: _____

True or false?

0.75 rounded to the nearest tenth is 0.8.

A. True

B. False

Round 2.963 to the nearest tenth.

- A.** 2.8
- B.** 2.96
- C.** 2.9
- D.** 3.0

Round 5.096 to the nearest hundredth.

Rounded: _____

True or false?

9.04 rounded to the nearest tenth is 9.1.

A. True

B. False

Add within 1000 using place value strategies and algorithms

MAIN	Grade 3 main question	3.NBT.A.2
BACKWARD 1	Add within 100	2.NBT.B.5
BACKWARD 2	Add within 1000 by adding hundreds, tens, and ones	2.NBT.B.7
FORWARD 1	Add multi-digit whole numbers with the standard algorithm	4.NBT.B.4
FORWARD 2	Add decimals to hundredths	5.NBT.B.7

SET 24 • 3.NBT.A.2

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.2

Add. Show your work using place value or the standard algorithm.

$$358 + 276$$

Sum: _____

Add.

$$407 + 395$$

Sum: _____

$$529 + 263 = ?$$

A. 782

B. 792

C. 266

D. 702

True or false?

$$186 + 614 = 800$$

A. True

B. False

True or false?

$$475 + 348 = 813$$

A. True

B. False

SET 24 • 3.NBT.A.2

BACKWARD 1

Add within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$58 + 36$$

Sum: _____

True or false?

$$47 + 25 = 72$$

A. True

B. False

$$29 + 64 = ?$$

A. 94

B. 83

C. 93

D. 35

Add.

$$75 + 18$$

Sum: _____

True or false?

$$66 + 27 = 83$$

A. True

B. False

SET 24 • 3.NBT.A.2

BACKWARD 2

Add within 1000 by adding hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

Add using hundreds, tens, and ones.

$$234 + 152$$

Sum: _____

True or false?

$$300 + 400 = 700$$

A. True

B. False

Which shows the sum $245 + 132$ by place value?

A. $200 + 70 + 7$

B. $300 + 70 + 7$

C. $300 + 70 + 5$

D. $300 + 60 + 7$

Add.

$$420 + 160$$

Sum: _____

True or false?

$$513 + 241 = 744$$

A. True

B. False

SET 24 • 3.NBT.A.2

FORWARD 1

Add multi-digit whole numbers with the standard algorithm • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Use the standard algorithm to add. Show your work.

$$3,768 + 2,457$$

Sum: _____

Add.

$$45,092 + 18,659$$

Sum: _____

$$2,806 + 4,395 = ?$$

A. 7,201

B. 7,191

C. 7,101

D. 6,201

True or false?

$$9,999 + 1 = 10,000$$

A. True

B. False

True or false?

$$6,485 + 3,729 = 10,114$$

A. True

B. False

SET 24 • 3.NBT.A.2

FORWARD 2

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$3.45 + 2.8$$

Sum: _____

True or false?

$$0.6 + 0.47 = 1.07$$

A. True

B. False

Add.

$$12.5 + 3.75$$

A. 16.25

B. 15.80

C. 16.20

D. 4.00

Add.

$$4.09 + 5.91$$

Sum: _____

True or false?

$$2.7 + 1.35 = 3.12$$

A. True

B. False

Subtract within 1000 using place value strategies and algorithms

MAIN	Grade 3 main question	3.NBT.A.2
BACKWARD 1	Subtract within 100	2.NBT.B.5
BACKWARD 2	Subtract within 1000 by subtracting hundreds, tens, and ones	2.NBT.B.7
FORWARD 1	Subtract multi-digit whole numbers with the standard algorithm	4.NBT.B.4
FORWARD 2	Subtract decimals to hundredths	5.NBT.B.7

SET 25 • 3.NBT.A.2

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.2

Subtract. Show your work using place value or the standard algorithm.

$$623 - 358$$

Difference: _____

Subtract.

$$700 - 246$$

Difference: _____

$$815 - 379 = ?$$

- A.** 546
- B.** 1,194
- C.** 436
- D.** 444

True or false?

$$902 - 518 = 384$$

A. True

B. False

True or false?

$$560 - 287 = 383$$

A. True

B. False

SET 25 • 3.NBT.A.2

BACKWARD 1

Subtract within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$82 - 45$$

Difference: _____

True or false?

$$60 - 23 = 37$$

A. True

B. False

$$74 - 39 = ?$$

- A.** 35
- B.** 25
- C.** 113
- D.** 45

Subtract.

$$53 - 17$$

Difference: _____

True or false?

$$91 - 58 = 43$$

A. True

B. False

SET 25 • 3.NBT.A.2

BACKWARD 2

Subtract within 1000 by subtracting hundreds, tens, and ones • Grade 2 •
Number & Operations in Base Ten • 2.NBT.B.7

Subtract using hundreds, tens, and ones.

$$587 - 243$$

Difference: _____

True or false?

$$800 - 300 = 500$$

A. True

B. False

$$465 - 120 = ?$$

A. 245

B. 355

C. 345

D. 585

Subtract.

$$690 - 250$$

Difference: _____

True or false?

$$758 - 426 = 342$$

A. True

B. False

SET 25 • 3.NBT.A.2

FORWARD 1

Subtract multi-digit whole numbers with the standard algorithm • Grade 4 •
Number & Operations in Base Ten • 4.NBT.B.4

Use the standard algorithm to subtract. Show your work.

$$7,203 - 4,568$$

Difference: _____

Subtract.

$$50,000 - 23,418$$

Difference: _____

$$8,412 - 3,756 = ?$$

A. 4,656

B. 5,344

C. 4,666

D. 4,756

True or false?

$$6,001 - 2,999 = 3,002$$

A. True

B. False

True or false?

$$9,350 - 4,875 = 4,575$$

A. True

B. False

SET 25 • 3.NBT.A.2

FORWARD 2

Subtract decimals to hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Subtract.

$$6.4 - 2.75$$

Difference: _____

True or false?

$$5 - 1.25 = 3.75$$

A. True

B. False

Subtract.

$$10.3 - 4.68$$

- A.** 5.72
- B.** 14.98
- C.** 6.38
- D.** 5.62

Subtract.

$$8.06 - 3.9$$

Difference: _____

True or false?

$$7.5 - 2.25 = 5.35$$

A. True

B. False

Use the relationship between addition and subtraction to subtract and to check answers

MAIN	Grade 3 main question	3.NBT.A.2
BACKWARD 1	Use addition facts to subtract within 20	1.OA.C.6
BACKWARD 2	Use the relationship between addition and subtraction within 100	2.NBT.B.5
FORWARD 1	Check multi-digit subtraction with addition	4.NBT.B.4
FORWARD 2	Check decimal subtraction with addition	5.NBT.B.7

SET 26 • 3.NBT.A.2

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.2

Find $500 - 287$ by counting up from 287.

Difference: _____

Which addition checks $642 - 275 = 367$?

A. $275 + 642 = 367$

B. $367 + 275 = 642$

C. $367 + 642 = 275$

D. $642 + 275 = 367$

True or false?

Because $318 + 457 = 775$, you know that $775 - 457 = 318$.

- A. True
- B. False

Use addition to check: is $803 - 426 = 387$ correct?
Show the addition you used.

Correct? _____

Check: _____

True or false?

Because $245 + 380 = 625$, you know that $625 - 245 = 385$.

- A. True
- B. False

SET 26 • 3.NBT.A.2

BACKWARD 1

Use addition facts to subtract within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6

Find $16 - 9$ by thinking $9 + ? = 16$.

Difference: _____

True or false?

$11 - 4 = 7$ because $4 + 7 = 11$.

A. True

B. False

Which addition fact checks $15 - 6 = 9$?

A. $9 + 6 = 15$

B. $9 - 6 = 3$

C. $6 + 6 = 12$

D. $15 + 6 = 21$

Write the addition fact that checks $18 - 9 = 9$.

Fact: _____

True or false?

$12 - 8 = 5$ because $8 + 5 = 12$.

A. True

B. False

SET 26 • 3.NBT.A.2

BACKWARD 2

Use the relationship between addition and subtraction within 100 • Grade 2 •
Number & Operations in Base Ten • 2.NBT.B.5

Find $70 - 45$ by counting up from 45.

Difference: _____

True or false?

Because $36 + 28 = 64$, you know that $64 - 28 = 36$.

- A. True
- B. False

Which addition checks $90 - 37 = 53$?

A. $90 + 37 = 53$

B. $53 + 90 = 37$

C. $37 + 90 = 53$

D. $53 + 37 = 90$

Is $81 - 34 = 47$ correct? Write the addition you used to check.

Correct? _____

Check: _____

True or false?

Because $42 + 19 = 61$, you know that $61 - 42 = 29$.

- A. True
- B. False

SET 26 • 3.NBT.A.2

FORWARD 1

Check multi-digit subtraction with addition • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Use addition to check: is $6,000 - 2,348 = 3,652$ correct?

Correct? _____

Check: _____

True or false?

Because $4,519 + 2,786 = 7,305$, you know that $7,305 - 2,786 = 4,519$.

A. True

B. False

Which addition checks $9,104 - 5,677 = 3,427$?

A. $3,427 + 9,104 = 5,677$

B. $3,427 + 5,677 = 9,104$

C. $5,677 + 9,104 = 3,427$

D. $9,104 + 5,677 = 3,427$

Find $8,000 - 3,995$ by counting up from 3,995.

Difference: _____

True or false?

Because $12,450 + 6,780 = 19,230$, you know that $19,230 - 12,450 = 6,870$.

A. True

B. False

SET 26 • 3.NBT.A.2

FORWARD 2

Check decimal subtraction with addition • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Use addition to check: is $5.2 - 1.75 = 3.45$ correct?

Correct? _____

Check: _____

True or false?

Because $2.6 + 3.85 = 6.45$, you know that $6.45 - 3.85 = 2.6$.

A. True

B. False

Which addition checks $9.3 - 4.56 = 4.74$?

A. $9.3 + 4.56 = 4.74$

B. $4.74 + 4.56 = 9.3$

C. $4.74 + 9.3 = 4.56$

D. $4.56 + 9.3 = 4.74$

Find $4 - 2.35$ by counting up from 2.35.

Difference: _____

True or false?

Because $1.4 + 2.75 = 4.15$, you know that $4.15 - 1.4 = 2.65$.

- A. True
- B. False

Multiply one-digit whole numbers by multiples of 10

MAIN	Grade 3 main question	3.NBT.A.3
BACKWARD 1	A hundred is ten tens	2.NBT.A.1.a
BACKWARD 2	Skip-count by 10s	2.NBT.A.2
FORWARD 1	Multiply by multiples of 10, 100, and 1,000	4.NBT.B.5
FORWARD 2	Multiply by powers of 10 and explain the pattern in the zeros	5.NBT.A.2

SET 27 • 3.NBT.A.3

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

Multiply.

$$4 \times 70$$

Product: _____

True or false?

$$9 \times 20 = 160$$

A. True

B. False

$$6 \times 90 = ?$$

A. 5,400

B. 540

C. 54

D. 96

True or false?

$$7 \times 40 = 280$$

A. True

B. False

Explain how you can use $3 \times 8 = 24$ to find 3×80 .

Product: _____

Explanation: _____

SET 27 • 3.NBT.A.3

BACKWARD 1

A hundred is ten tens • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1.a

How many tens make 1 hundred?

Tens: _____

True or false?

10 tens and 1 hundred are the same amount.

A. True

B. False

Which is the same amount as 1 hundred?

- A.** 10 tens
- B.** 10 ones
- C.** 100 tens
- D.** 1 ten

Fill in the blank.

10 tens = ____

Blank: _____

True or false?

A bundle of 10 tens is the same amount as 10.

A. True

B. False

SET 27 • 3.NBT.A.3

BACKWARD 2

Skip-count by 10s • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 10s from 10.

What is the 6th number?

Number: _____

True or false?

Counting by 10s from 0, 8 jumps of 10 land on 80.

A. True

B. False

Count by 10s.

30, 40, 50, ____

A. 60

B. 70

C. 51

D. 500

Skip-count by 10s.

150, 160, 170, __, __

Next two: _____

True or false?

Counting by 10s from 0, 5 jumps of 10 land on 15.

A. True

B. False

SET 27 • 3.NBT.A.3

FORWARD 1

Multiply by multiples of 10, 100, and 1,000 • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$6 \times 400$$

Product: _____

True or false?

$$8 \times 3,000 = 24,000$$

A. True

B. False

$$5 \times 800 = ?$$

- A.** 4,000
- B.** 40,000
- C.** 805
- D.** 400

Multiply.

$$30 \times 40$$

Product: _____

True or false?

$$7 \times 600 = 420$$

A. True

B. False

SET 27 • 3.NBT.A.3

FORWARD 2

Multiply by powers of 10 and explain the pattern in the zeros • Grade 5 •
Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$36 \times 10^3$$

Product: _____

True or false?

$$4.2 \times 100 = 420$$

A. True

B. False

Which is equal to 5×10^2 ?

A. 5,000

B. 52

C. 500

D. 50

Explain why $7 \times 1,000$ ends in three zeros.

Explanation: _____

True or false?

$$0.6 \times 10 = 0.06$$

A. True

B. False

Understand a unit fraction as one part of a whole split into equal parts

MAIN	Grade 3 main question	3.NF.A.1
BACKWARD 1	Split shapes into halves, thirds, and fourths	2.G.A.3
BACKWARD 2	Halves and quarters of circles and rectangles	1.G.A.3
FORWARD 1	Write a fraction as a sum of unit fractions	4.NF.B.3.b
FORWARD 2	Find a unit fraction of a unit fraction	5.NF.B.4.a

SET 28 • 3.NF.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

The bar is split into 6 equal parts. One part is shaded.
What fraction of the bar is shaded?



Fraction: _____

A pizza is cut into 8 equal slices.

What fraction of the pizza is one slice?

A. $\frac{8}{1}$

B. $\frac{1}{7}$

C. $\frac{1}{8}$

D. $\frac{7}{8}$

True or false?

$\frac{1}{3}$ is one part when a whole is split into 3 equal parts.

- A.** True
- B.** False

Shade $\frac{1}{4}$ of the bar. Then explain why the part you shaded is $\frac{1}{4}$ of the bar.



Explanation: _____

True or false?

If a whole is cut into 2 parts of different sizes, each part is $\frac{1}{2}$ of the whole.

- A. True
- B. False

SET 28 • 3.NF.A.1

BACKWARD 1

Split shapes into halves, thirds, and fourths • Grade 2 • Geometry • 2.G.A.3

A rectangle is split into 3 equal shares.

What is each share called?

Name: _____

True or false?

Four fourths make one whole.

A. True

B. False

A circle is split into 2 equal shares.

How many halves make the whole circle?

A. 4

B. 2

C. 1

D. 3

A square is split into 4 equal shares.

What is each share called?

Name: _____

True or false?

A rectangle split into 3 equal shares is split into fourths.

- A.** True
- B.** False

SET 28 • 3.NF.A.1

BACKWARD 2

Halves and quarters of circles and rectangles • Grade 1 • Geometry • 1.G.A.3

True or false?

Cutting a circle into 4 equal parts makes smaller parts than cutting it into 2 equal parts.

A. True

B. False

A sandwich is cut into 2 equal parts.

What is each part called?

Name: _____

A pie is cut into 4 equal parts.

What is each part called?

- A.** A quarter
- B.** A whole
- C.** A half
- D.** A third

How many quarters make a whole circle?

Quarters: _____

True or false?

Halves of a rectangle are smaller than quarters of the same rectangle.

A. True

B. False

SET 28 • 3.NF.A.1

FORWARD 1

Write a fraction as a sum of unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{5}{6}$ as a sum of unit fractions.

Sum: _____

True or false?

$$\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

A. True

B. False

Which sum is equal to $\frac{4}{5}$?

A. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

B. $\frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5}$

C. $\frac{4}{5} + \frac{1}{5}$

D. $\frac{2}{5} + \frac{1}{5}$

Write $\frac{7}{8}$ as a sum of two fractions with a denominator of 8.

Sum: _____

True or false?

$$\frac{2}{3} = \frac{1}{3} + \frac{1}{2}$$

A. True

B. False

SET 28 • 3.NF.A.1

FORWARD 2

Find a unit fraction of a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

What is $\frac{1}{2}$ of $\frac{1}{4}$?

Answer: _____

True or false?

$$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

A. True

B. False

Multiply.

$$\frac{1}{4} \times \frac{1}{3}$$

A. $\frac{1}{7}$

B. $\frac{1}{12}$

C. $\frac{4}{3}$

D. $\frac{2}{7}$

A ribbon is $\frac{1}{2}$ yard long. Ana uses $\frac{1}{5}$ of it.

What fraction of a yard does she use?

Answer: _____

True or false?

$\frac{1}{2}$ of $\frac{1}{2}$ is 1.

A. True

B. False

Understand a fraction a/b as a parts of size $1/b$

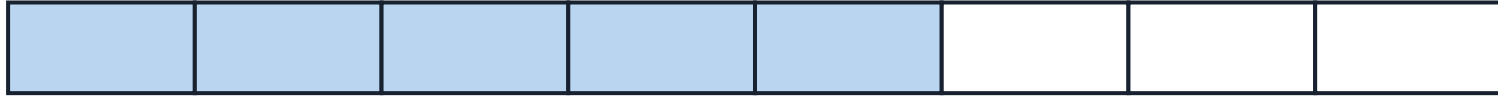
MAIN	Grade 3 main question	3.NF.A.1
BACKWARD 1	Count halves, thirds, and fourths of a shape	2.G.A.3
BACKWARD 2	Count halves and quarters of a shape	1.G.A.3
FORWARD 1	Understand a fraction a/b as a multiple of $1/b$	4.NF.B.4.a
FORWARD 2	Find a fraction of a quantity by first finding the unit fraction of it	5.NF.B.4.a

SET 29 • 3.NF.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

What fraction of the bar is shaded?



Fraction: _____

Which fraction is 3 parts of size $\frac{1}{4}$?

A. $\frac{1}{3}$

B. $\frac{4}{3}$

C. $\frac{3}{8}$

D. $\frac{3}{4}$

True or false?

$\frac{4}{6}$ is 4 parts of size $\frac{1}{6}$.

A. True

B. False

Shade $\frac{3}{8}$ of the bar.

How many parts of size $\frac{1}{8}$ did you shade?

--	--	--	--	--	--	--	--

Parts: _____

True or false?

$\frac{5}{6}$ is 6 parts of size $\frac{1}{5}$.

A. True

B. False

SET 29 • 3.NF.A.1

BACKWARD 1

Count halves, thirds, and fourths of a shape • Grade 2 • Geometry • 2.G.A.3

A rectangle is cut into 3 equal shares, and 2 shares are shaded.

How many thirds are shaded?

Thirds: _____

True or false?

A whole is 3 thirds.

A. True

B. False

A circle is cut into 4 equal shares.

How many fourths make the whole circle?

A. 2

B. 3

C. 4

D. 1

A rectangle is cut into 2 equal shares, and both are shaded.

How many halves are shaded?

Halves: _____

True or false?

A whole is 3 halves.

A. True

B. False

SET 29 • 3.NF.A.1

BACKWARD 2

Count halves and quarters of a shape • Grade 1 • Geometry • 1.G.A.3

A pizza is cut into 4 equal parts, and 3 parts are eaten.
How many quarters are eaten?

Quarters: _____

True or false?

Two halves make one whole.

A. True

B. False

A rectangle is cut into 4 equal parts.

How many parts make the whole?

A. 8

B. 2

C. 1

D. 4

A pie is cut into 2 equal parts, and 1 part is eaten.
What is the part that was eaten called?

Name: _____

True or false?

Four quarters make two wholes.

A. True

B. False

SET 29 • 3.NF.A.1

FORWARD 1

Understand a fraction a/b as a multiple of $1/b$ • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{5}{6}$ as a whole number times $\frac{1}{6}$.

Expression: _____

True or false?

$$\frac{7}{4} = 7 \times \frac{1}{4}$$

A. True

B. False

Which equation shows $\frac{3}{8}$ as a multiple of $\frac{1}{8}$?

A. $\frac{3}{8} = \frac{1}{8} \div 3$

B. $\frac{3}{8} = 3 \times \frac{1}{8}$

C. $\frac{3}{8} = 8 \times \frac{1}{3}$

D. $\frac{3}{8} = 3 + \frac{1}{8}$

Fill in the blank.

$$\frac{9}{10} = \underline{\hspace{1cm}} \times \frac{1}{10}$$

Blank: _____

True or false?

$$\frac{4}{5} = 5 \times \frac{1}{4}$$

A. True

B. False

SET 29 • 3.NF.A.1

FORWARD 2

Find a fraction of a quantity by first finding the unit fraction of it • Grade 5 •
Number & Operations–Fractions • 5.NF.B.4.a

Find $\frac{3}{5}$ of 20 by first finding $\frac{1}{5}$ of 20.

$\frac{1}{5}$ of 20: _____

$\frac{3}{5}$ of 20: _____

True or false?

$\frac{2}{3}$ of 9 is 2 parts of size 3, which is 6.

A. True

B. False

What is $\frac{5}{8}$ of 16?

A. 80

B. 8

C. 10

D. 2

Multiply.

$$\frac{4}{7} \times 14$$

Product: _____

True or false?

$\frac{3}{4}$ of 20 is 3 parts of size 4, which is 12.

A. True

B. False

Represent a unit fraction on a number line

MAIN	Grade 3 main question	3.NF.A.2.a
BACKWARD 1	Whole numbers as lengths from 0 on a number line	2.MD.B.6
BACKWARD 2	Measure lengths in whole inches with a ruler	2.MD.A.1
FORWARD 1	Write tenths and hundredths as decimals and locate them on a n...	4.NF.C.6
FORWARD 2	Divide a unit fraction by a whole number	5.NF.B.7.a

SET 30 • 3.NF.A.2.a

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.2.a

The space from 0 to 1 is split into 4 equal parts.

What fraction is at point A?



A = _____

Mark and label $\frac{1}{3}$ on the number line.



Into how many equal parts must the space from 0 to 1 be split to show $\frac{1}{6}$?

A. 1

B. 5

C. 6

D. 7

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

A. True

B. False

True or false?

On a number line from 0 to 1 split into 8 equal parts, the first tick mark after 0 is $\frac{1}{7}$.

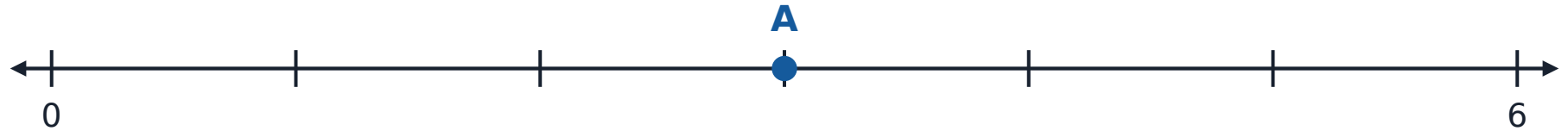
- A. True
- B. False

SET 30 • 3.NF.A.2.a

BACKWARD 1

Whole numbers as lengths from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line, 5 is 5 equal lengths of 1 from 0.

A. True

B. False

A jump from 0 to 4 on a number line is how many lengths of 1?

A. 1

B. 4

C. 3

D. 5

What number is at point B?



B = _____

True or false?

On a number line, the distance from 0 to 7 is 6 lengths of 1.

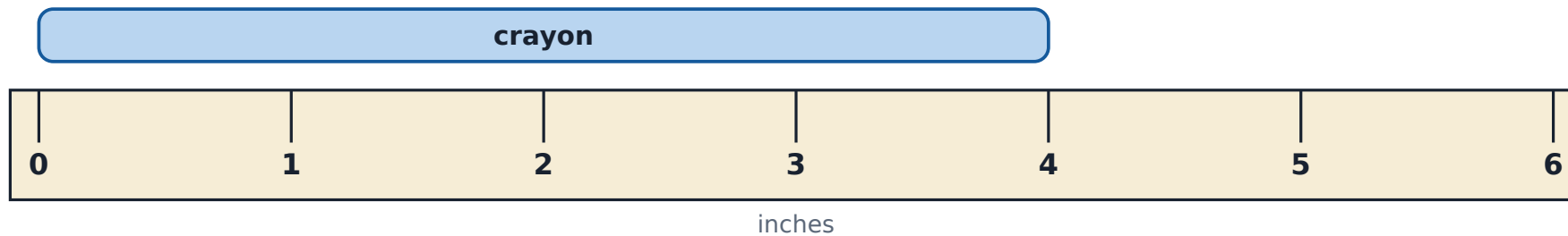
- A. True
- B. False

SET 30 • 3.NF.A.2.a

BACKWARD 2

Measure lengths in whole inches with a ruler • Grade 2 • Measurement & Data
• 2.MD.A.1

How long is the crayon?



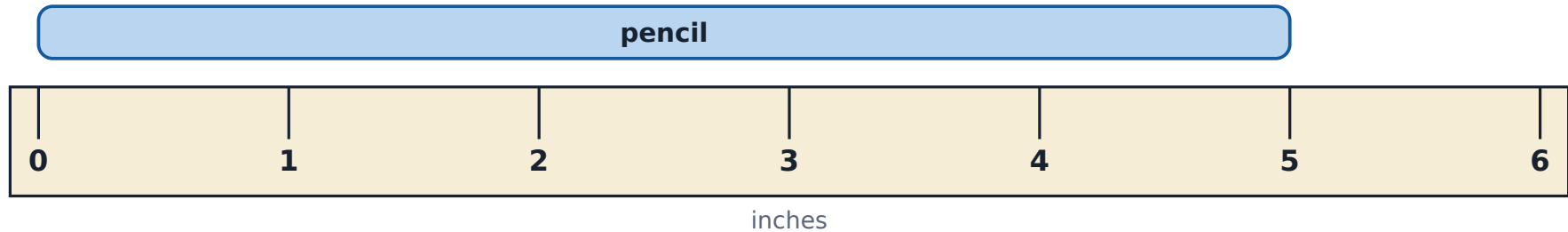
Length: _____

True or false?

To measure with a ruler, line up one end of the object with 0.

- A.** True
- B.** False

How long is the pencil?



A. 6 inches

B. 5 inches

C. 1 inch

D. 4 inches

A leaf reaches from 0 to 3 on an inch ruler.
How long is the leaf?

Length: _____

True or false?

A stick that reaches from 0 to 6 on an inch ruler is 7 inches long.

- A.** True
- B.** False

SET 30 • 3.NF.A.2.a

FORWARD 1

Write tenths and hundredths as decimals and locate them on a number line •
Grade 4 • Number & Operations–Fractions • 4.NF.C.6

The space from 0 to 1 is split into 10 equal parts.

Write the decimal at point A.



A = _____

True or false?

$$\frac{1}{10} = 0.1$$

A. True

B. False

Which decimal is equal to $\frac{1}{100}$?

A. 0.1

B. 0.01

C. 10.0

D. 1.01

Write $\frac{1}{100}$ as a decimal.

Decimal: _____

True or false?

On a number line, 0.1 is between 0 and $\frac{1}{100}$.

A. True

B. False

SET 30 • 3.NF.A.2.a

FORWARD 2

Divide a unit fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a

$\frac{1}{2}$ is split into 3 equal parts.

What fraction is each part?

Fraction: _____

True or false?

$$\frac{1}{4} \div 2 = \frac{1}{8}$$

A. True

B. False

$$\frac{1}{3} \div 2 = ?$$

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{5}$

D. 6

Divide.

$$\frac{1}{2} \div 4$$

Quotient: _____

True or false?

$$\frac{1}{5} \div 3 = \frac{3}{5}$$

A. True

B. False

Represent a fraction a/b on a number line by marking off a lengths of $1/b$

MAIN	Grade 3 main question	3.NF.A.2.b
BACKWARD 1	Mark whole numbers as lengths from 0 on a number line	2.MD.B.6
BACKWARD 2	Show sums as jumps on a number line	2.MD.B.6
FORWARD 1	Multiply a fraction by a whole number with jumps on a number li...	4.NF.B.4.b
FORWARD 2	Add fractions with unlike denominators	5.NF.A.1

SET 31 • 3.NF.A.2.b

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.2.b

What fraction is at point B?



B = _____

Start at 0 and mark off 5 lengths of $\frac{1}{6}$. Mark and label the point $\frac{5}{6}$.



A point is 3 lengths of $\frac{1}{8}$ from 0.

What fraction is at the point?

A. $\frac{1}{3}$

B. $\frac{3}{1}$

C. $\frac{3}{8}$

D. $\frac{8}{3}$

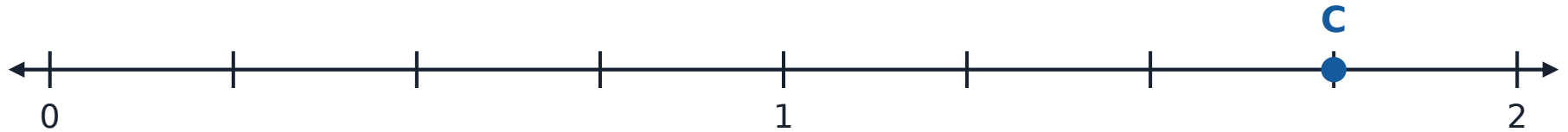
True or false?

The point $\frac{2}{3}$ is 2 lengths of $\frac{1}{3}$ from 0.

A. True

B. False

What fraction is at point C? Write it as a fraction with a denominator of 4.



C = _____

SET 31 • 3.NF.A.2.b

BACKWARD 1

Mark whole numbers as lengths from 0 on a number line • Grade 2 •
Measurement & Data • 2.MD.B.6

Mark and label 6 on the number line.



True or false?

The point 3 is 3 lengths of 1 from 0.

A. True

B. False

A point is 7 lengths of 1 from 0.

What number is it?

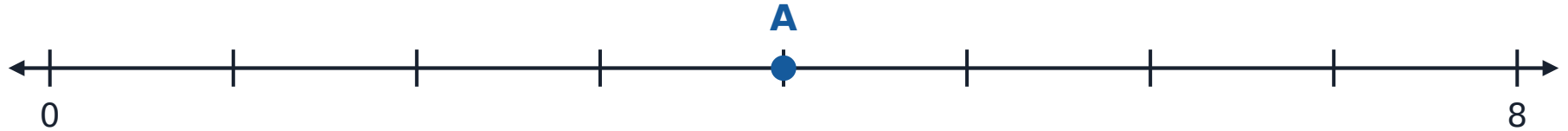
A. 8

B. 6

C. 1

D. 7

What number is at point A?



A = _____

True or false?

The point 9 is 10 lengths of 1 from 0.

A. True

B. False

SET 31 • 3.NF.A.2.b

BACKWARD 2

Show sums as jumps on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

Start at 20 on a number line and jump 15 more.

Where do you land?

Number: _____

True or false?

Starting at 0, a jump of 30 and then a jump of 20 lands on 50.

A. True

B. False

Start at 10 and make three jumps of 10.

Where do you land?

A. 40

B. 13

C. 30

D. 50

Start at 25 and jump 40 more.

Where do you land?

Number: _____

True or false?

Starting at 40, a jump of 15 lands on 65.

A. True

B. False

SET 31 • 3.NF.A.2.b

FORWARD 1

Multiply a fraction by a whole number with jumps on a number line • Grade 4 •
Number & Operations–Fractions • 4.NF.B.4.b

Start at 0 and make 4 jumps of $\frac{2}{3}$ on a number line.

Where do you land?

Number: _____

True or false?

3 jumps of $\frac{3}{4}$ from 0 land on $\frac{9}{4}$.

A. True

B. False

$$5 \times \frac{2}{6} = ?$$

A. $\frac{7}{6}$

B. $\frac{10}{30}$

C. $\frac{2}{30}$

D. $\frac{10}{6}$

Multiply.

$$6 \times \frac{3}{8}$$

Product: _____

True or false?

2 jumps of $\frac{5}{6}$ from 0 land on $\frac{10}{12}$.

A. True

B. False

SET 31 • 3.NF.A.2.b

FORWARD 2

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{2} + \frac{1}{3}$$

Sum: _____

True or false?

$$\frac{3}{4} + \frac{1}{8} = \frac{7}{8}$$

A. True

B. False

Add.

$$\frac{2}{3} + \frac{1}{6}$$

A. $\frac{5}{6}$

B. $\frac{3}{6}$

C. $\frac{3}{9}$

D. $\frac{2}{18}$

Add.

$$\frac{3}{8} + \frac{1}{4}$$

Sum: _____

True or false?

$$\frac{1}{4} + \frac{1}{3} = \frac{2}{7}$$

A. True

B. False

Understand equivalent fractions as the same size or the same point on a number line

MAIN	Grade 3 main question	3.NF.A.3.a
BACKWARD 1	Measure one length with two different units	2.MD.A.2
BACKWARD 2	Equal shares of the same whole	2.G.A.3
FORWARD 1	Explain equivalent fractions with visual models	4.NF.A.1
FORWARD 2	Use equivalent fractions to add fractions with unlike denominators	5.NF.A.1

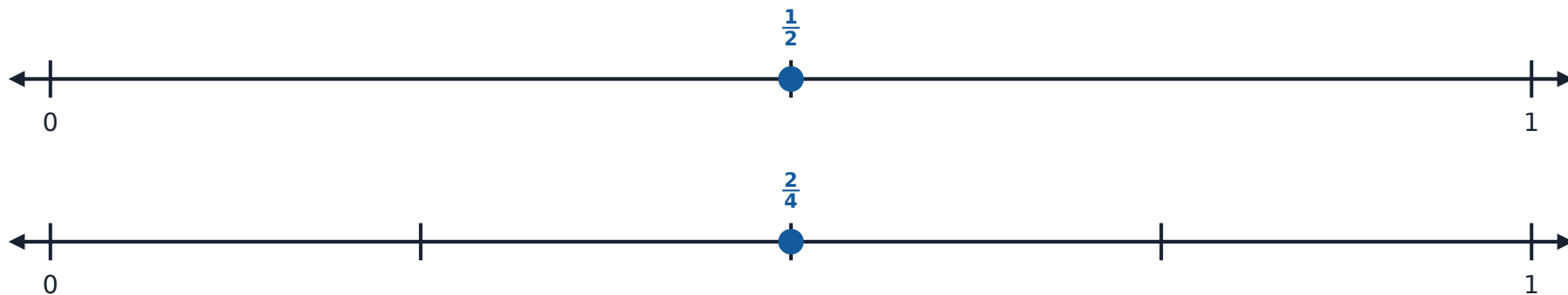
SET 32 • 3.NF.A.3.a

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.a

True or false?

The number lines are the same length. $\frac{1}{2}$ and $\frac{2}{4}$ are at the same point.

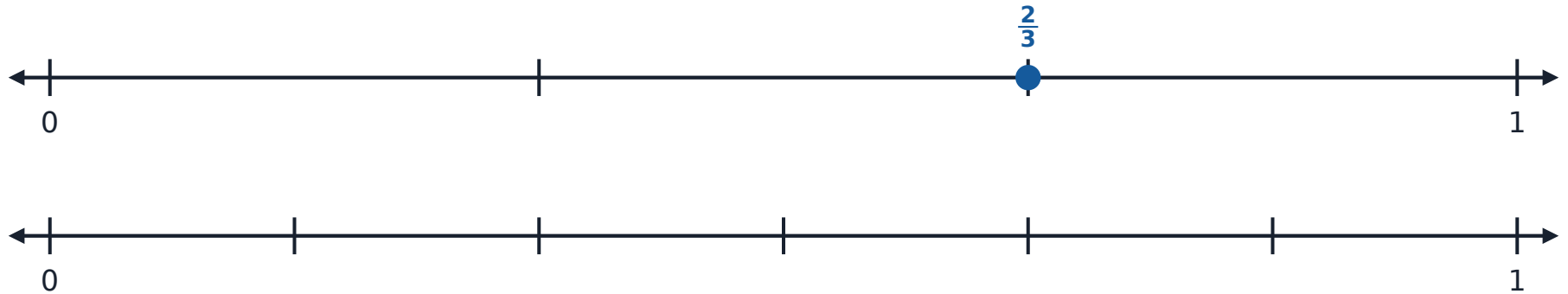


A. True

B. False

The number lines are the same length.

Which fraction is at the same point as $\frac{2}{3}$?



A. $\frac{2}{6}$

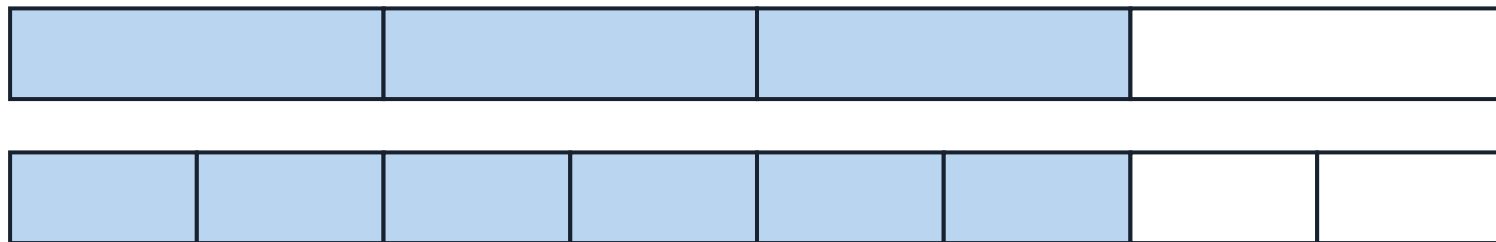
B. $\frac{3}{6}$

C. $\frac{5}{6}$

D. $\frac{4}{6}$

The two bars are the same size. The top bar shows $\frac{3}{4}$ shaded and the bottom bar shows $\frac{6}{8}$ shaded.

Are the fractions equivalent? Explain.



Answer: _____

Explanation: _____

True or false?

$\frac{1}{3}$ and $\frac{2}{6}$ are at different points on a number line from 0 to 1.

- A.** True
- B.** False

Name a fraction in eighths that is at the same point as $\frac{1}{4}$ on a number line.

Fraction: _____

SET 32 • 3.NF.A.3.a

BACKWARD 1

Measure one length with two different units • Grade 2 • Measurement & Data •
2.MD.A.2

True or false?

A table measured in centimeters gives a larger number than the same table measured in meters.

A. True

B. False

A pencil is 6 paper clips long and 3 crayons long.

Which unit is longer?

- A.** You cannot tell
- B.** The crayon
- C.** The paper clip
- D.** They are the same length

A shelf is 10 large blocks long and 20 small blocks long.
Did the shelf's length change?

Answer: _____

A ribbon is 4 feet long. Is its length in inches a number greater than or less than 4?

Answer: _____

True or false?

When you measure with a bigger unit, you get a bigger number.

- A.** True
- B.** False

SET 32 • 3.NF.A.3.a

BACKWARD 2

Equal shares of the same whole • Grade 2 • Geometry • 2.G.A.3

True or false?

Two halves of the same rectangle can have different shapes and still be equal shares.

- A. True
- B. False

A square is cut into 2 equal triangles. Another square of the same size is cut into 2 equal rectangles.

Is a triangle the same size as a rectangle?

Answer: _____

A circle is cut into 4 equal shares.

What is each share called?

- A.** A third
- B.** A fourth
- C.** A whole
- D.** A half

How many halves make one whole?

Halves: _____

True or false?

A third of a rectangle is larger than a half of the same rectangle.

A. True

B. False

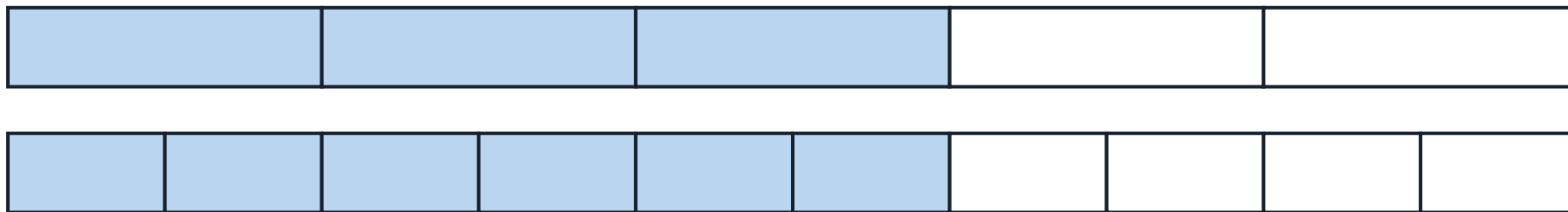
SET 32 • 3.NF.A.3.a

FORWARD 1

Explain equivalent fractions with visual models • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

The bars show $\frac{3}{5}$ and $\frac{6}{10}$ of the same whole.

Explain why the fractions are equal.



Explanation: _____

True or false?

$\frac{2}{5} = \frac{4}{10}$ because the numerator and the denominator are both multiplied by 2.

- A.** True
- B.** False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{9}{12}$

B. $\frac{3}{12}$

C. $\frac{6}{4}$

D. $\frac{4}{3}$

Find the missing number.

$$\frac{4}{5} = \frac{?}{10}$$

Missing number: _____

True or false?

$$\frac{1}{3} = \frac{3}{6}$$

A. True

B. False

SET 32 • 3.NF.A.3.a

FORWARD 2

Use equivalent fractions to add fractions with unlike denominators • Grade 5 •
Number & Operations–Fractions • 5.NF.A.1

Rewrite $\frac{1}{2}$ and $\frac{1}{5}$ with a common denominator. Then add.

Rewritten: _____

Sum: _____

True or false?

$$\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12}$$

A. True

B. False

Add.

$$\frac{1}{6} + \frac{3}{4}$$

A. $\frac{3}{24}$

B. $\frac{4}{12}$

C. $\frac{4}{10}$

D. $\frac{11}{12}$

Add.

$$\frac{2}{5} + \frac{1}{10}$$

Sum: _____

True or false?

$$\frac{1}{3} + \frac{1}{2} = \frac{2}{5}$$

A. True

B. False

Recognize and generate simple equivalent fractions, and explain why they are equivalent

MAIN	Grade 3 main question	3.NF.A.3.b
BACKWARD 1	Halves, thirds, and fourths that make a whole	2.G.A.3
BACKWARD 2	Write even numbers as doubles	2.OA.C.3
FORWARD 1	Generate equivalent fractions by multiplying the numerator and th...	4.NF.A.1
FORWARD 2	Explain equivalent fractions as multiplying by a fraction equal to 1	5.NF.B.5.b

SET 33 • 3.NF.A.3.b

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

Find the missing number.

$$\frac{1}{3} = \frac{?}{6}$$

Missing number: _____

Which fraction is equivalent to $\frac{3}{4}$?

A. $\frac{4}{3}$

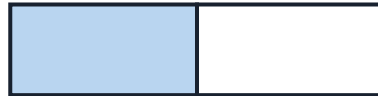
B. $\frac{4}{6}$

C. $\frac{6}{8}$

D. $\frac{3}{8}$

The bar shows $\frac{1}{2}$ shaded. Draw lines to split each half into 4 equal parts.

Then write the equivalent fraction in eighths.



Fraction: _____

True or false?

$$\frac{2}{6} = \frac{1}{3}$$

A. True

B. False

Explain why $\frac{2}{8} = \frac{1}{4}$. You may use a model.

Explanation: _____

SET 33 • 3.NF.A.3.b

BACKWARD 1

Halves, thirds, and fourths that make a whole • Grade 2 • Geometry • 2.G.A.3

How many fourths make a whole?

Fourths: _____

True or false?

Three thirds make one whole.

A. True

B. False

Which describes one whole?

A. 3 fourths

B. 3 halves

C. 2 thirds

D. 2 halves

A rectangle is cut into 3 equal shares.

How many of the shares make the whole rectangle?

Shares: _____

True or false?

Four halves make one whole.

A. True

B. False

SET 33 • 3.NF.A.3.b

BACKWARD 2

Write even numbers as doubles • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.3

Write 8 as the sum of two equal addends.

Sum: _____

True or false?

$$6 = 3 + 3$$

A. True

B. False

Which number is the double of 4?

- A.** 8
- B.** 6
- C.** 4
- D.** 16

What is the double of 3?

Double: _____

True or false?

$10 = 6 + 4$ shows 10 as the sum of two equal addends.

A. True

B. False

FORWARD 1

Generate equivalent fractions by multiplying the numerator and the denominator by the same number • Grade 4 • Number & Operations–Fractions
• 4.NF.A.1

Find the missing number.

$$\frac{3}{5} = \frac{?}{15}$$

Missing number: _____

True or false?

$$\frac{4}{6} = \frac{8}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{2}{5}$?

A. $\frac{3}{6}$

B. $\frac{5}{2}$

C. $\frac{4}{10}$

D. $\frac{2}{10}$

Write two fractions that are equivalent to $\frac{3}{4}$.

Fractions: _____

True or false?

$$\frac{5}{8} = \frac{10}{24}$$

A. True

B. False

SET 33 • 3.NF.A.3.b

FORWARD 2

Explain equivalent fractions as multiplying by a fraction equal to 1 • Grade 5 •
Number & Operations–Fractions • 5.NF.B.5.b

True or false?

$$\frac{2}{3} \times \frac{4}{4} = \frac{8}{12}, \text{ and } \frac{8}{12} = \frac{2}{3} \text{ because } \frac{4}{4} = 1.$$

A. True

B. False

To rewrite $\frac{3}{5}$ as $\frac{9}{15}$, by which whole number must you multiply BOTH its numerator and its denominator?

Which fraction equal to 1 is that the same as multiplying by?

Whole number: _____

Fraction equal to 1: _____

Which equation shows that $\frac{5}{6} = \frac{10}{12}$?

A. $\frac{5}{6} \times \frac{2}{1} = \frac{10}{12}$

B. $\frac{5}{6} + \frac{5}{6} = \frac{10}{12}$

C. $\frac{5}{6} \times \frac{2}{2} = \frac{10}{12}$

D. $\frac{5}{6} \times 2 = \frac{10}{12}$

Explain why multiplying $\frac{1}{4}$ by $\frac{3}{3}$ does not change its value.

Explanation: _____

True or false?

$\frac{1}{2} \times \frac{2}{3} = \frac{2}{6}$ shows that $\frac{1}{2} = \frac{2}{6}$.

A. True

B. False

Express whole numbers as fractions and recognize fractions that are equal to whole numbers

MAIN	Grade 3 main question	3.NF.A.3.c
BACKWARD 1	Two halves or four quarters make a whole	1.G.A.3
BACKWARD 2	Count lengths of 1 on a number line	2.MD.B.6
FORWARD 1	Write mixed numbers as fractions and fractions as mixed numbers	4.NF.B.3.b
FORWARD 2	Interpret a fraction as the numerator divided by the denominator	5.NF.B.3

SET 34 • 3.NF.A.3.c

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

Write 5 as a fraction with a denominator of 1.

Fraction: _____

Which fraction is equal to 2?

A. $\frac{4}{8}$

B. $\frac{2}{4}$

C. $\frac{2}{8}$

D. $\frac{8}{4}$

True or false?

$$\frac{6}{6} = 6$$

A. True

B. False

The number line from 0 to 3 is marked in thirds.

What fraction in thirds names the point 2?



Fraction: _____

True or false?

$$\frac{3}{3} = 3$$

A. True

B. False

SET 34 • 3.NF.A.3.c

BACKWARD 1

Two halves or four quarters make a whole • Grade 1 • Geometry • 1.G.A.3

How many halves make one whole pizza?

Halves: _____

True or false?

4 quarters of a sandwich make the whole sandwich.

A. True

B. False

A pie is cut into 4 equal parts. All 4 parts are eaten.

How much of the pie is eaten?

- A.** The whole pie
- B.** Half of the pie
- C.** A quarter of the pie
- D.** None of the pie

How many quarters make one whole?

Quarters: _____

True or false?

2 quarters make a whole circle.

A. True

B. False

SET 34 • 3.NF.A.3.c

BACKWARD 2

Count lengths of 1 on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

How many lengths of 1 are there from 0 to 3 on a number line?

Lengths: _____

True or false?

From 0 to 5 on a number line there are 5 lengths of 1.

A. True

B. False

Which number is 2 lengths of 1 from 0?

A. 2

B. 0

C. 1

D. 3

What number is at point A?



A = _____

True or false?

From 0 to 4 on a number line there are 3 lengths of 1.

A. True

B. False

SET 34 • 3.NF.A.3.c

FORWARD 1

Write mixed numbers as fractions and fractions as mixed numbers • Grade 4 •
Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$1\frac{3}{4} = \frac{4}{4} + \frac{3}{4}$$

A. True

B. False

Which mixed number is equal to $\frac{11}{4}$?

A. $2\frac{3}{4}$

B. $1\frac{3}{4}$

C. $2\frac{1}{4}$

D. $3\frac{1}{4}$

Write $\frac{9}{2}$ as a mixed number.

Mixed number: _____

True or false?

$$3\frac{2}{5} = \frac{32}{5}$$

A. True

B. False

SET 34 • 3.NF.A.3.c

FORWARD 2

Interpret a fraction as the numerator divided by the denominator • Grade 5 •
Number & Operations–Fractions • 5.NF.B.3

Write $\frac{20}{4}$ as a division. Then find its value.

Division: _____

Value: _____

True or false?

$$\frac{15}{3} = 15 \div 3 = 5$$

A. True

B. False

Which division is equal to $\frac{7}{2}$?

A. $7 - 2$

B. 7×2

C. $7 \div 2$

D. $2 \div 7$

Find the value of $\frac{42}{6}$.

Value: _____

True or false?

$$\frac{5}{10} = 10 \div 5$$

A. True

B. False

Compare two fractions with the same denominator

MAIN	Grade 3 main question	3.NF.A.3.d
BACKWARD 1	Compare two- and three-digit numbers	2.NBT.A.4
BACKWARD 2	Order objects by length	1.MD.A.1
FORWARD 1	Compare fractions with different numerators and different denomi...	4.NF.A.2
FORWARD 2	Use benchmark fractions to estimate sums of fractions	5.NF.A.2

SET 35 • 3.NF.A.3.d

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{3}{8} \text{ — } \frac{5}{8}$$

Symbol: _____

True or false?

$$\frac{5}{6} > \frac{1}{6}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{1}{4}$

B. $\frac{3}{4}$

C. $\frac{0}{4}$

D. $\frac{2}{4}$

Compare $\frac{2}{3}$ and $\frac{1}{3}$. Write $>$, $=$, or $<$, and explain how you know.

Comparison: _____

Explanation: _____

True or false?

$$\frac{4}{8} < \frac{3}{8}$$

A. True

B. False

SET 35 • 3.NF.A.3.d

BACKWARD 1

Compare two- and three-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $=$, or $<$.

57 75

Symbol: _____

True or false?

$$130 > 103$$

A. True

B. False

Which number is least?

A. 902

B. 209

C. 920

D. 290

Write $>$, $=$, or $<$.

408 480

Symbol: _____

True or false?

$$615 < 561$$

A. True

B. False

SET 35 • 3.NF.A.3.d

BACKWARD 2

Order objects by length • Grade 1 • Measurement & Data • 1.MD.A.1

A red pencil is longer than a blue pencil. The blue pencil is longer than a green pencil.

Which pencil is longest?

Answer: _____

True or false?

If rope A is longer than rope B, then rope B is shorter than rope A.

A. True

B. False

Ty's string is shorter than Ann's string. Ann's string is shorter than Bo's string.

Whose string is shortest?

- A.** Bo's
- B.** Ann's
- C.** Ty's
- D.** They are the same length.

A bat is longer than a stick.

Is the stick longer or shorter than the bat?

Answer: _____

True or false?

If a crayon is shorter than a marker, then the marker is shorter than the crayon.

A. True

B. False

SET 35 • 3.NF.A.3.d

FORWARD 1

Compare fractions with different numerators and different denominators •
Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Write $>$, $=$, or $<$.

$$\frac{2}{3} \text{ — } \frac{5}{8}$$

Symbol: _____

True or false?

$$\frac{3}{5} > \frac{1}{2}$$

A. True

B. False

Which comparison is true?

A. $\frac{3}{4} < \frac{7}{10}$

B. $\frac{3}{4} = \frac{7}{10}$

C. $\frac{3}{4} > \frac{7}{10}$

D. $\frac{7}{10} > \frac{3}{4}$

Compare $\frac{5}{6}$ and $\frac{7}{9}$. Write $>$, $=$, or $<$.

Comparison: _____

True or false?

$$\frac{2}{5} > \frac{1}{2}$$

A. True

B. False

SET 35 • 3.NF.A.3.d

FORWARD 2

Use benchmark fractions to estimate sums of fractions • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Is $\frac{2}{5} + \frac{3}{8}$ more or less than 1? Explain using benchmark fractions.

Answer: _____

Reason: _____

True or false?

$\frac{4}{7} + \frac{3}{5} > 1$, because each fraction is greater than $\frac{1}{2}$.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{5}{6}$?

A. About 3

B. About $\frac{1}{2}$

C. About 1

D. About 2

Use benchmark fractions to estimate $\frac{1}{12} + \frac{6}{11}$.

Estimate: _____

True or false?

$\frac{1}{8} + \frac{1}{9}$ is about 1.

A. True

B. False

Compare two fractions with the same numerator

MAIN	Grade 3 main question	3.NF.A.3.d
BACKWARD 1	More equal shares make smaller shares	1.G.A.3
BACKWARD 2	Compare halves, thirds, and fourths of the same shape	2.G.A.3
FORWARD 1	Compare fractions by finding a common numerator or a common...	4.NF.A.2
FORWARD 2	Compare the size of a product with the size of one factor without...	5.NF.B.5.a

SET 36 • 3.NF.A.3.d

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{1}{3} \text{ — } \frac{1}{6}$$

Symbol: _____

True or false?

$$\frac{2}{8} < \frac{2}{4}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{1}{6}$

B. $\frac{1}{8}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

Compare $\frac{2}{3}$ and $\frac{2}{6}$ of the same whole. Write $>$, $=$, or $<$, and explain how you know.

Comparison: _____

Explanation: _____

True or false?

$$\frac{5}{8} > \frac{5}{6}$$

A. True

B. False

SET 36 • 3.NF.A.3.d

BACKWARD 1

More equal shares make smaller shares • Grade 1 • Geometry • 1.G.A.3

True or false?

A pizza cut into 4 equal parts has larger parts than the same pizza cut into 2 equal parts.

- A. True
- B. False

Which is bigger: half of a sandwich or a quarter of the same sandwich?

Answer: _____

The same cake is cut in two different ways.

Which way makes the smaller pieces?

- A.** 4 equal pieces
- B.** You cannot tell
- C.** Both make the same size
- D.** 2 equal pieces

A ribbon is cut into 2 equal parts. Another ribbon of the same length is cut into 4 equal parts.

Which ribbon has the longer parts?

Answer: _____

True or false?

Halves of a circle are larger than quarters of the same circle.

A. True

B. False

SET 36 • 3.NF.A.3.d

BACKWARD 2

Compare halves, thirds, and fourths of the same shape • Grade 2 • Geometry •
2.G.A.3

Which is larger: a third or a fourth of the same rectangle?

Answer: _____

True or false?

A half of a circle is larger than a third of the same circle.

A. True

B. False

Which share of the same square is smallest?

- A.** A fourth
- B.** The whole
- C.** A half
- D.** A third

Ann eats a half of a pie. Bo eats a third of a pie of the same size.

Who eats more?

Answer: _____

True or false?

A fourth of a rectangle is larger than a half of the same rectangle.

A. True

B. False

FORWARD 1

Compare fractions by finding a common numerator or a common denominator

• Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Rewrite $\frac{3}{5}$ with a numerator of 6. Then compare it with $\frac{6}{12}$ using $>$, $=$, or $<$.

Rewritten: _____

Comparison: _____

True or false?

$$\frac{4}{7} > \frac{4}{9}$$

A. True

B. False

Which comparison is true?

A. $\frac{2}{3} > \frac{4}{10}$

B. $\frac{2}{3} = \frac{4}{10}$

C. $\frac{4}{10} > \frac{2}{3}$

D. $\frac{2}{3} < \frac{4}{10}$

Write $>$, $=$, or $<$.

$$\frac{5}{12} \text{ — } \frac{5}{9}$$

Symbol: _____

True or false?

$$\frac{3}{10} > \frac{3}{5}$$

A. True

B. False

FORWARD 2

Compare the size of a product with the size of one factor without multiplying •
Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

True or false?

$\frac{3}{4} \times 10$ is less than 10.

A. True

B. False

Without multiplying, is $\frac{5}{3} \times 12$ greater than, less than, or equal to 12?

Answer: _____

Which product is less than 8?

A. $\frac{3}{2} \times 8$

B. $\frac{4}{4} \times 8$

C. $\frac{2}{3} \times 8$

D. 2×8

Without multiplying, which is greater: $\frac{1}{3} \times 9$ or $\frac{1}{6} \times 9$?

Answer: _____

True or false?

$\frac{7}{8} \times 5$ is greater than 5.

A. True

B. False

Compare fractions only when they refer to the same whole, and justify the comparison with a model

MAIN	Grade 3 main question	3.NF.A.3.d
BACKWARD 1	Find how much longer one object is than another	2.MD.A.4
BACKWARD 2	Equal shares of same-size wholes	2.G.A.3
FORWARD 1	Compare fractions with the benchmark $\frac{1}{2}$ and explain the comp...	4.NF.A.2
FORWARD 2	Explain why multiplying by a fraction greater or less than 1 makes...	5.NF.B.5.b

SET 37 • 3.NF.A.3.d

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.d

True or false?

Ana ate $\frac{1}{2}$ of a small pizza. Ben ate $\frac{1}{2}$ of a large pizza. They ate the same amount of pizza.

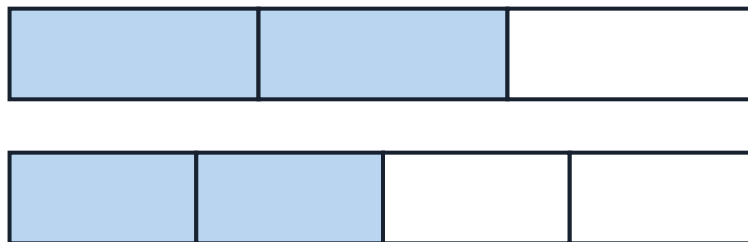
- A.** True
- B.** False

Explain why $\frac{1}{4}$ of a large cake can be more cake than $\frac{1}{2}$ of a small cake.

Explanation: _____

The two bars are the same size.

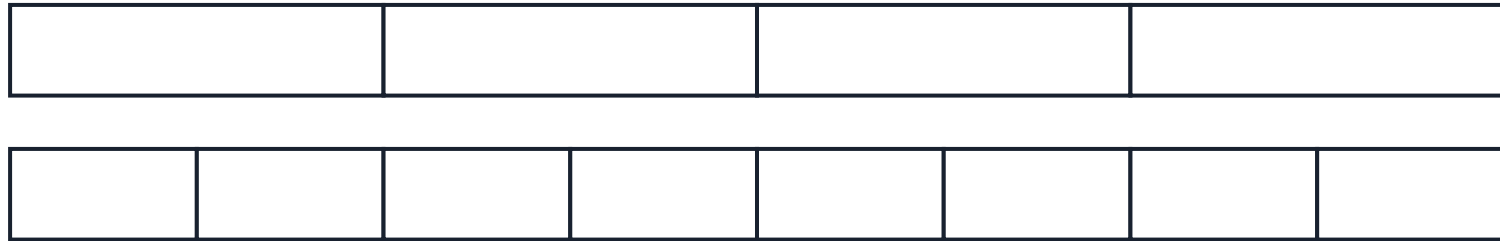
Which comparison do they show?



- A.** $\frac{2}{3} > \frac{2}{4}$ **B.** $\frac{2}{3} < \frac{2}{4}$ **C.** $\frac{2}{4} > \frac{2}{3}$ **D.** $\frac{2}{3} = \frac{2}{4}$

The bars are the same size. Shade $\frac{3}{4}$ of the top bar and $\frac{3}{8}$ of the bottom bar.

Then write $>$, $=$, or $<$ to compare $\frac{3}{4}$ and $\frac{3}{8}$.



Comparison: _____

True or false?

To compare $\frac{2}{6}$ and $\frac{5}{6}$ with models, both models must show the same whole.

- A. True
- B. False

SET 37 • 3.NF.A.3.d

BACKWARD 1

Find how much longer one object is than another • Grade 2 • Measurement & Data • 2.MD.A.4

A pencil is 7 inches long. A crayon is 4 inches long.
How much longer is the pencil?

Answer: _____

True or false?

A 10-cm stick is 6 cm longer than a 4-cm stick.

A. True

B. False

A book is 12 inches long and a card is 5 inches long.

How much longer is the book?

- A.** 17 inches
- B.** 12 inches
- C.** 5 inches
- D.** 7 inches

A rope is 9 feet long and a string is 6 feet long.
How much longer is the rope?

Answer: _____

True or false?

An 8-inch ribbon is 5 inches longer than a 2-inch ribbon.

- A. True
- B. False

SET 37 • 3.NF.A.3.d

BACKWARD 2

Equal shares of same-size wholes • Grade 2 • Geometry • 2.G.A.3

True or false?

Halves of a big circle and halves of a small circle are the same size.

A. True

B. False

Two same-size rectangles are each cut in half.
Are all four halves the same size?

Answer: _____

Which pair shows the same amount?

- A.** A third of a sheet and a half of the same sheet
- B.** A fourth of a sheet and a third of the same sheet
- C.** Half of a big sheet and half of a small sheet
- D.** Half of a sheet of paper and half of a same-size sheet

A small pizza and a large pizza are each cut into fourths.
Which pizza has the larger fourths?

Answer: _____

True or false?

Equal shares of the same rectangle are the same size.

- A.** True
- B.** False

SET 37 • 3.NF.A.3.d

FORWARD 1

Compare fractions with the benchmark $\frac{1}{2}$ and explain the comparison • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Compare $\frac{2}{5}$ and $\frac{5}{8}$ by comparing each one with $\frac{1}{2}$. Write $>$, $=$, or $<$.

Comparison: _____

Reason: _____

True or false?

$$\frac{7}{12} > \frac{1}{2}$$

A. True

B. False

Which fraction is less than $\frac{1}{2}$?

A. $\frac{3}{7}$

B. $\frac{5}{8}$

C. $\frac{6}{10}$

D. $\frac{4}{6}$

Explain why $\frac{3}{4}$ of a 1-meter rope is longer than $\frac{3}{4}$ of a 1-foot rope.

Explanation: _____

True or false?

$$\frac{5}{12} > \frac{1}{2}$$

A. True

B. False

FORWARD 2

Explain why multiplying by a fraction greater or less than 1 makes a product larger or smaller • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.b

True or false?

Multiplying 6 by $\frac{5}{4}$ gives a number greater than 6.

A. True

B. False

Explain why $\frac{2}{3} \times 15$ is less than 15.

Explanation: _____

Which product is greater than 20?

A. $\frac{1}{2} \times 20$

B. $\frac{5}{5} \times 20$

C. $\frac{6}{5} \times 20$

D. $\frac{4}{5} \times 20$

Explain why $\frac{4}{4} \times 7 = 7$.

Explanation: _____

True or false?

Multiplying 9 by $\frac{3}{8}$ gives a number greater than 9.

- A. True
- B. False

Tell and write time to the nearest minute

MAIN	Grade 3 main question	3.MD.A.1
BACKWARD 1	Tell time to the nearest five minutes	2.MD.C.7
BACKWARD 2	Skip-count by 5s	2.NBT.A.2
FORWARD 1	Express hours in minutes and minutes in seconds	4.MD.A.1
FORWARD 2	Convert between units of time, including fractions of an hour	5.MD.A.1

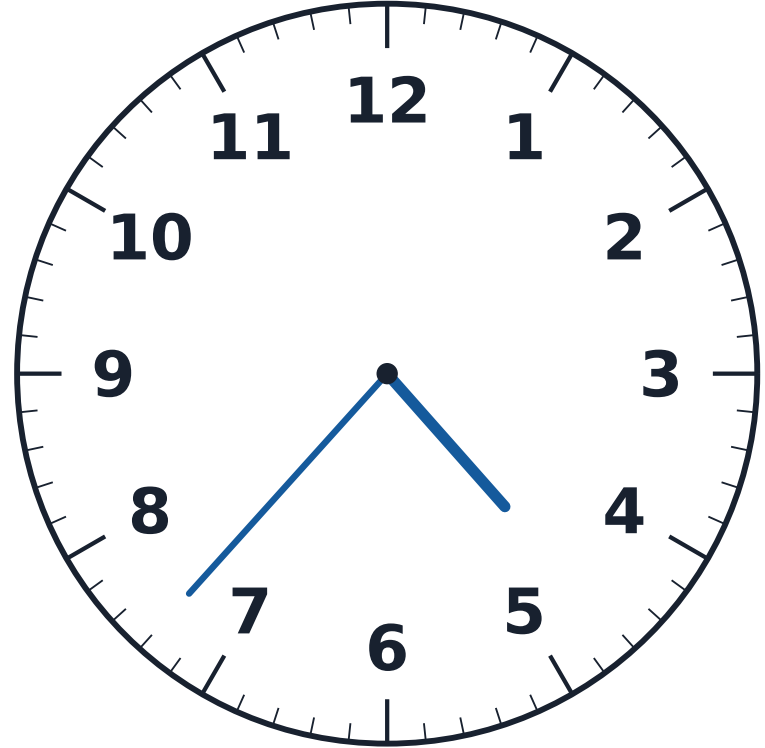
SET 38 • 3.MD.A.1

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.1

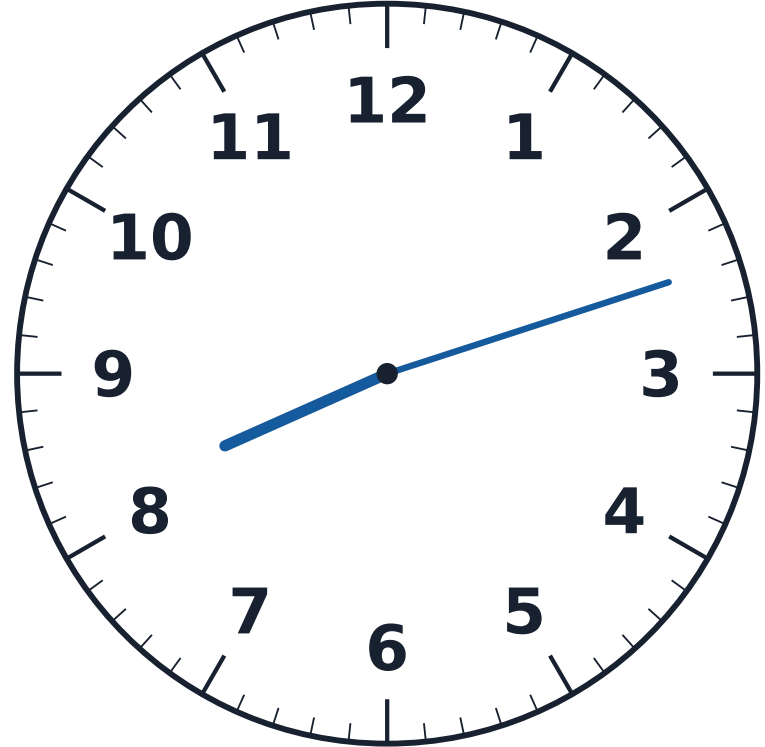
What time does the clock show?

Time: _____

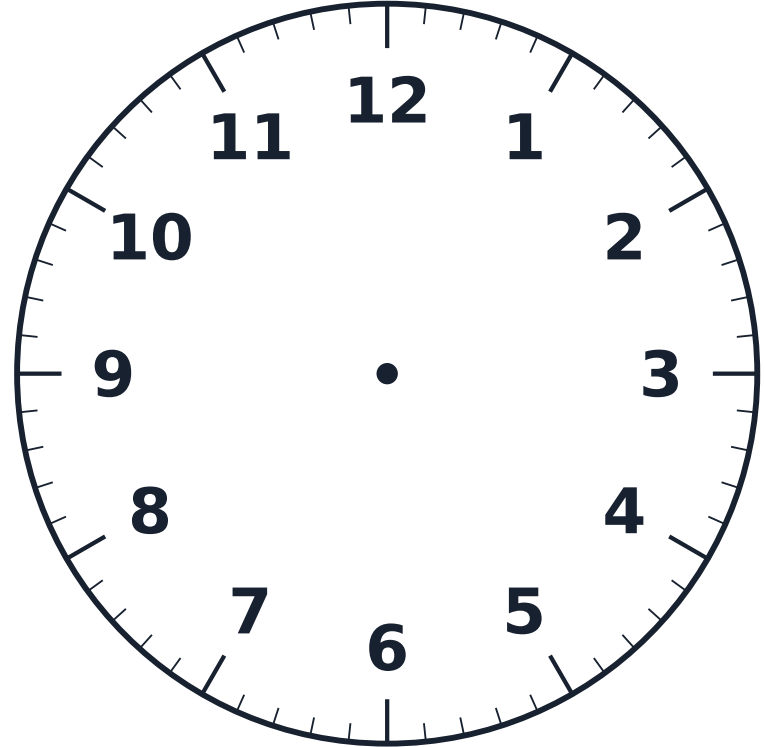


What time does the clock show?

- A.** 8:12
- B.** 8:14
- C.** 2:40
- D.** 3:40



Draw the hands on the clock to show 10:53.

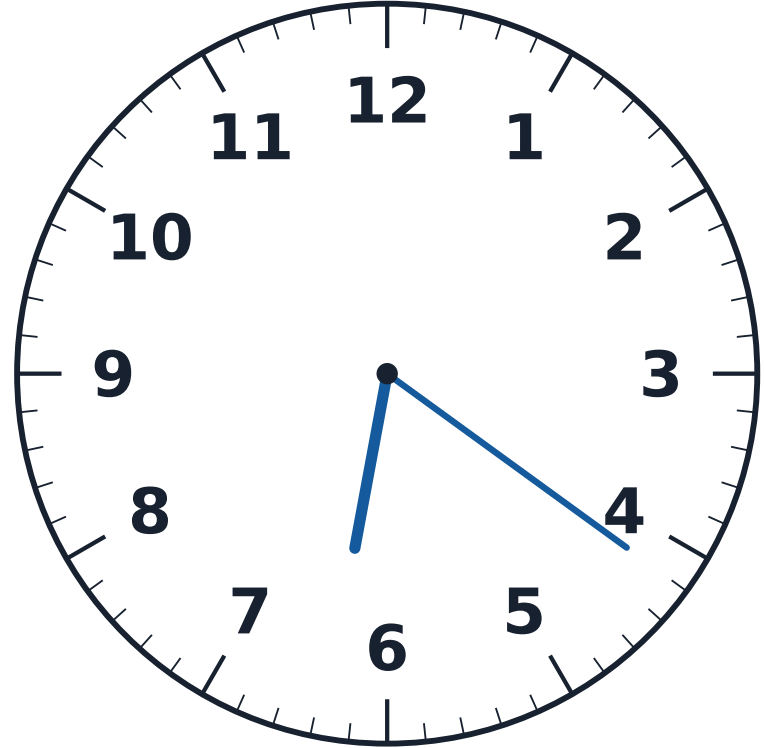


True or false?

The clock shows 6:21.

A. True

B. False



True or false?

The clock shows 1:48.

A. True

B. False



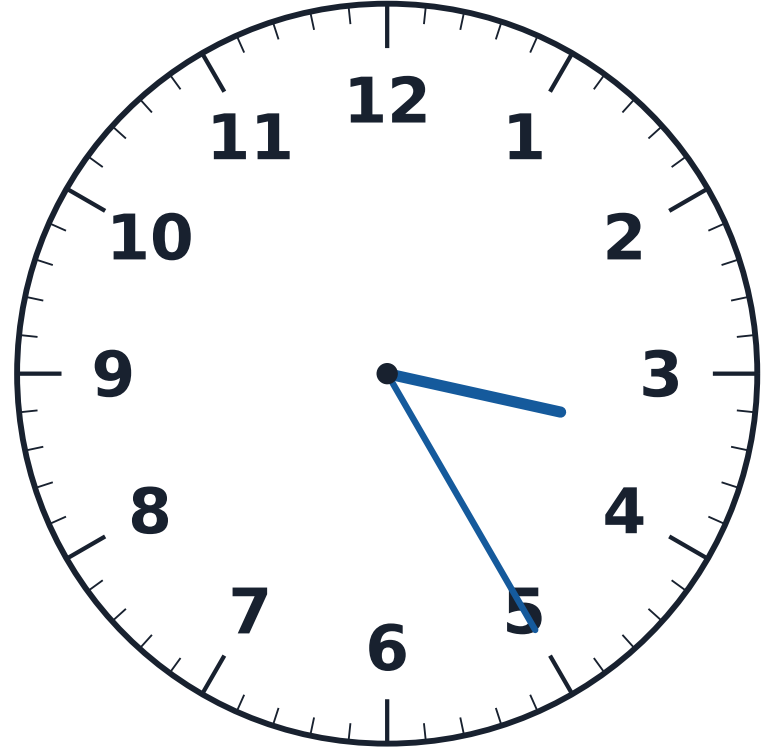
SET 38 • 3.MD.A.1

BACKWARD 1

Tell time to the nearest five minutes • Grade 2 • Measurement & Data •
2.MD.C.7

What time does the clock show?

Time: _____

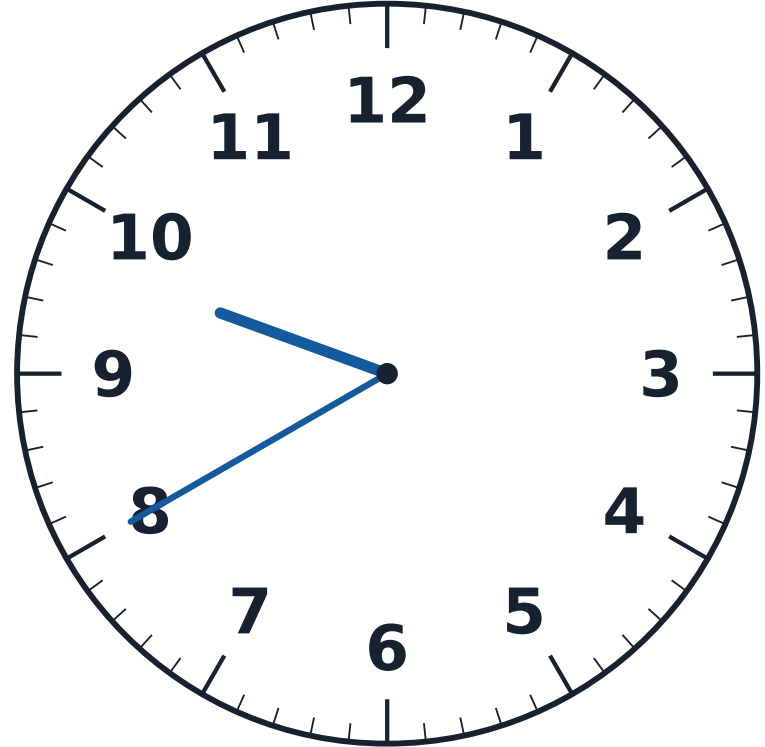


True or false?

The clock shows 9:40.

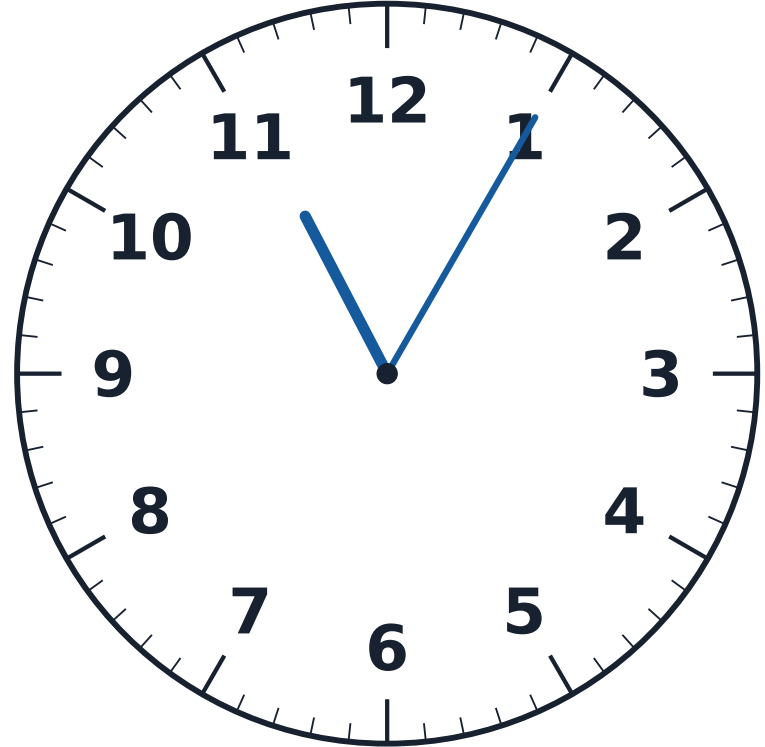
A. True

B. False



What time does the clock show?

- A.** 11:01
- B.** 12:05
- C.** 11:05
- D.** 1:55



When the minute hand points to the 6, how many minutes after the hour is it?

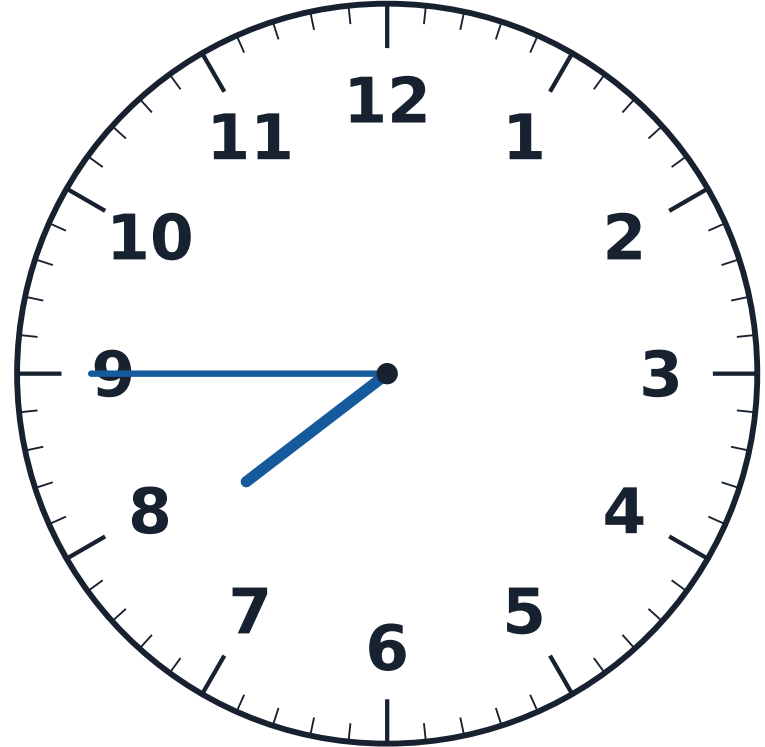
Minutes: _____

True or false?

The clock shows 7:15.

A. True

B. False



SET 38 • 3.MD.A.1

BACKWARD 2

Skip-count by 5s • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 5s from 0 to 35.

How many 5s did you count?

Fives: _____

True or false?

Counting by 5s: 40, 45, 50, 55.

A. True

B. False

Count by 5s.

15, 20, 25, ____

A. 30

B. 35

C. 50

D. 26

Skip-count by 5s.

35, 40, 45, __, __

Next two: _____

True or false?

Counting by 5s from 5, the 9th number is 40.

- A. True
- B. False

SET 38 • 3.MD.A.1

FORWARD 1

Express hours in minutes and minutes in seconds • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

True or false?

5 minutes = 300 seconds

A. True

B. False

1 hour 20 minutes = ____ minutes

- A.** 140
- B.** 80
- C.** 70
- D.** 120

Complete the table.

Minutes	1	2	3	4
Seconds	60	?	180	?

Missing values: _____

True or false?

2 hours = 200 minutes

A. True

B. False

SET 38 • 3.MD.A.1

FORWARD 2

Convert between units of time, including fractions of an hour • Grade 5 •
Measurement & Data • 5.MD.A.1

How many hours is 150 minutes?

Hours: _____

True or false?

90 seconds = $1\frac{1}{2}$ minutes

A. True

B. False

A movie is 135 minutes long.

How long is it in hours and minutes?

- A.** 1 hour 35 minutes
- B.** 2 hours 15 minutes
- C.** 2 hours 35 minutes
- D.** 13 hours 5 minutes

How many minutes are in $\frac{3}{4}$ of an hour?

Minutes: _____

True or false?

$3\frac{1}{2}$ hours = 330 minutes

A. True

B. False

Solve word problems about time intervals in minutes

MAIN	Grade 3 main question	3.MD.A.1
BACKWARD 1	Tell time and use a.m. and p.m.	2.MD.C.7
BACKWARD 2	Add two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Solve word problems about time intervals in hours and minutes	4.MD.A.2
FORWARD 2	Solve multistep time problems with conversions and fractions of a...	5.MD.A.1

SET 39 • 3.MD.A.1

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.1

A soccer game starts at 2:15 and ends at 3:05.

How many minutes long is the game?

Minutes: _____

Mia starts reading at 6:40. She reads for 35 minutes.
What time does she stop?

Time: _____

A bus ride takes 25 minutes and ends at 8:10.

What time did the ride start?

A. 7:55

B. 8:35

C. 7:35

D. 7:45

True or false?

Ben practices piano for 20 minutes and then guitar for 15 minutes. He practices for 35 minutes in all.

- A.** True
- B.** False

True or false?

A class starts at 10:50 and lasts 45 minutes. It ends at 11:25.

- A.** True
- B.** False

SET 39 • 3.MD.A.1

BACKWARD 1

Tell time and use a.m. and p.m. • Grade 2 • Measurement & Data • 2.MD.C.7

True or false?

Times from midnight until noon are a.m. times.

- A.** True
- B.** False

Which time is in the afternoon?

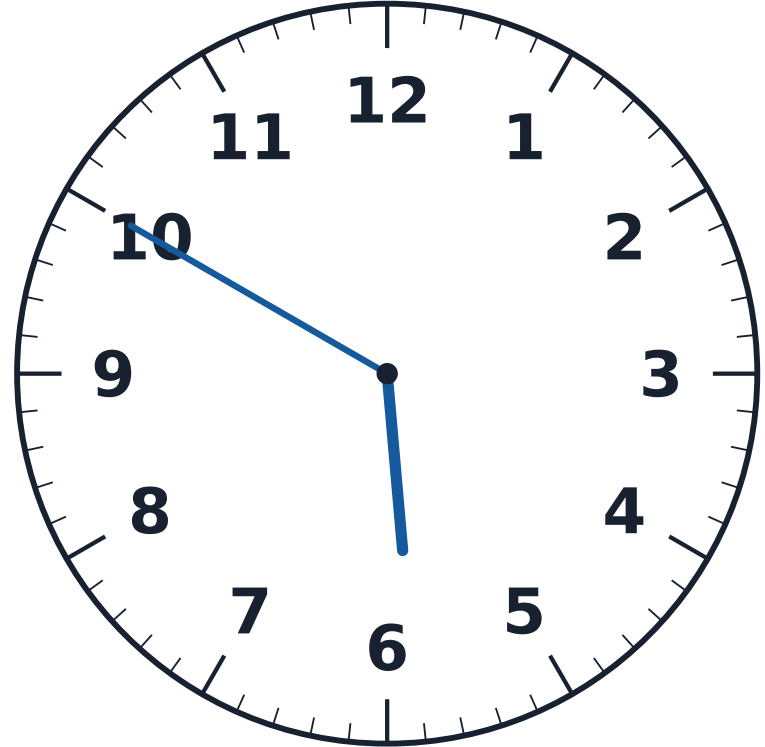
- A.** 11:00 a.m.
- B.** 3:00 a.m.
- C.** 6:00 a.m.
- D.** 3:00 p.m.

Is 9:00 at night an a.m. time or a p.m. time?

Answer: _____

What time does the clock show?

Time: _____



True or false?

Noon is written as 12:00 a.m.

A. True

B. False

SET 39 • 3.MD.A.1

BACKWARD 2

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$35 + 25$$

Sum: _____

True or false?

$$45 + 15 = 60$$

A. True

B. False

$$20 + 55 = ?$$

A. 35

B. 75

C. 85

D. 65

Add.

$$40 + 35$$

Sum: _____

True or false?

$$25 + 50 = 85$$

A. True

B. False

SET 39 • 3.MD.A.1

FORWARD 1

Solve word problems about time intervals in hours and minutes • Grade 4 •
Measurement & Data • 4.MD.A.2

A train leaves at 9:45 a.m. and arrives at 1:20 p.m.

How long is the trip?

Time: _____

True or false?

A movie starts at 6:50 p.m. and lasts 1 hour 45 minutes. It ends at 8:35 p.m.

- A.** True
- B.** False

A hike takes 2 hours 40 minutes and ends at 4:10 p.m.

When did it start?

- A.** 2:30 p.m.
- B.** 6:50 p.m.
- C.** 1:50 p.m.
- D.** 1:30 p.m.

Ty practices for 35 minutes each day for 4 days.

How long does he practice in all? Write the answer in hours and minutes.

Time: _____

True or false?

A 50-minute lesson that starts at 11:30 a.m. ends at 12:10 p.m.

- A. True
- B. False

SET 39 • 3.MD.A.1

FORWARD 2

Solve multistep time problems with conversions and fractions of a minute or hour • Grade 5 • Measurement & Data • 5.MD.A.1

A race has 3 laps. Each lap takes $1\frac{1}{2}$ minutes.

How many seconds does the race take?

Seconds: _____

True or false?

A trip of 2 hours 15 minutes is 2.25 hours long.

A. True

B. False

Ana runs for 0.75 hour and then walks for 20 minutes.

How many minutes is that in all?

- A.** 95
- B.** 65
- C.** 75
- D.** 55

A video is $2\frac{1}{4}$ minutes long.

How many seconds is that?

Seconds: _____

True or false?

$\frac{1}{3}$ of an hour is 30 minutes.

A. True

B. False

Represent time-interval problems on a number line diagram

MAIN	Grade 3 main question	3.MD.A.1
BACKWARD 1	Show sums and differences within 100 on a number line	2.MD.B.6
BACKWARD 2	Read clocks to the nearest five minutes	2.MD.C.7
FORWARD 1	Use number line diagrams for time problems in hours and minutes	4.MD.A.2
FORWARD 2	Use fractions and decimals of an hour in time problems	5.MD.A.1

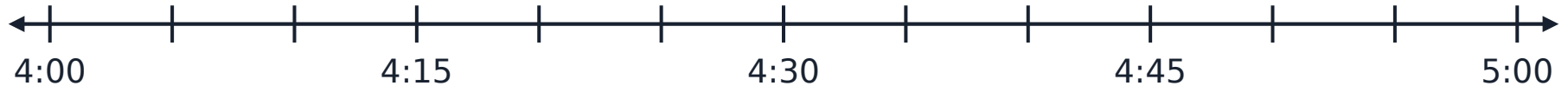
SET 40 • 3.MD.A.1

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.1

Lu starts her homework at 4:10 and works for 40 minutes.

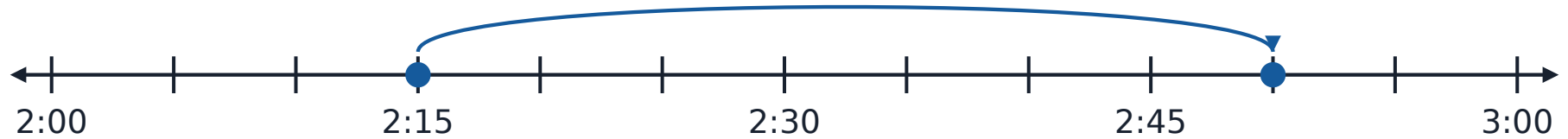
Show the time with jumps on the number line. What time does she finish?



Time: _____

The number line shows a trip.

How many minutes long is the trip?



Minutes: _____

A number line from 3:00 to 4:00 is marked every 5 minutes. Kim jumps from 3:15 to 3:40.

How many minutes is the jump?

A. 25

B. 55

C. 5

D. 40

True or false?

On a number line from 9:00 to 10:00, 9:30 is halfway between 9:00 and 10:00.

- A. True
- B. False

True or false?

On a number line, the jump from 7:20 to 8:05 is 35 minutes.

A. True

B. False

SET 40 • 3.MD.A.1

BACKWARD 1

Show sums and differences within 100 on a number line • Grade 2 •
Measurement & Data • 2.MD.B.6

Start at 30 on a number line and jump 25.

Where do you land?

Number: _____

True or false?

A jump from 15 to 45 on a number line is a jump of 30.

A. True

B. False

A jump from 40 to 75 on a number line is a jump of how much?

- A.** 115
- B.** 35
- C.** 25
- D.** 45

Show $20 + 15$ with jumps on the number line.



True or false?

Starting at 10 on a number line, a jump of 30 lands on 50.

- A.** True
- B.** False

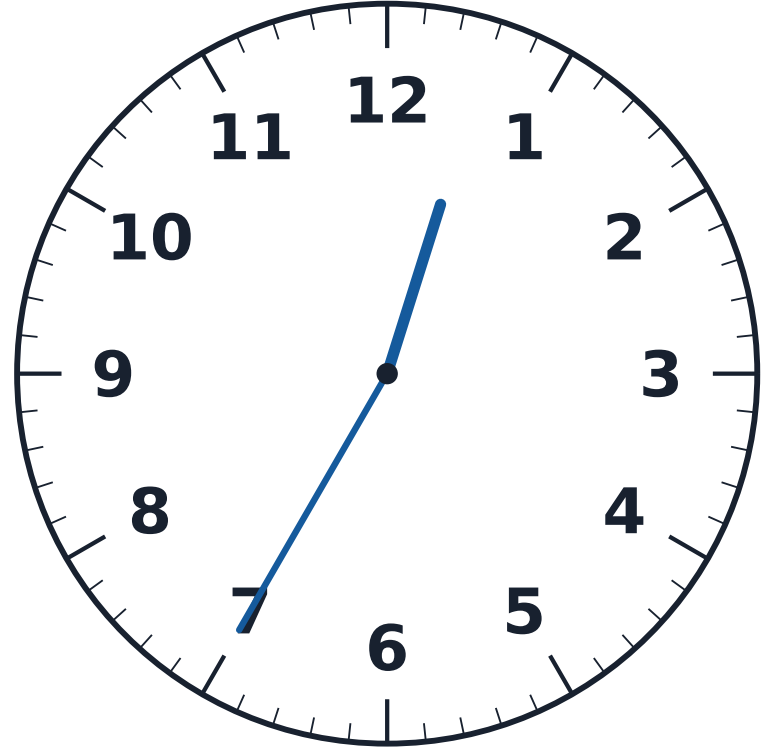
SET 40 • 3.MD.A.1

BACKWARD 2

Read clocks to the nearest five minutes • Grade 2 • Measurement & Data •
2.MD.C.7

What time does the clock show?

Time: _____

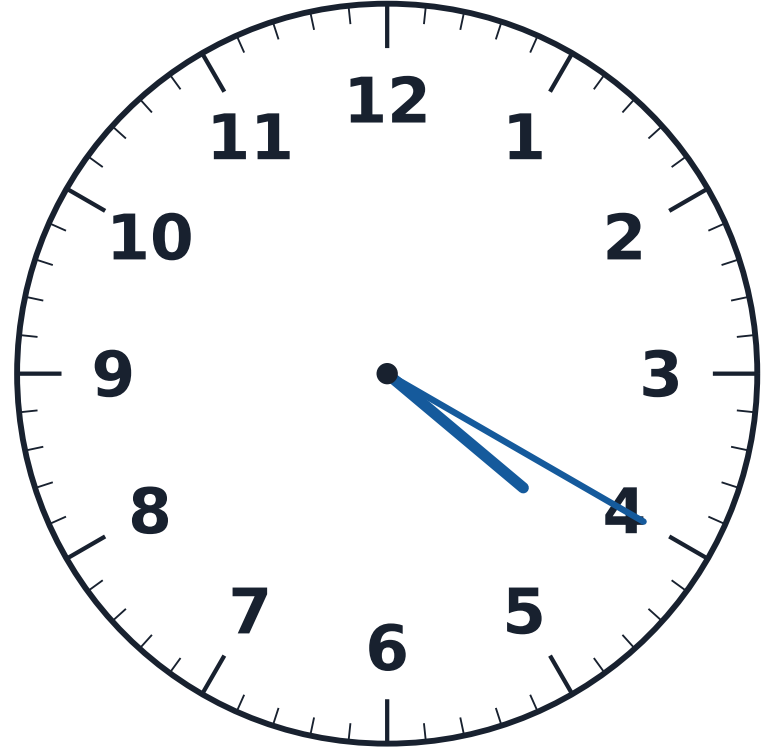


True or false?

The clock shows 4:20.

A. True

B. False



What time does the clock show?

- A.** 10:10
- B.** 11:10
- C.** 10:02
- D.** 2:50



At 8:45, which number does the minute hand point to?

Number: _____

True or false?

At 3:30, the minute hand points to the 3.

- A. True
- B. False

SET 40 • 3.MD.A.1

FORWARD 1

Use number line diagrams for time problems in hours and minutes • Grade 4 •
Measurement & Data • 4.MD.A.2

A field trip goes from 9:40 a.m. to 12:15 p.m.
Use a number line to find how long the trip is.

Time: _____

True or false?

On a number line, the jump from 10:45 a.m. to 12:30 p.m. is 1 hour 45 minutes.

- A.** True
- B.** False

A jump of 1 hour 20 minutes starts at 2:50 p.m. on a number line.

Where does it end?

A. 3:70 p.m.

B. 4:10 p.m.

C. 3:10 p.m.

D. 4:20 p.m.

A show starts at 7:35 p.m. and lasts 1 hour 40 minutes.

Show the jumps on the number line. When does the show end?



Ends at: _____

True or false?

On a number line, the jump from 11:50 a.m. to 1:05 p.m. is 1 hour 25 minutes.

- A. True
- B. False

SET 40 • 3.MD.A.1

FORWARD 2

Use fractions and decimals of an hour in time problems • Grade 5 •
Measurement & Data • 5.MD.A.1

How many minutes are in $\frac{1}{4}$ of an hour?

Minutes: _____

True or false?

$1\frac{1}{2}$ hours is 90 minutes.

A. True

B. False

A trip takes $2\frac{1}{4}$ hours.

How many minutes is that?

- A.** 135
- B.** 150
- C.** 225
- D.** 125

Mia reads for 0.5 hour and then for 0.75 hour.
How many minutes does she read in all?

Minutes: _____

True or false?

$\frac{2}{3}$ of an hour is 45 minutes.

A. True

B. False

Measure and estimate liquid volumes and masses in liters, grams, and kilograms

MAIN	Grade 3 main question	3.MD.A.2
BACKWARD 1	Estimate lengths in inches, feet, centimeters, and meters	2.MD.A.3
BACKWARD 2	Choose and use tools to measure length	2.MD.A.1
FORWARD 1	Know the relative sizes of grams and kilograms, and of milliliters...	4.MD.A.1
FORWARD 2	Convert metric units of mass and liquid volume with decimals	5.MD.A.1

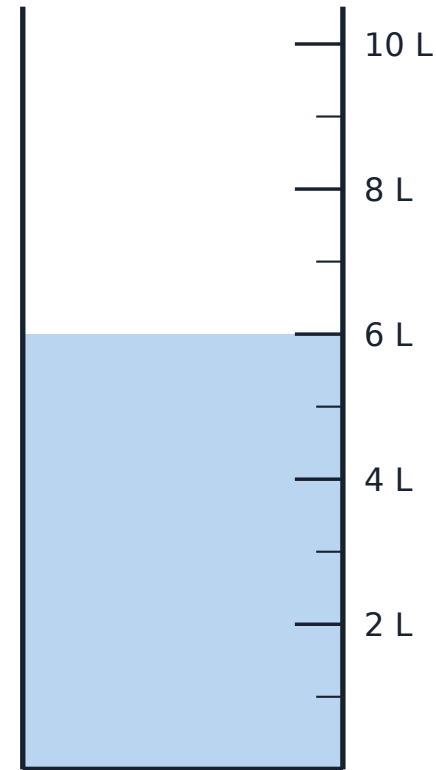
SET 41 • 3.MD.A.2

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.2

How many liters of water are
in the container?

Liters: _____



Which object has a mass of about 1 kilogram?

- A.** A grape
- B.** A paper clip
- C.** A pineapple
- D.** A car

True or false?

A bathtub holds more than 1 liter of water.

A. True

B. False

Would you measure the mass of a bean in grams or in kilograms?

Unit: _____

Shade the container to show
3 liters.



SET 41 • 3.MD.A.2

BACKWARD 1

Estimate lengths in inches, feet, centimeters, and meters • Grade 2 •
Measurement & Data • 2.MD.A.3

About how long is a new pencil: 7 inches or 7 feet?

Answer: _____

True or false?

A door is about 2 meters tall.

A. True

B. False

Which is the best estimate of the length of a car?

- A.** 40 meters
- B.** 4 inches
- C.** 4 centimeters
- D.** 4 meters

About how long is a paper clip: 3 centimeters or 3 meters?

Answer: _____

True or false?

A school bus is about 12 inches long.

A. True

B. False

SET 41 • 3.MD.A.2

BACKWARD 2

Choose and use tools to measure length • Grade 2 • Measurement & Data •
2.MD.A.1

True or false?

A meter stick is a good tool for measuring the length of a classroom.

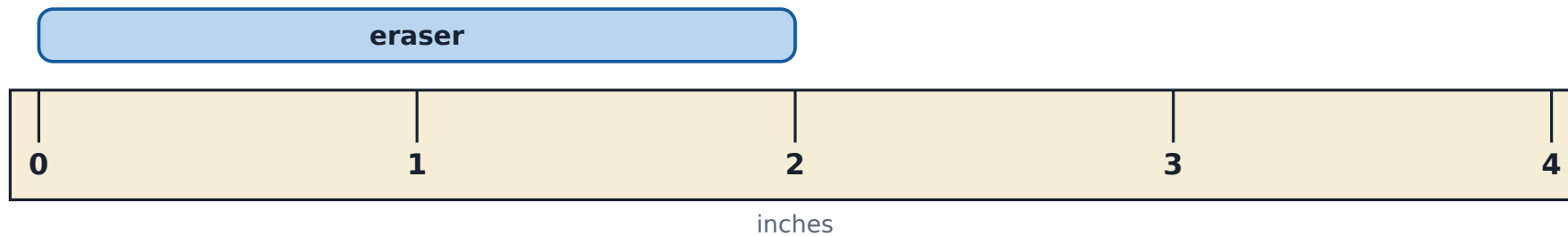
A. True

B. False

Which tool is best for measuring the length of a crayon?

- A.** A ruler
- B.** A clock
- C.** A scale
- D.** A measuring cup

How long is the eraser?



Length: _____

Name a tool you can use to measure the length of a hallway.

Tool: _____

True or false?

A clock is a tool for measuring length.

A. True

B. False

SET 41 • 3.MD.A.2

FORWARD 1

Know the relative sizes of grams and kilograms, and of milliliters and liters •

Grade 4 • Measurement & Data • 4.MD.A.1

How many grams are in 3 kilograms?

Grams: _____

True or false?

1 liter = 1,000 milliliters

A. True

B. False

Which is heaviest?

- A.** 200 grams
- B.** 2 kilograms
- C.** 1,500 grams
- D.** 20 grams

Complete the table.

Liters	1	2	3	4
Milliliters	1,000	?	3,000	?

Missing values: _____

True or false?

5 kilograms = 500 grams

A. True

B. False

SET 41 • 3.MD.A.2

FORWARD 2

Convert metric units of mass and liquid volume with decimals • Grade 5 •
Measurement & Data • 5.MD.A.1

Convert.

2.5 kilograms = ____ grams

Grams: _____

True or false?

750 milliliters = 0.75 liter

A. True

B. False

Convert.

3,200 grams = ____ kilograms

A. 32

B. 320

C. 3.2

D. 0.32

Convert.

1.25 liters = ____ milliliters

Milliliters: _____

True or false?

400 grams = 4 kilograms

A. True

B. False

Solve one-step word problems about masses or liquid volumes

MAIN	Grade 3 main question	3.MD.A.2
BACKWARD 1	Subtract lengths in one-step word problems within 100	2.MD.B.5
BACKWARD 2	Add two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Solve liquid volume and mass problems that need a unit conversi...	4.MD.A.2
FORWARD 2	Solve multistep problems with metric conversions and decimals	5.MD.A.1

SET 42 • 3.MD.A.2

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.2

A pot holds 9 liters of soup. 4 liters are served.
How many liters are left?

Liters: _____

Each bag of rice has a mass of 5 kilograms.

What is the mass of 6 bags?

Mass: _____

A 36-liter tank is filled with a 4-liter bucket.
How many full buckets fill the tank?

- A.** 8
- B.** 9
- C.** 40
- D.** 32

True or false?

A 450-gram melon and a 300-gram apple have a total mass of 850 grams.

- A. True
- B. False

True or false?

24 liters of juice are shared equally among 3 jugs.
Each jug gets 6 liters.

A. True

B. False

SET 42 • 3.MD.A.2

BACKWARD 1

Subtract lengths in one-step word problems within 100 • Grade 2 •
Measurement & Data • 2.MD.B.5

A rope is 85 cm long. 38 cm is cut off.

How long is the rope now?

Length: _____

True or false?

An 80-inch shelf with 35 inches cut off is 45 inches long.

- A. True
- B. False

A path is 64 m long. Ana has walked 29 m.

How far does she have left?

A. 25 m

B. 93 m

C. 35 m

D. 45 m

A stick is 74 inches long. Lu cuts off 26 inches.
How long is the stick now?

Length: _____

True or false?

A 90-cm board with 45 cm cut off is 55 cm long.

A. True

B. False

SET 42 • 3.MD.A.2

BACKWARD 2

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$45 + 30$$

Sum: _____

True or false?

$$60 + 25 = 85$$

A. True

B. False

$$38 + 27 = ?$$

A. 11

B. 55

C. 75

D. 65

Add.

$$54 + 19$$

Sum: _____

True or false?

$$29 + 46 = 65$$

A. True

B. False

SET 42 • 3.MD.A.2

FORWARD 1

Solve liquid volume and mass problems that need a unit conversion • Grade 4 •
Measurement & Data • 4.MD.A.2

A jug holds 2 liters. Ella pours out 750 milliliters.
How many milliliters are left?

Milliliters: _____

True or false?

Four 250-gram bags of nuts have a total mass of 1 kilogram.

A. True

B. False

A 3-kilogram bag of flour is split into 6 equal bags.
What is the mass of each bag?

- A.** 2 grams
- B.** 50 grams
- C.** 500 grams
- D.** 5,000 grams

A box has a mass of 1 kg 200 g. Another box has a mass of 850 g.

What is their total mass in grams?

Grams: _____

True or false?

A 5-liter jug fills exactly 8 cups that each hold 500 milliliters.

A. True

B. False

SET 42 • 3.MD.A.2

FORWARD 2

Solve multistep problems with metric conversions and decimals • Grade 5 •
Measurement & Data • 5.MD.A.1

A 1.5-liter bottle fills 6 equal cups.

How many milliliters go in each cup?

Milliliters: _____

True or false?

3 bags that each have a mass of 0.4 kg have a total mass of 1,200 grams.

A. True

B. False

A 2.4-kg cheese is cut into 8 equal pieces.

What is the mass of each piece?

- A.** 0.03 kilogram
- B.** 3 kilograms
- C.** 300 grams
- D.** 30 grams

A tank has 4.5 liters of water. 1,750 mL leaks out.
How many liters are left?

Liters: _____

True or false?

Five 350-mL glasses hold 1.5 liters in all.

A. True

B. False

Draw a scaled picture graph to represent data

MAIN

Grade 3 main question

3.MD.B.3

BACKWARD 1

Draw and read a picture graph with a single-unit scale

2.MD.D.10

BACKWARD 2

Skip-count by 5s and 10s

2.NBT.A.2

FORWARD 1

Make a line plot of measurements in fractions of a unit

4.MD.B.4

FORWARD 2

Use operations on fractions to solve problems with line plot data

5.MD.B.2

SET 43 • 3.MD.B.3

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.3

Make a picture graph of the data. Each ● stands for 2 votes.

Apples: 6 votes, Bananas: 10 votes, Grapes: 4 votes

Favorite fruit	
Apples	
Bananas	
Grapes	

Key: ● = 2 votes

In a picture graph, each ● stands for 5 books. Ty read 20 books.

How many ● should be drawn for Ty?

A. 4

B. 20

C. 5

D. 15

A picture graph has the key ● = 3 cars. The row for red cars has 5 ●.

How many red cars are there?

Red cars: _____

True or false?

If ● = 4 students, then 7 ● stand for 28 students.

- A. True
- B. False

Data: Dogs 8, Cats 12, Fish 4. You will make a picture graph with ● = 4 pets.

How many ● go in each row?

Dogs: _____

Cats: _____

Fish: _____

SET 43 • 3.MD.B.3

BACKWARD 1

Draw and read a picture graph with a single-unit scale • Grade 2 •
Measurement & Data • 2.MD.D.10

Make a picture graph. Draw one ● for each pet.

Dogs: 4, Cats: 3, Birds: 2

Pets	
Dogs	
Cats	
Birds	

Key: ● = 1 pet

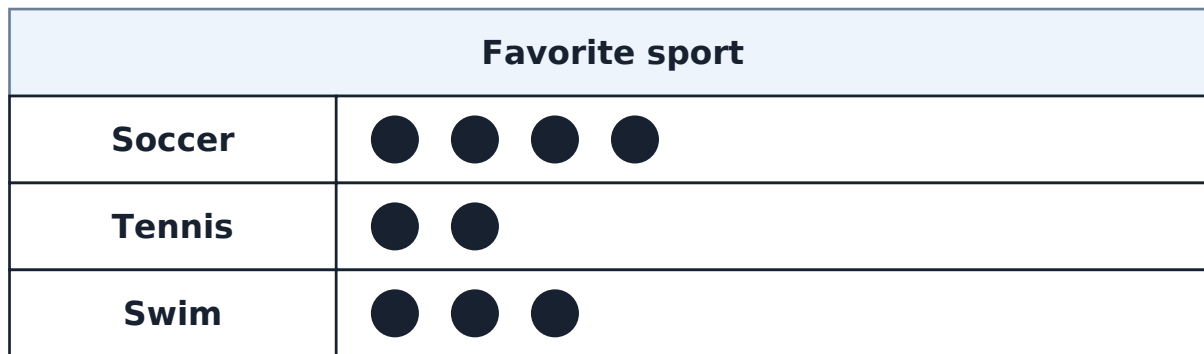
True or false?

In a picture graph where ● = 1 child, 5 ● stand for 5 children.

- A. True
- B. False

Use the picture graph.

How many children chose soccer?



Key: ● = 1 child

A. 5

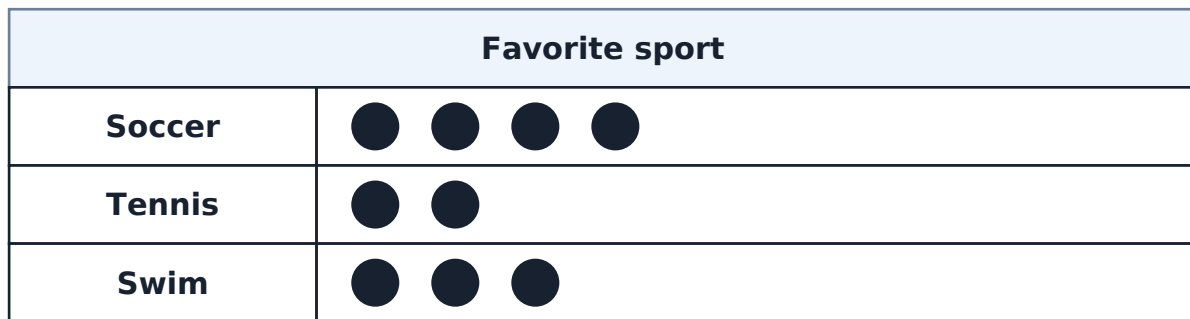
B. 4

C. 2

D. 3

Use the picture graph.

How many more children chose swim than tennis?

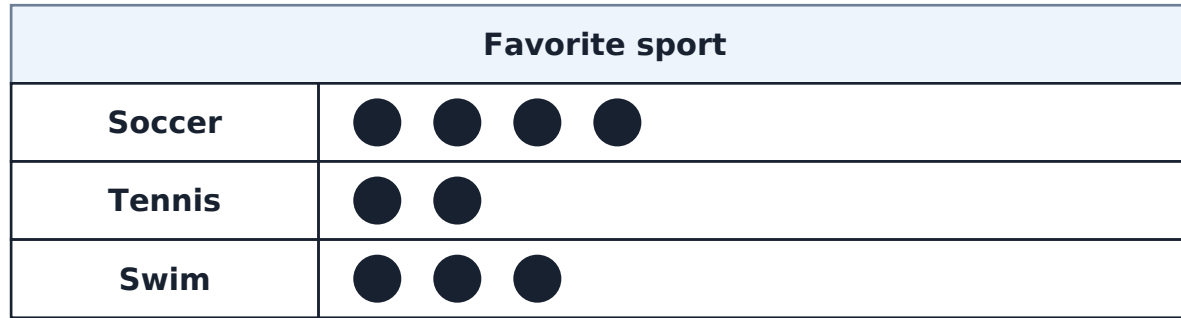


Key: ● = 1 child

More: _____

True or false?

In the picture graph, tennis has the most votes.



Key: ● = 1 child

A. True

B. False

SET 43 • 3.MD.B.3

BACKWARD 2

Skip-count by 5s and 10s • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.2

Skip-count by 5s from 5.

What is the 4th number?

Number: _____

True or false?

Skip-counting by 10s: 10, 20, 30, 40, 50.

A. True

B. False

Skip-count by 5s.

5, 10, 15, 20, ____

A. 21

B. 30

C. 35

D. 25

Skip-count by 10s from 10.

What is the 7th number?

Number: _____

True or false?

Skip-counting by 5s: 5, 10, 15, 25.

A. True

B. False

SET 43 • 3.MD.B.3

FORWARD 1

Make a line plot of measurements in fractions of a unit • Grade 4 •
Measurement & Data • 4.MD.B.4

Make a line plot of the data.

Bean lengths (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, 1



True or false?

A line plot shows each data value as an X above a number line.

- A.** True
- B.** False

Which number line works best for a line plot of $\frac{1}{8}$, $\frac{3}{8}$, and $\frac{5}{8}$ inch?

- A.** A line from 0 to 1 marked in halves
- B.** A line from 0 to 1 marked in eighths
- C.** A line from 0 to 8 marked in ones
- D.** A line from 1 to 2 marked in fourths

Lengths (inches): $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, $\frac{1}{2}$

In a line plot of the data, how many X's go above $\frac{1}{2}$?

X's: _____

True or false?

In a line plot of $\frac{1}{4}$, $\frac{1}{4}$, and $\frac{3}{4}$, there are 3 X's above $\frac{1}{4}$.

A. True

B. False

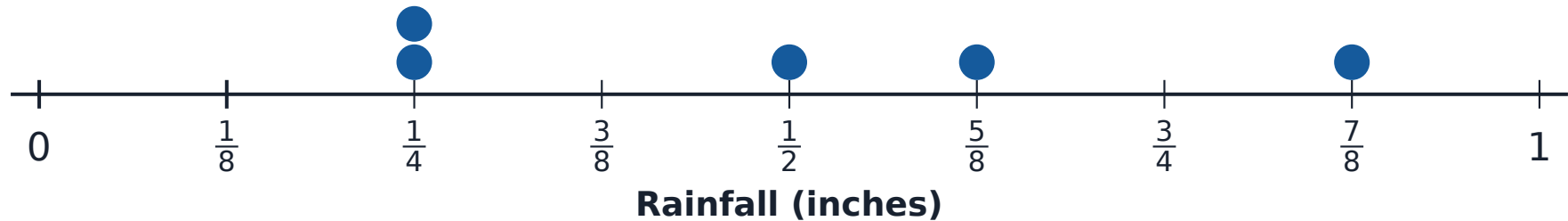
SET 43 • 3.MD.B.3

FORWARD 2

Use operations on fractions to solve problems with line plot data • Grade 5 •
Measurement & Data • 5.MD.B.2

The line plot shows rainfall on 5 days.

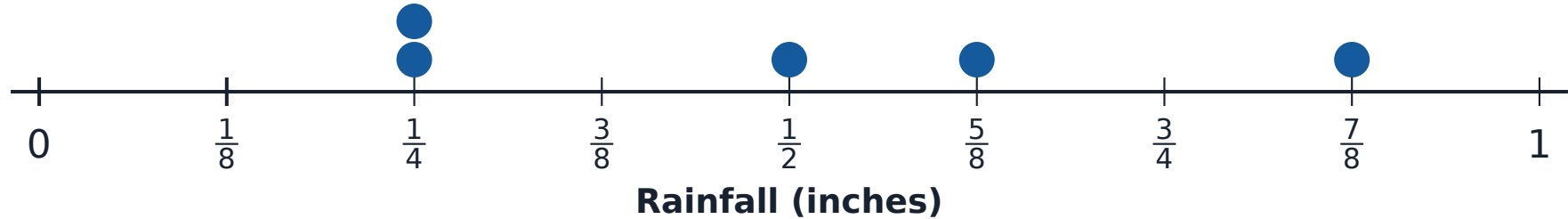
What is the total rainfall?



Total: _____

True or false?

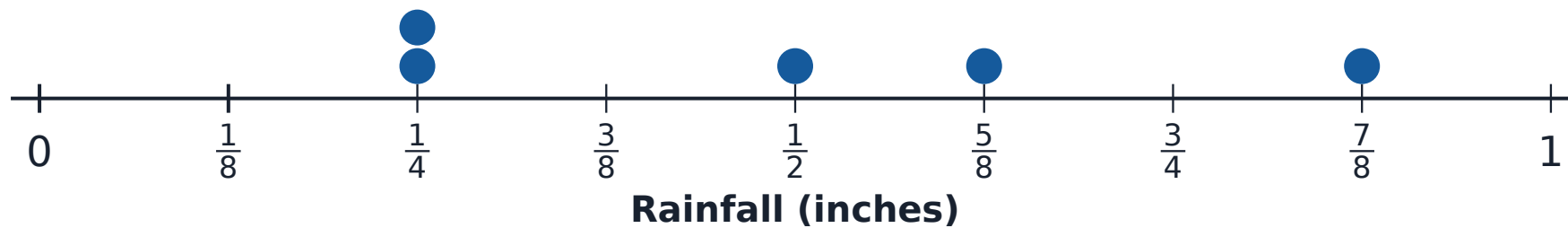
The two days with $\frac{1}{4}$ inch of rain had $\frac{1}{2}$ inch in all.



A. True

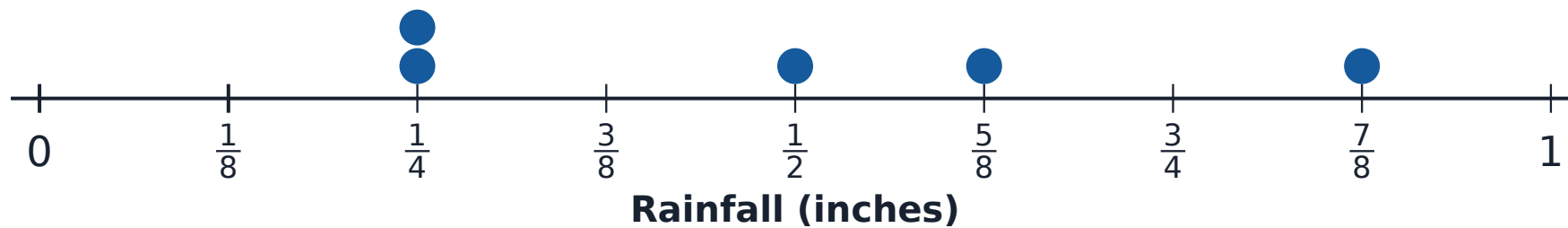
B. False

What is the difference between the greatest and the least rainfall?



- A.** $\frac{3}{8}$ inch **B.** $\frac{1}{2}$ inch **C.** $\frac{5}{8}$ inch **D.** $\frac{7}{8}$ inch

If the total rainfall had been spread equally over the 5 days, how much would have fallen each day?



Each day: _____

True or false?

The total of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ is $\frac{3}{14}$.

A. True

B. False

Draw a scaled bar graph to represent data

MAIN	Grade 3 main question	3.MD.B.3
BACKWARD 1	Draw and read a bar graph with a single-unit scale	2.MD.D.10
BACKWARD 2	Read whole numbers on a number line scale	2.MD.B.6
FORWARD 1	Use a bar graph to solve multiplicative comparison problems — ne...	4.OA.A.2
FORWARD 2	Graph data as points in the first quadrant — nearest related (no di...	5.G.A.2

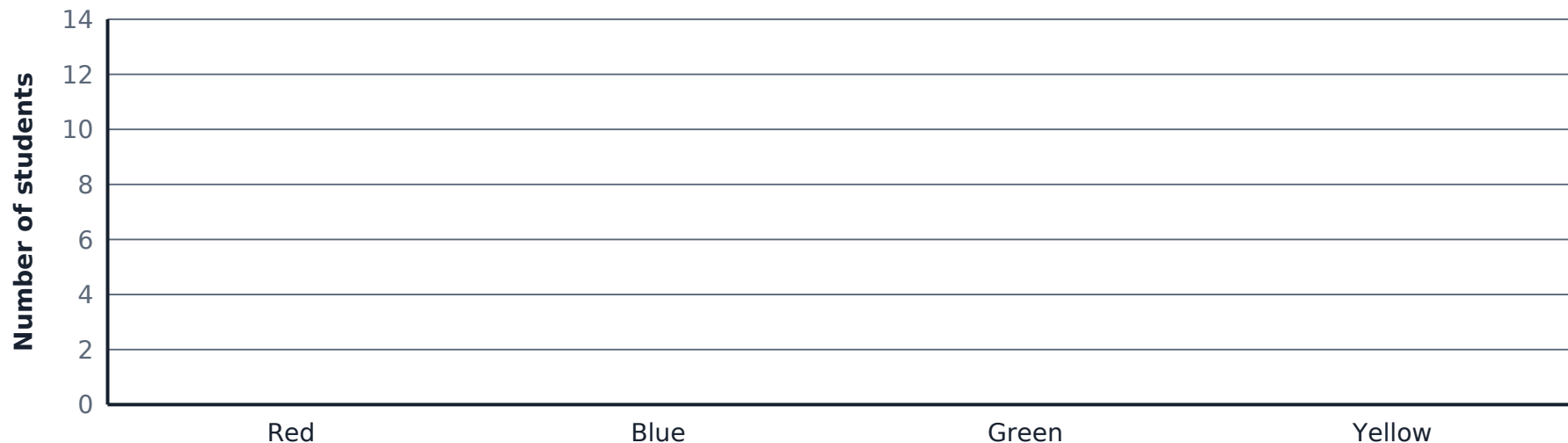
SET 44 • 3.MD.B.3

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.3

Make a bar graph of the data. The scale counts by 2s.

Red: 8, Blue: 12, Green: 6, Yellow: 10



A bar graph has a scale that counts by 2s. A bar ends halfway between 6 and 8.

What number does the bar show?

A. 14

B. 7

C. 8

D. 6

True or false?

On a bar graph with a scale that counts by 10s, a bar for 40 ends on the fourth line above 0.

- A.** True
- B.** False

Which scale is better for a bar graph of 10, 25, 40, and 35: counting by 1s or counting by 5s? Explain.

Scale: _____

Explanation: _____

True or false?

On a bar graph with a scale that counts by 2s, a bar for 9 ends exactly on a labeled line.

- A. True
- B. False

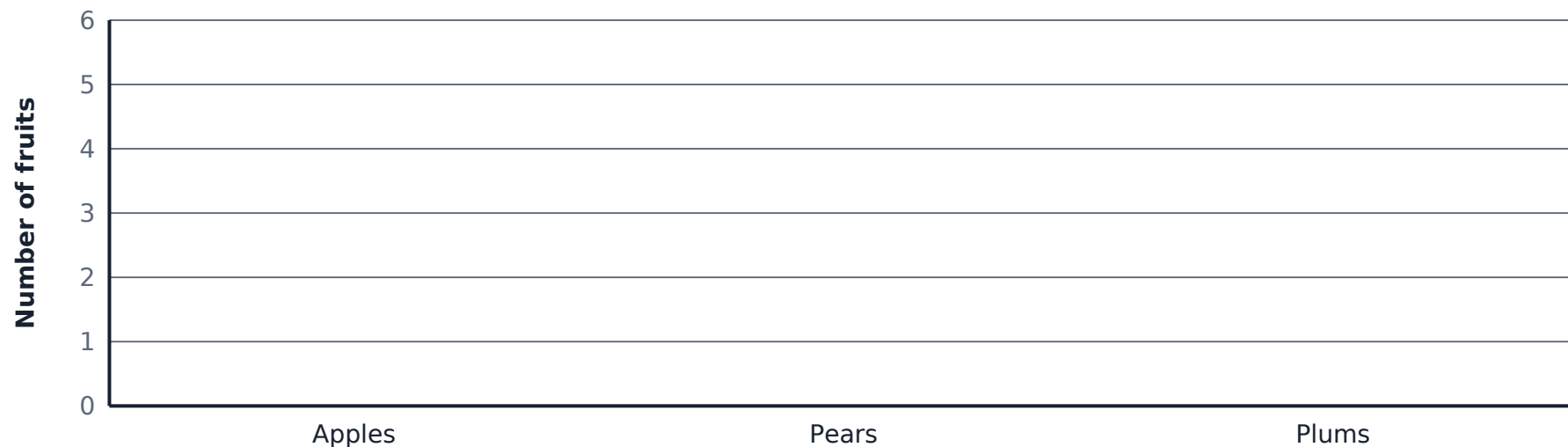
SET 44 • 3.MD.B.3

BACKWARD 1

Draw and read a bar graph with a single-unit scale • Grade 2 • Measurement & Data • 2.MD.D.10

Make a bar graph.

Apples: 4, Pears: 2, Plums: 5



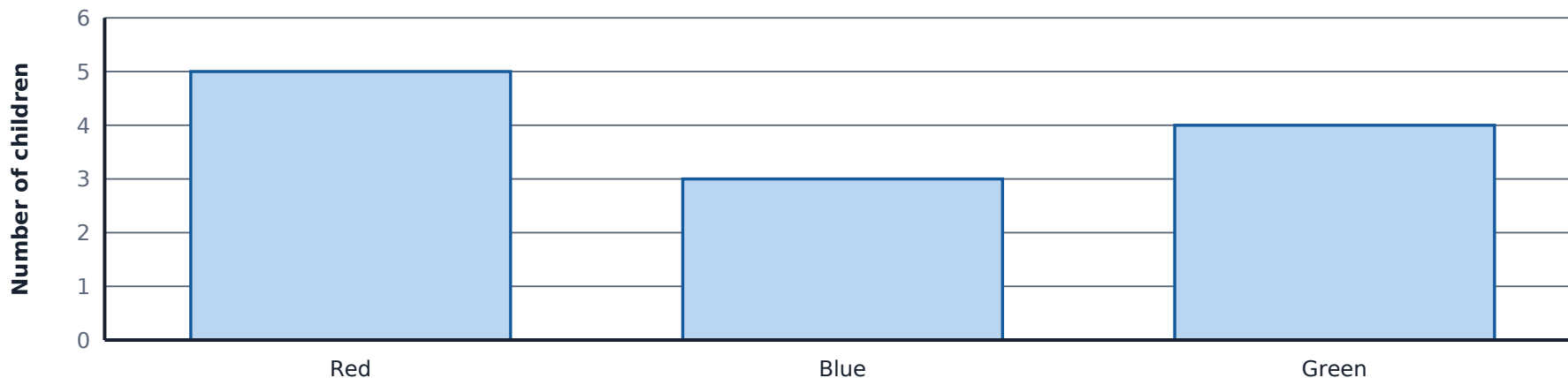
True or false?

In a bar graph, a taller bar shows a greater number.

- A.** True
- B.** False

Use the bar graph.

How many children chose red?



A. 4

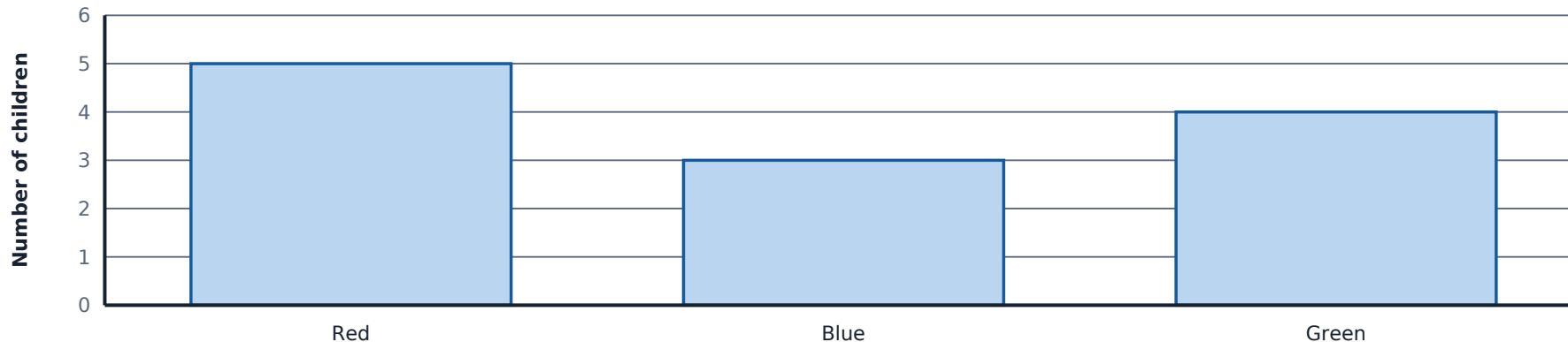
B. 5

C. 3

D. 6

Use the bar graph.

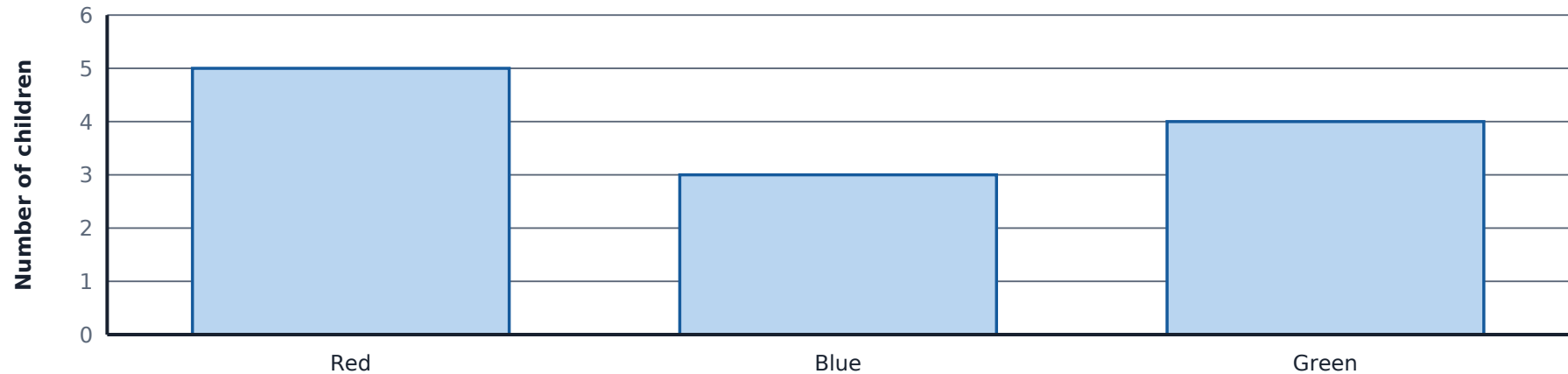
How many more children chose red than blue?



More: _____

True or false?

In the bar graph, green has the most votes.



A. True

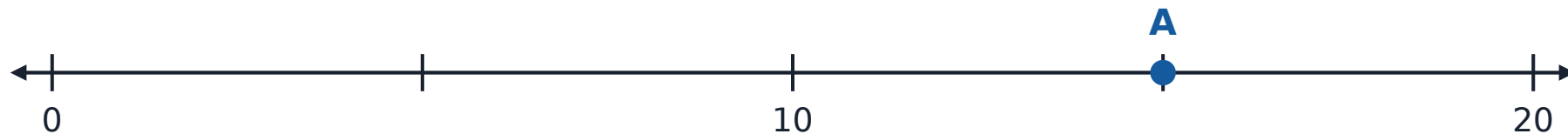
B. False

SET 44 • 3.MD.B.3

BACKWARD 2

Read whole numbers on a number line scale • Grade 2 • Measurement & Data
• 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line that counts by 10s, 30 comes right after 20.

A. True

B. False

A number line counts by 2s: 0, 2, 4, 6, 8.

What number is halfway between 4 and 6?

- A.** 5
- B.** 10
- C.** 7
- D.** 3

What number is at point B?



B = _____

True or false?

On a number line that counts by 5s, the mark after 25 is 35.

- A. True
- B. False

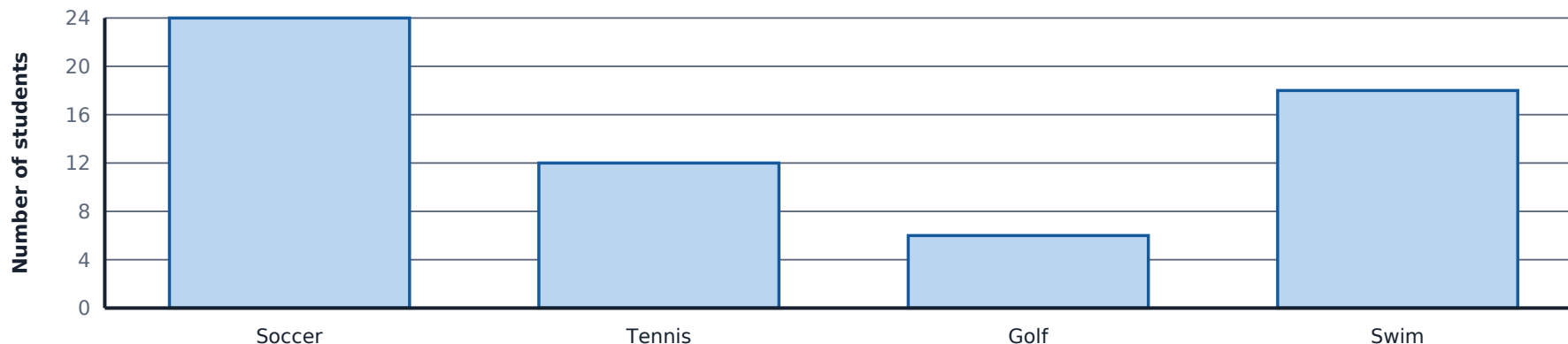
FORWARD 1

Use a bar graph to solve multiplicative comparison problems • Grade 4 •
Operations & Algebraic Thinking • 4.OA.A.2

► No Grade 4 standard directly continues this skill. This branch uses the nearest related Grade 4 standard.

Use the bar graph.

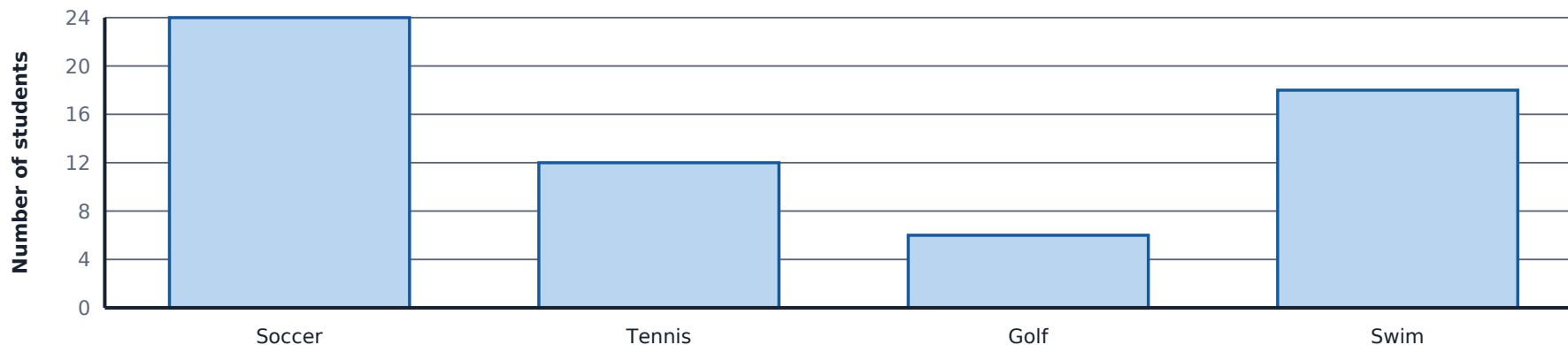
How many times as many students chose soccer as chose golf?



Times as many: _____

True or false?

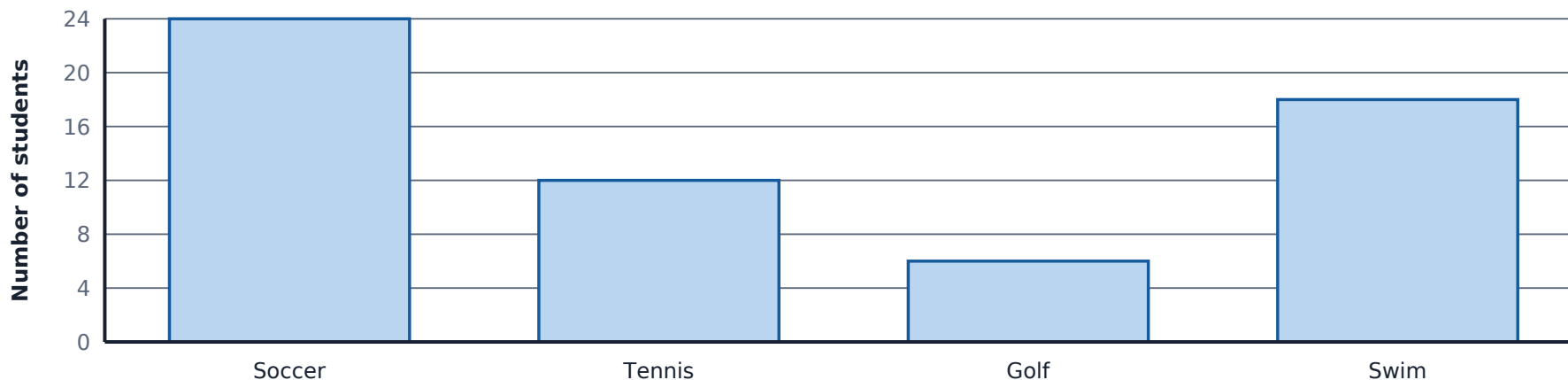
Use the bar graph. Twice as many students chose swim as chose golf.



- A. True
- B. False

Use the bar graph.

Which sport was chosen by 3 times as many students as golf?



A. Swim

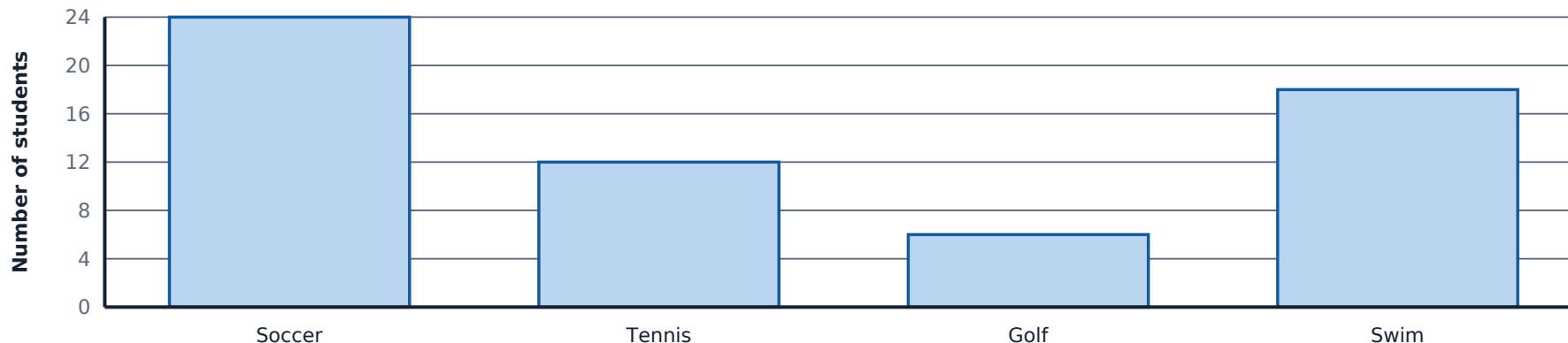
B. Golf

C. Soccer

D. Tennis

Use the bar graph.

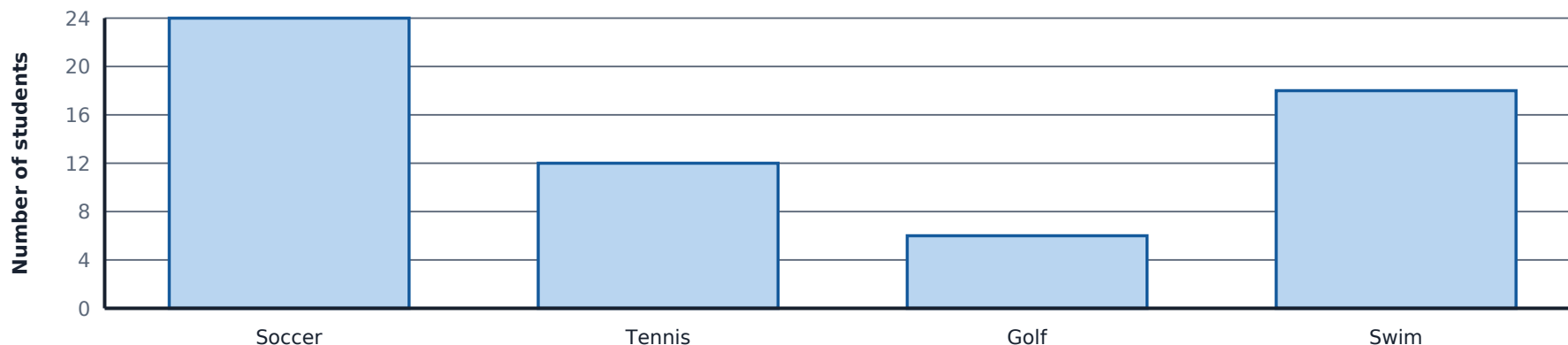
How many times as many students chose soccer as chose tennis?



Times as many: _____

True or false?

Use the bar graph. Soccer was chosen by 4 times as many students as golf.



A. True

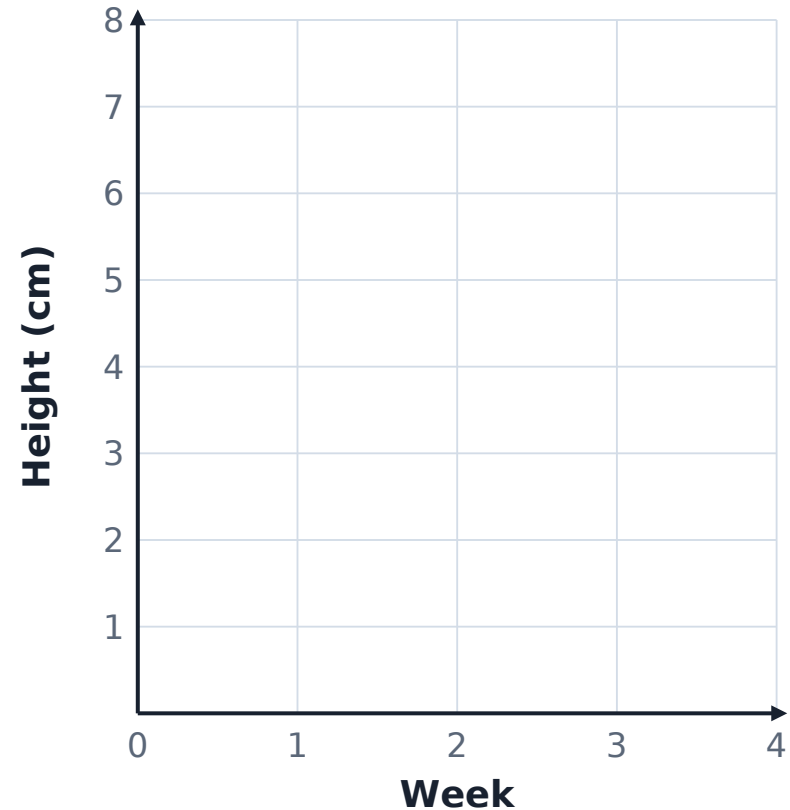
B. False

FORWARD 2

Graph data as points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A plant grows 2 cm each week. Graph the points (week, height): (0, 0), (1, 2), (2, 4), and (3, 6).

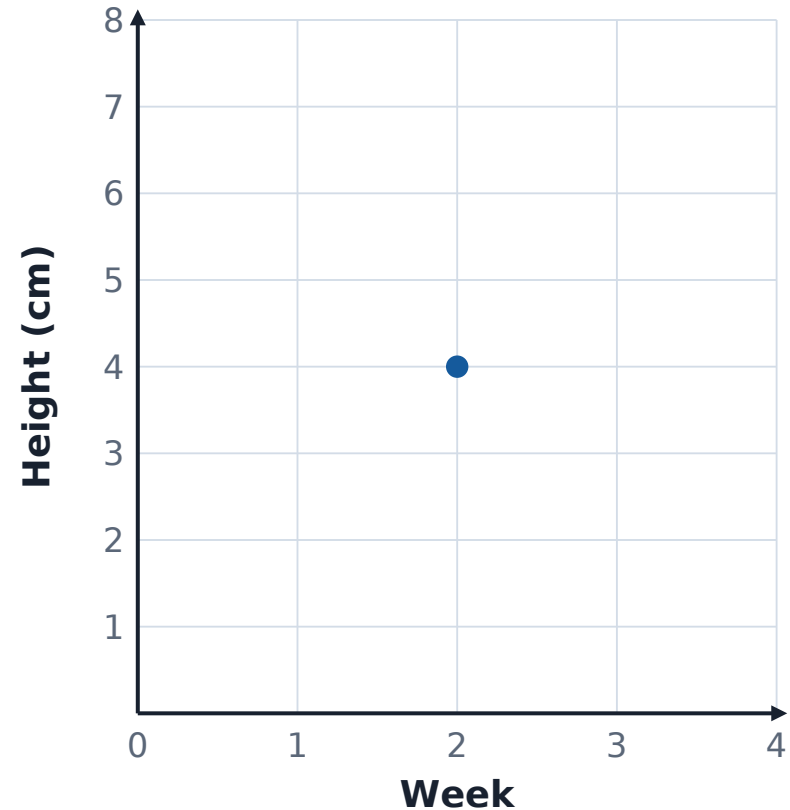


True or false?

On the graph, the point (2, 4) shows a height of 4 cm in week 2.

A. True

B. False



A store sells 5 hats each day.

Which point (days, hats) shows the total sold after 3 days?

A. (3, 5)

B. (15, 3)

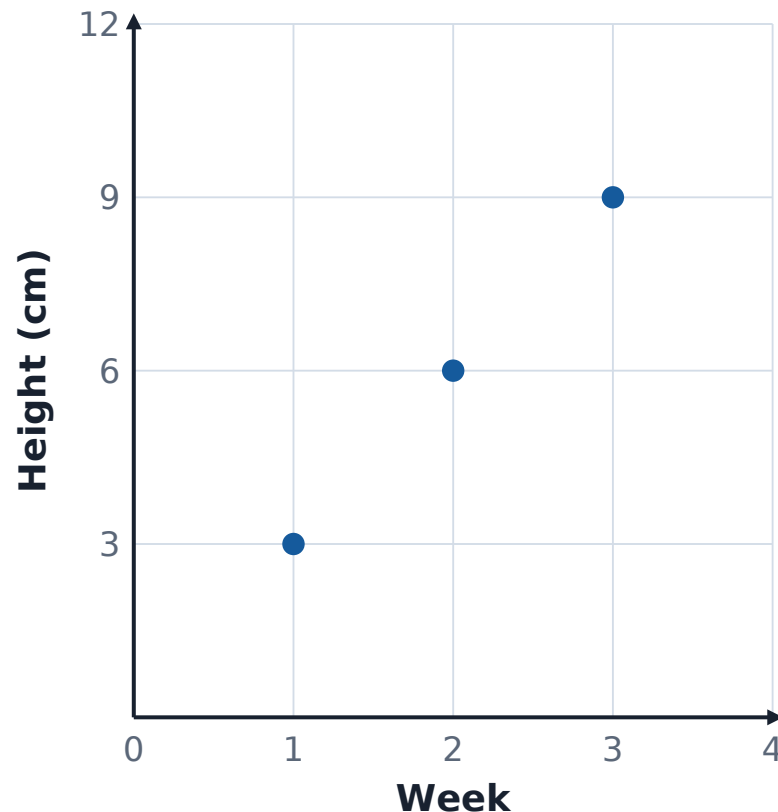
C. (3, 15)

D. (5, 3)

Use the graph.

What is the height in week 3?

Height: _____



True or false?

The point $(4, 0)$ is on the y -axis.

A. True

B. False

Solve "how many more" and "how many less" problems using scaled bar graphs and picture graphs

MAIN	Grade 3 main question	3.MD.B.3
BACKWARD 1	Solve compare problems with a single-unit bar graph	2.MD.D.10
BACKWARD 2	Solve compare word problems within 100	2.OA.A.1
FORWARD 1	Solve problems with data in a line plot of fractional measurements	4.MD.B.4
FORWARD 2	Find totals, differences, and equal shares of line plot data	5.MD.B.2

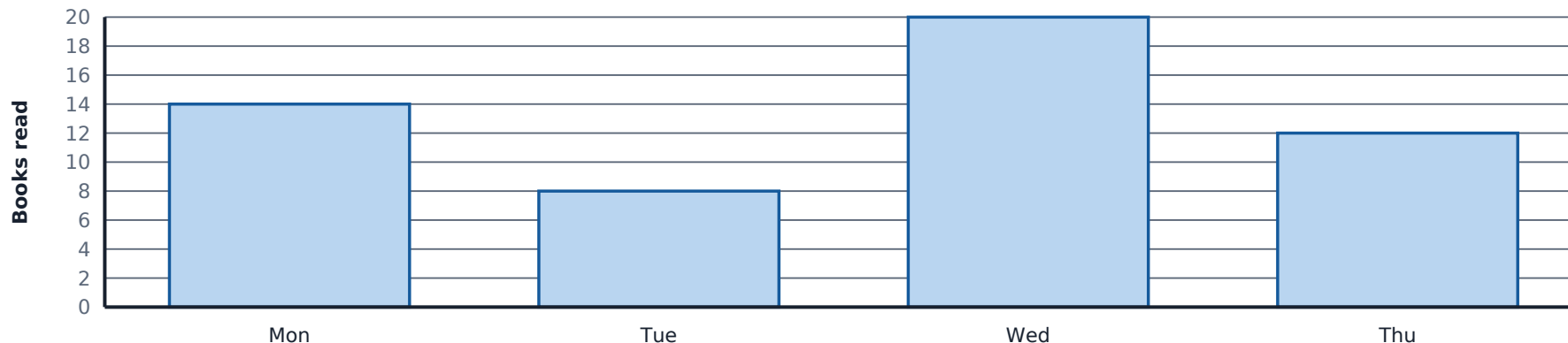
SET 45 • 3.MD.B.3

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.3

Use the bar graph.

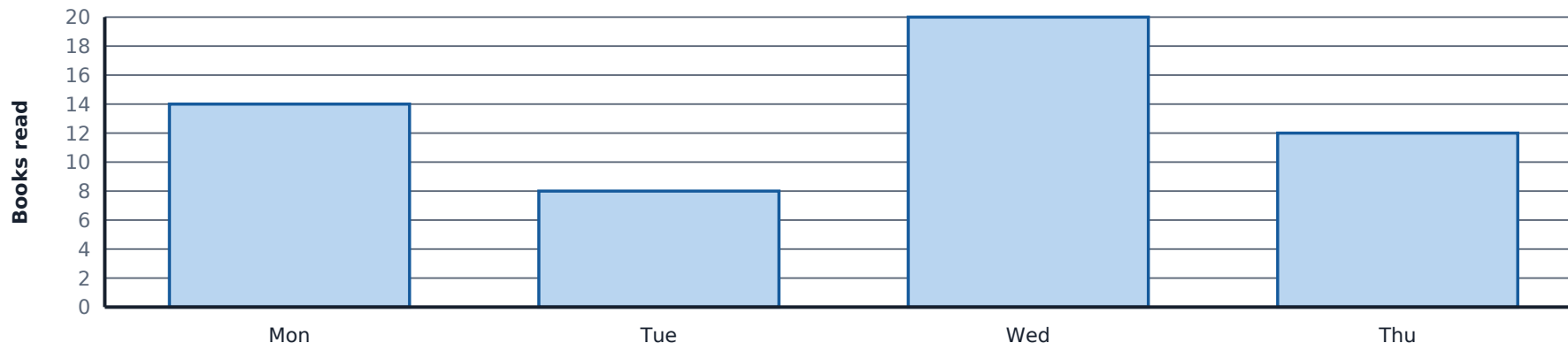
How many more books were read on Wednesday than on Tuesday?



More: _____

Use the bar graph.

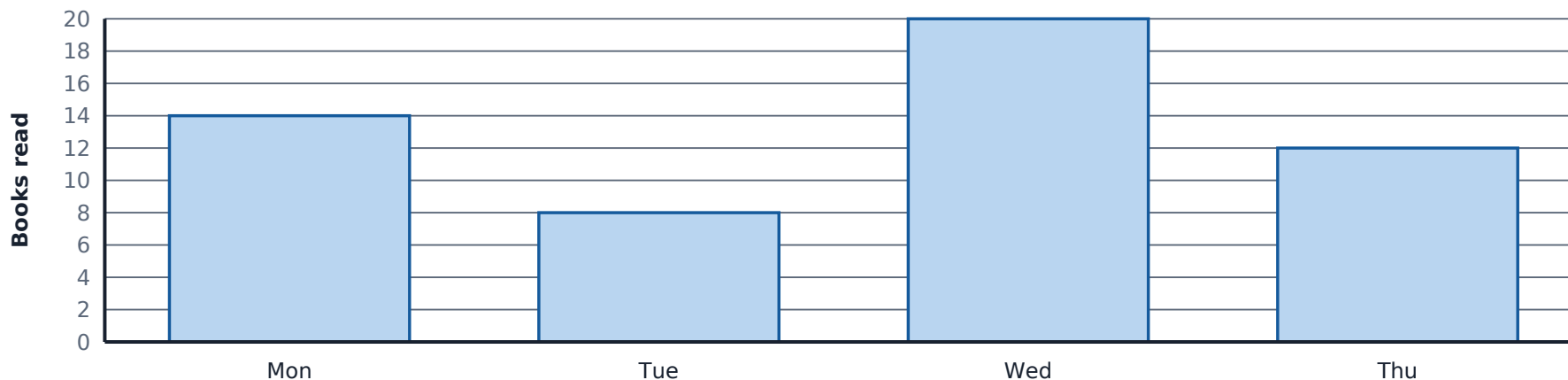
How many books were read on Monday and Thursday together?



Books: _____

Use the bar graph.

How many fewer books were read on Thursday than on Wednesday?



A. 32

B. 12

C. 8

D. 6

True or false?

Use the picture graph. Ana picked 6 more apples than Ben.

Apples picked	
Ana	● ● ● ● ●
Ben	● ● ●
Cy	● ● ● ●

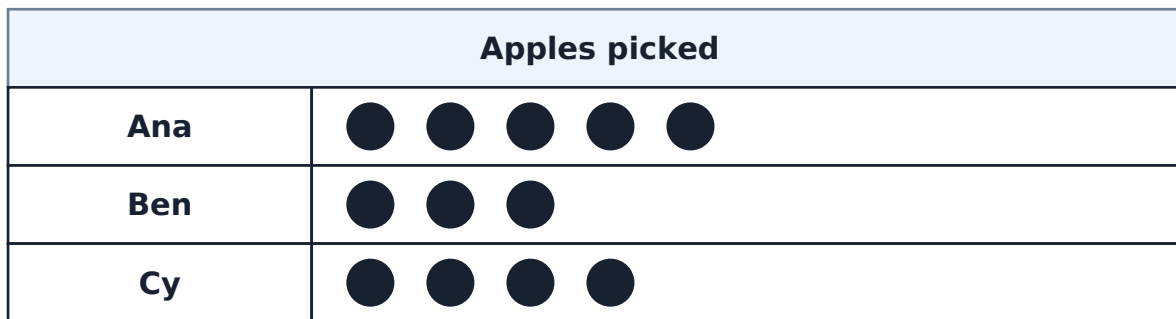
Key: ● = 3 apples

A. True

B. False

Use the picture graph.

How many apples did the three children pick in all?



Key: ● = 3 apples

Apples: _____

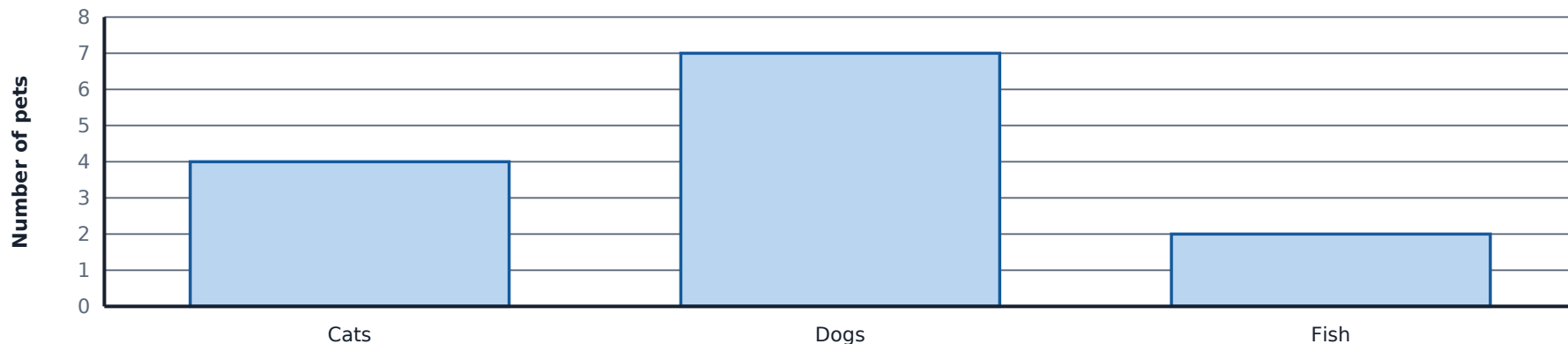
SET 45 • 3.MD.B.3

BACKWARD 1

Solve compare problems with a single-unit bar graph • Grade 2 • Measurement & Data • 2.MD.D.10

Use the bar graph.

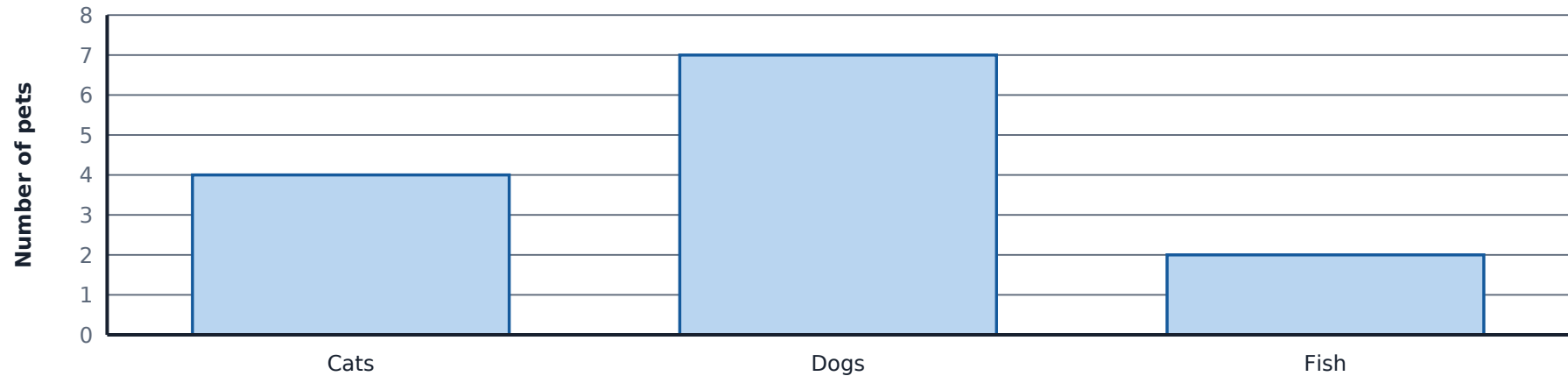
How many more dogs are there than cats?



More: _____

True or false?

Use the bar graph. There are 13 pets in all.

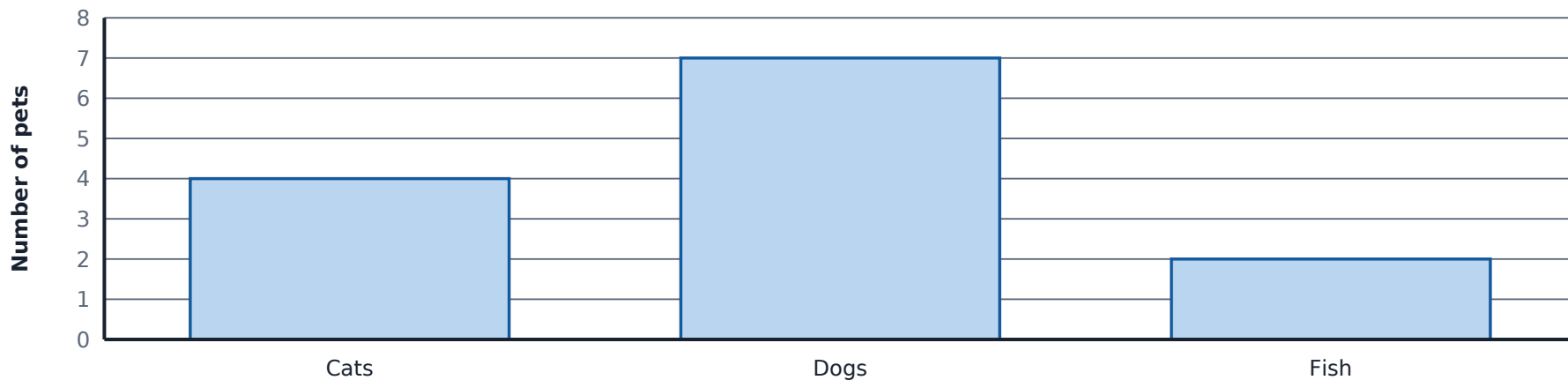


A. True

B. False

Use the bar graph.

How many fewer fish are there than dogs?



A. 9

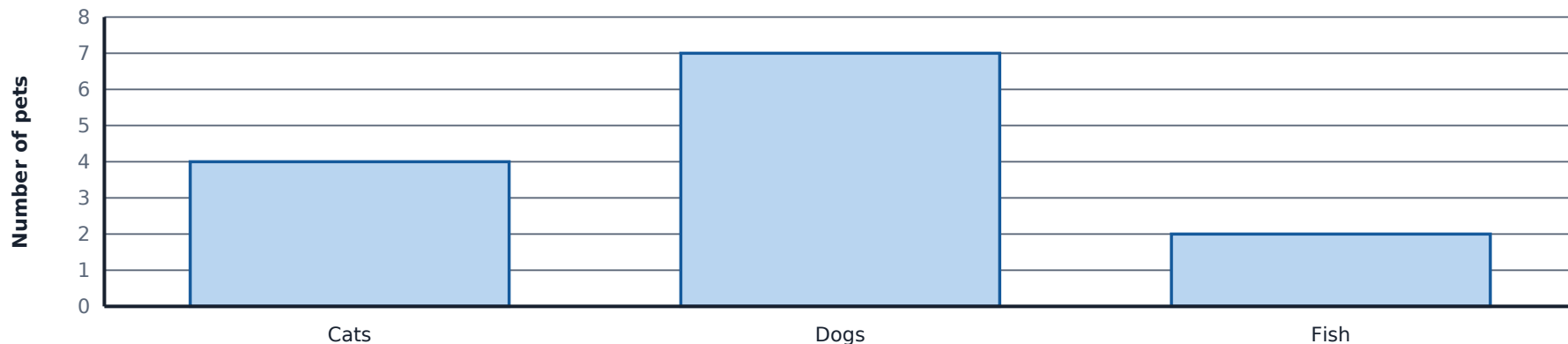
B. 2

C. 7

D. 5

Use the bar graph.

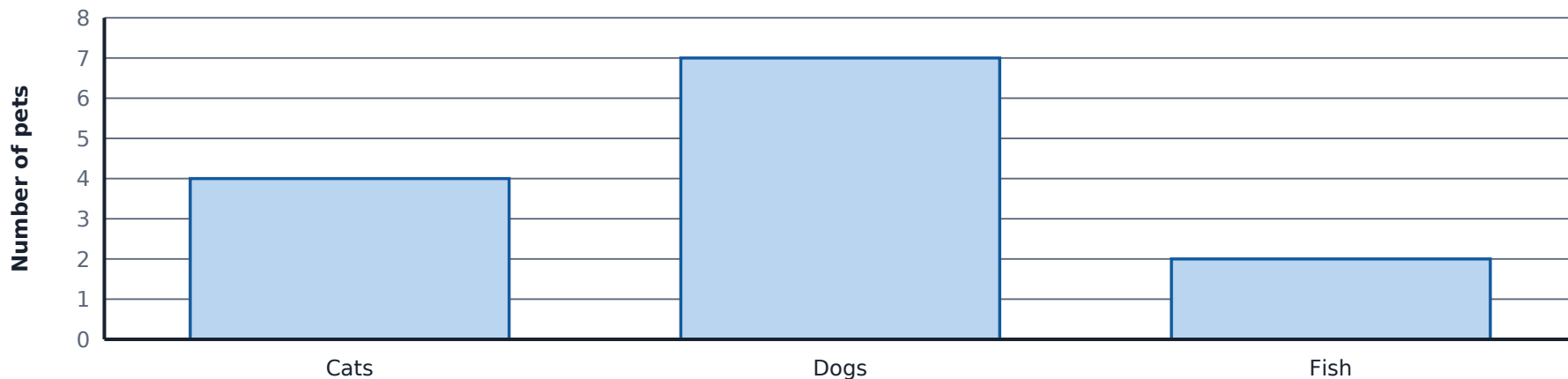
How many cats and fish are there together?



Pets: _____

True or false?

Use the bar graph. There are 3 more fish than cats.



A. True

B. False

SET 45 • 3.MD.B.3

BACKWARD 2

Solve compare word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Ty has 45 cards. Al has 28 cards.

How many more cards does Ty have?

More: _____

True or false?

A red ribbon is 50 cm long and a blue ribbon is 35 cm long. The red ribbon is 15 cm longer.

A. True

B. False

A school has 62 boys and 48 girls.

How many fewer girls than boys are there?

- A.** 24
- B.** 16
- C.** 14
- D.** 110

Kim read 33 pages. Jo read 19 more pages than Kim.
How many pages did Jo read?

Pages: _____

True or false?

Ann has 70 stamps. Bo has 25 fewer stamps than Ann. Bo has 95 stamps.

- A.** True
- B.** False

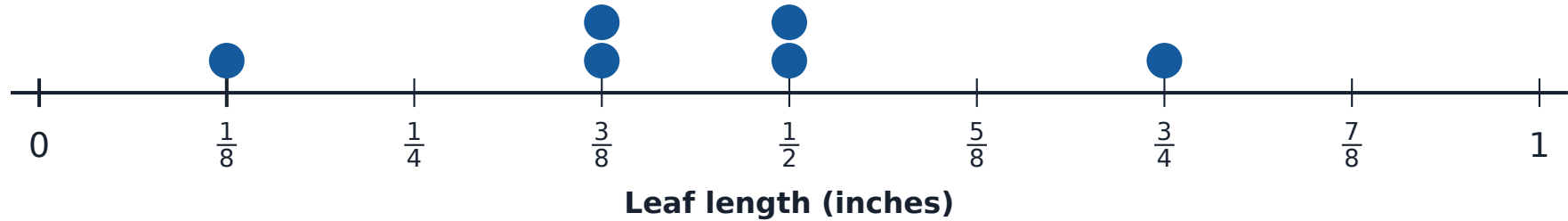
SET 45 • 3.MD.B.3

FORWARD 1

Solve problems with data in a line plot of fractional measurements • Grade 4 •
Measurement & Data • 4.MD.B.4

The line plot shows leaf lengths.

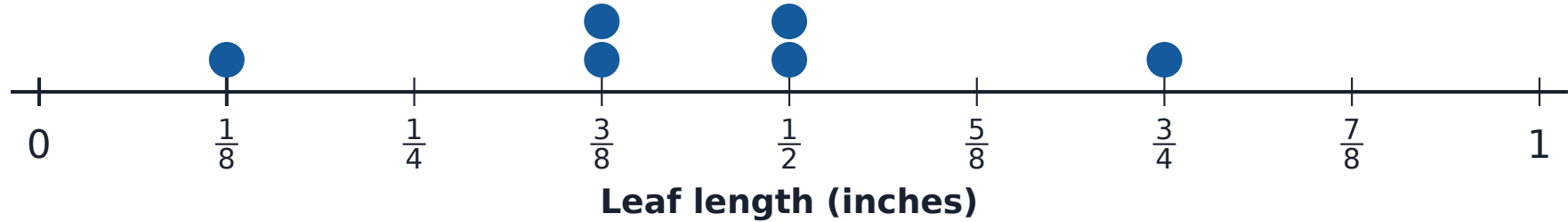
What is the difference between the longest and the shortest leaf?



Difference: _____

True or false?

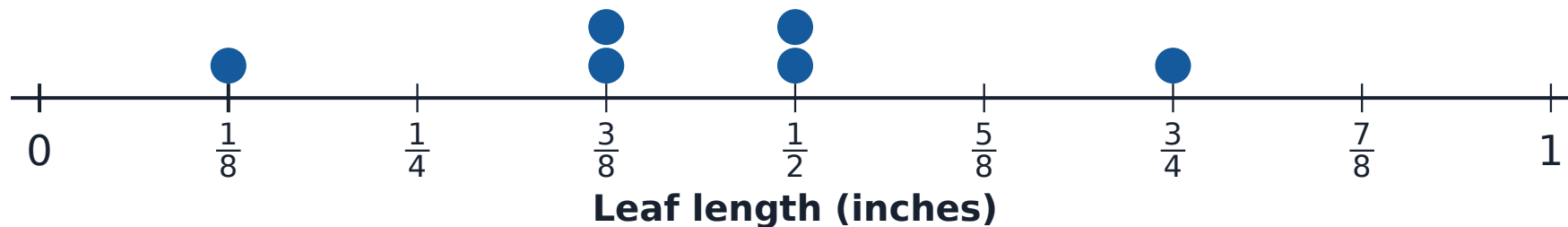
The two $\frac{3}{8}$ -inch leaves have a total length of $\frac{6}{8}$ inch.



A. True

B. False

How many leaves are longer than $\frac{3}{8}$ inch?



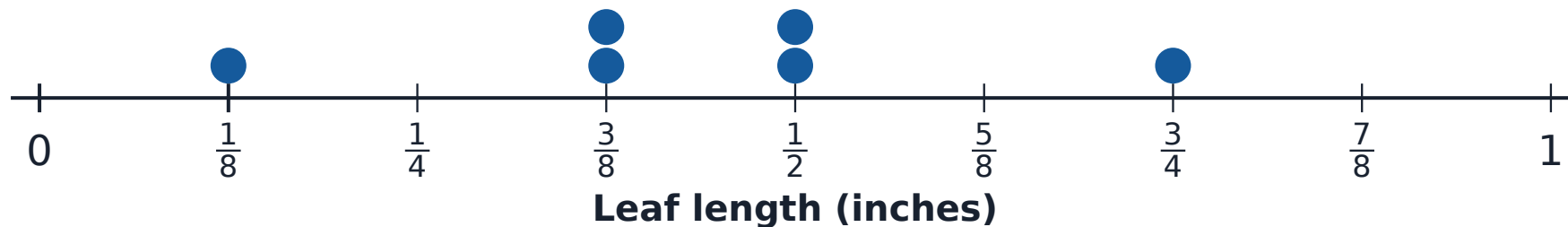
A. 4

B. 2

C. 1

D. 3

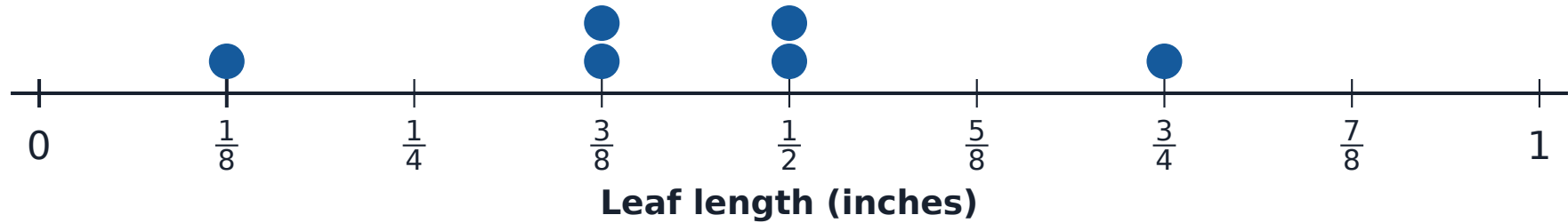
What is the total length of the two $\frac{1}{2}$ -inch leaves?



Total: _____

True or false?

The shortest leaf is $\frac{1}{4}$ inch long.



A. True

B. False

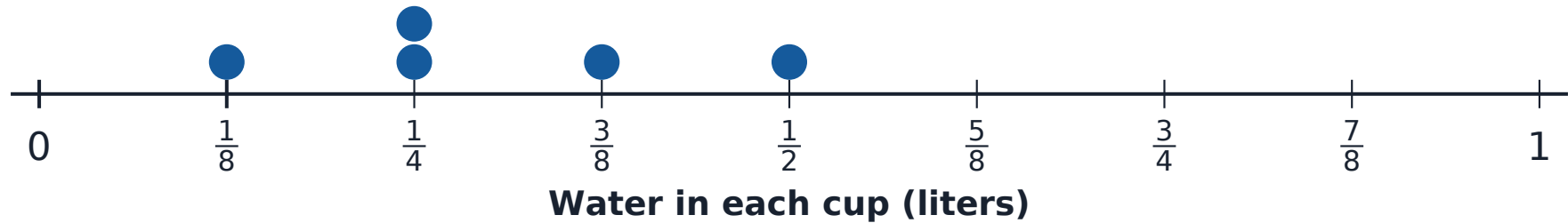
SET 45 • 3.MD.B.3

FORWARD 2

Find totals, differences, and equal shares of line plot data • Grade 5 •
Measurement & Data • 5.MD.B.2

The line plot shows the water in 5 cups.

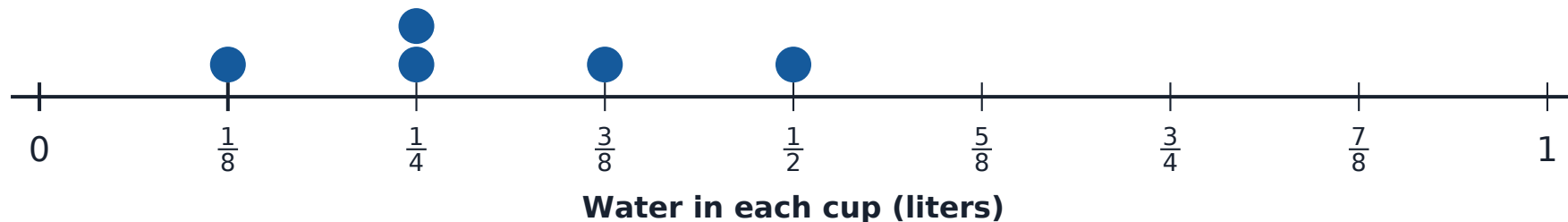
What is the total amount of water?



Total: _____

True or false?

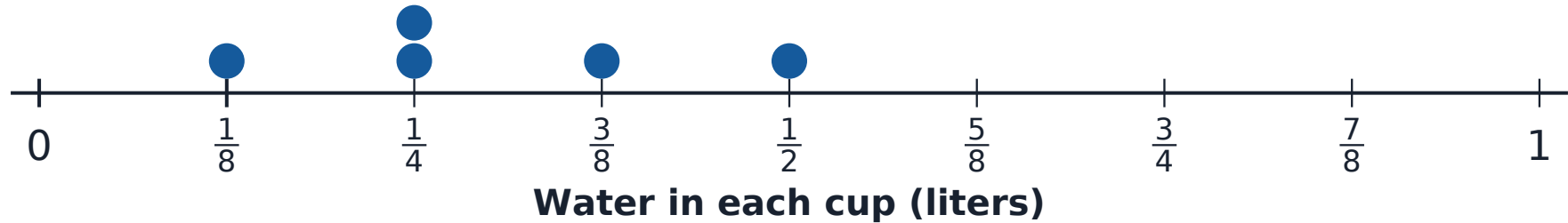
If all the water were poured into one jug, the jug would hold $1\frac{1}{2}$ liters.



A. True

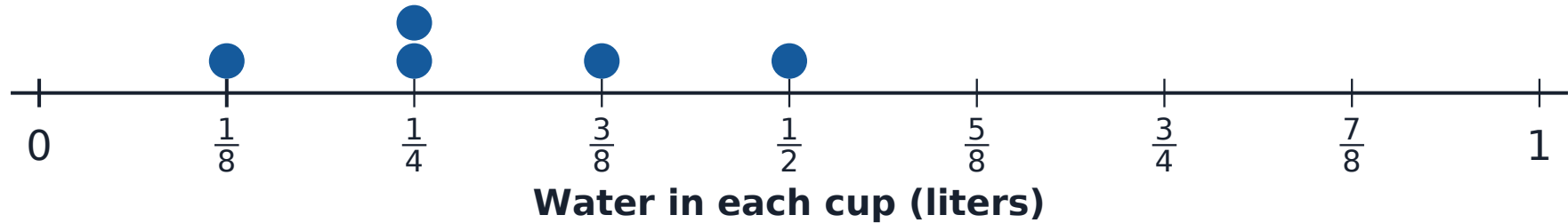
B. False

If the water were shared equally among the 5 cups, how much would each cup hold?



- A.** $\frac{3}{10}$ liter **B.** $1\frac{1}{2}$ liters **C.** $\frac{1}{4}$ liter **D.** $\frac{1}{5}$ liter

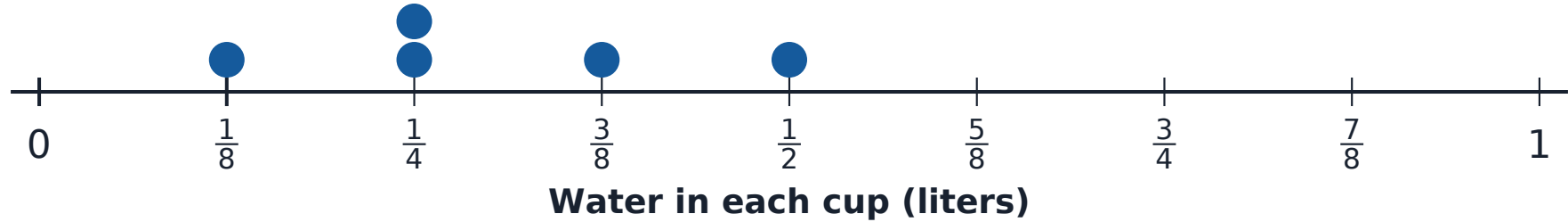
How much more water is in the fullest cup than in the cup with the least water?



More: _____

True or false?

The two cups with $\frac{1}{4}$ liter hold $\frac{1}{8}$ liter in all.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 45 • FORWARD 2 • Q5/5

Measure lengths to the nearest half inch and quarter inch

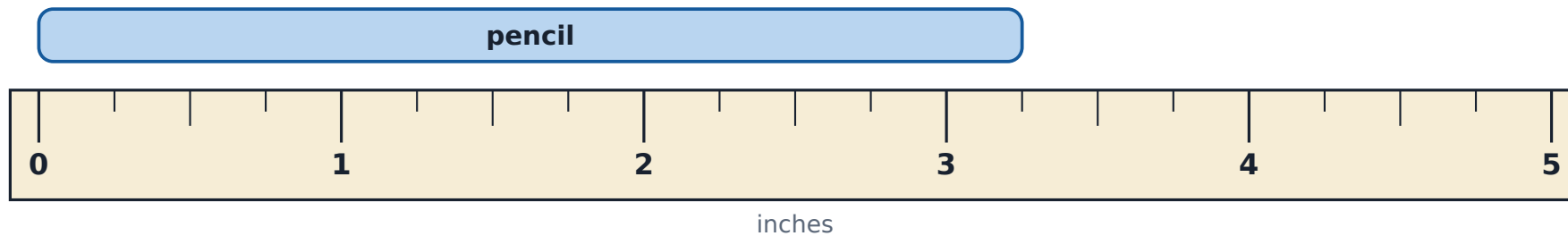
MAIN	Grade 3 main question	3.MD.B.4
BACKWARD 1	Measure lengths in whole inches and centimeters	2.MD.A.1
BACKWARD 2	Measure length by laying same-size units end to end	1.MD.A.2
FORWARD 1	Add mixed-number lengths in word problems	4.NF.B.3.d
FORWARD 2	Convert inches, feet, and yards with fractions	5.MD.A.1

SET 46 • 3.MD.B.4

MAIN

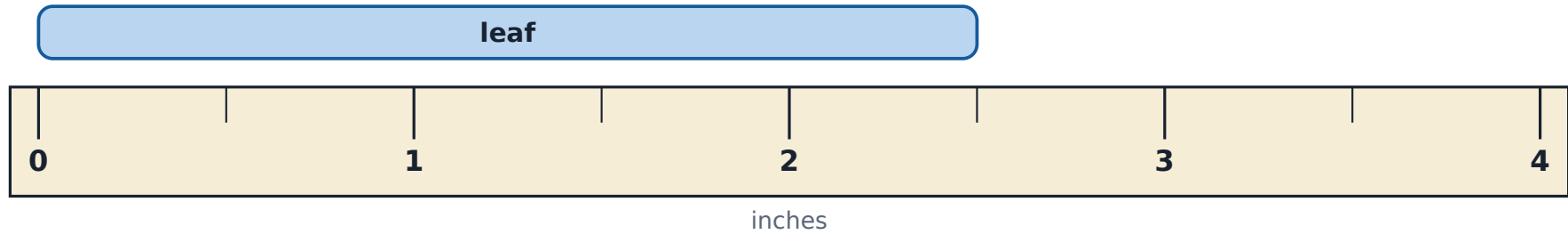
Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.4

How long is the pencil, to the nearest quarter inch?



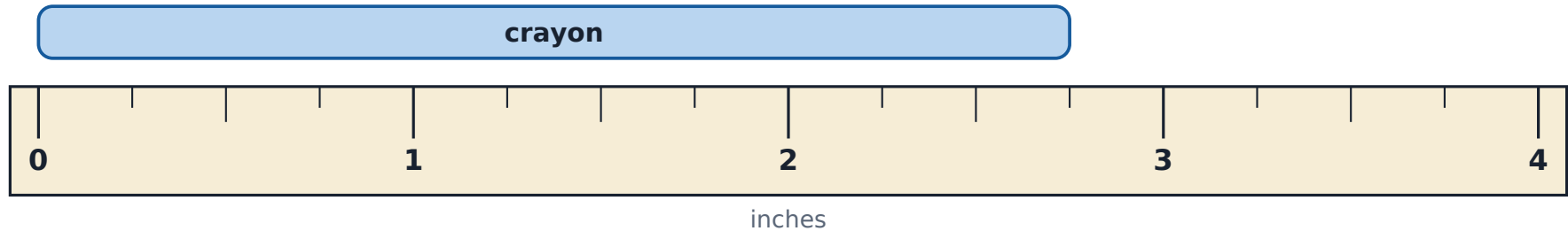
Length: _____

How long is the leaf, to the nearest half inch?



Length: _____

How long is the crayon, to the nearest quarter inch?



A. $3\frac{3}{4}$ inches

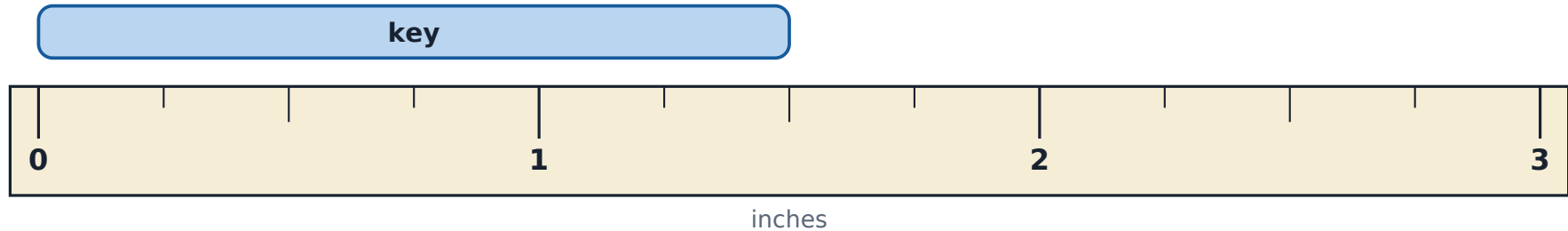
B. 3 inches

C. $2\frac{1}{2}$ inches

D. $2\frac{3}{4}$ inches

True or false?

The key is $1\frac{1}{2}$ inches long.

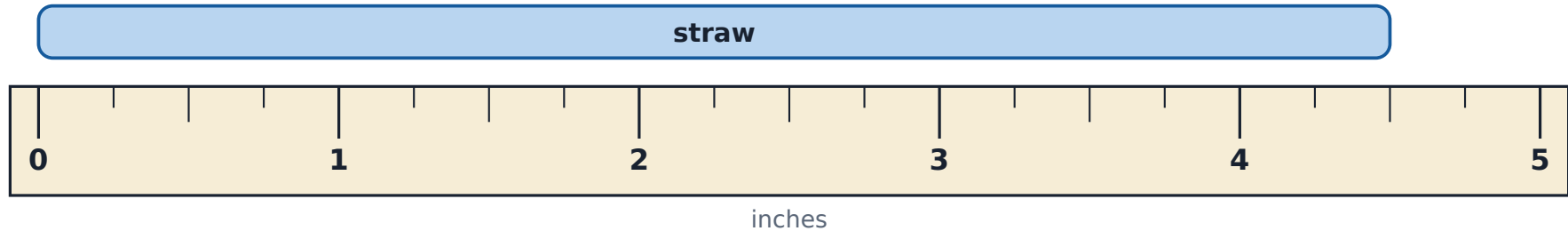


A. True

B. False

True or false?

The straw is $4\frac{1}{4}$ inches long.



A. True

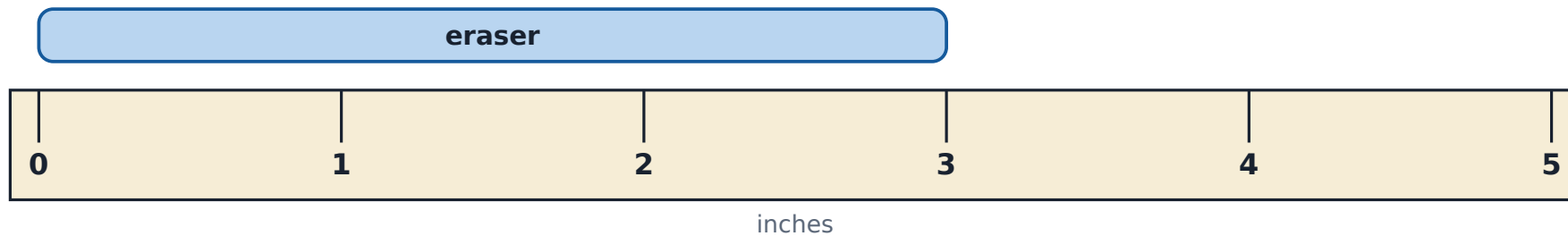
B. False

SET 46 • 3.MD.B.4

BACKWARD 1

Measure lengths in whole inches and centimeters • Grade 2 • Measurement & Data • 2.MD.A.1

How long is the eraser?



Length: _____

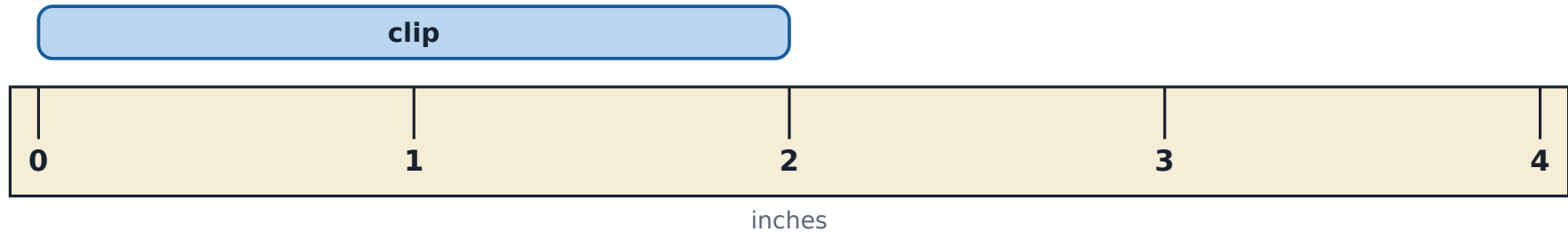
True or false?

A ruler can show inches or centimeters.

A. True

B. False

How long is the clip?



A. 2 inches

B. 1 inch

C. 4 inches

D. 3 inches

A string goes from 0 to 6 on a centimeter ruler.

How long is it?

Length: _____

True or false?

A nail that reaches from 0 to 5 on an inch ruler is 4 inches long.

- A. True
- B. False

SET 46 • 3.MD.B.4

BACKWARD 2

Measure length by laying same-size units end to end • Grade 1 • Measurement & Data • 1.MD.A.2

A pencil is as long as 5 paper clips laid end to end.
How many paper clips long is it?

Paper clips: _____

True or false?

When you measure with paper clips, there should be no gaps or overlaps.

- A. True
- B. False

A book is 8 cubes long. A box is 2 cubes longer.

How many cubes long is the box?

- A.** 6 cubes
- B.** 16 cubes
- C.** 8 cubes
- D.** 10 cubes

A rug is 9 shoes long. A mat is 3 shoes shorter.
How many shoes long is the mat?

Shoes: _____

True or false?

If the paper clips overlap when you measure, the count is still correct.

- A. True
- B. False

SET 46 • 3.MD.B.4

FORWARD 1

Add mixed-number lengths in word problems • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

A pencil is $3\frac{1}{4}$ inches long and a crayon is $2\frac{3}{4}$ inches long.

How long are they laid end to end?

Length: _____

True or false?

Ribbons of $2\frac{1}{2}$ inches and $1\frac{3}{4}$ inches laid end to end are $4\frac{1}{4}$ inches long.

A. True

B. False

A snail crawls $1\frac{1}{4}$ inches and then $2\frac{2}{4}$ inches.

How far does it crawl in all?

A. $3\frac{3}{4}$ inches

B. 4 inches

C. $3\frac{1}{4}$ inches

D. $1\frac{1}{4}$ inches

Three sticks are $\frac{1}{2}$ inch, $\frac{3}{4}$ inch, and $\frac{1}{4}$ inch long.

What is their total length?

Length: _____

True or false?

Three beads that are each $\frac{3}{4}$ inch long, in a row, are 2 inches long.

- A. True
- B. False

SET 46 • 3.MD.B.4

FORWARD 2

Convert inches, feet, and yards with fractions • Grade 5 • Measurement & Data
• 5.MD.A.1

How many feet is 30 inches?

Feet: _____

True or false?

18 inches = $1\frac{1}{2}$ feet

A. True

B. False

How many inches are in $2\frac{1}{4}$ feet?

A. 27

B. 24

C. 25

D. 29

A board is $1\frac{3}{4}$ yards long.

How many feet is that?

Feet: _____

True or false?

9 inches = $\frac{3}{4}$ yard

A. True

B. False

Make a line plot of measurement data in whole numbers, halves, or quarters

MAIN	Grade 3 main question	3.MD.B.4
BACKWARD 1	Make a line plot of whole-number lengths	2.MD.D.9
BACKWARD 2	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Make a line plot of measurements in halves, fourths, and eighths	4.MD.B.4
FORWARD 2	Use operations on fractions to solve problems with line plot data	5.MD.B.2

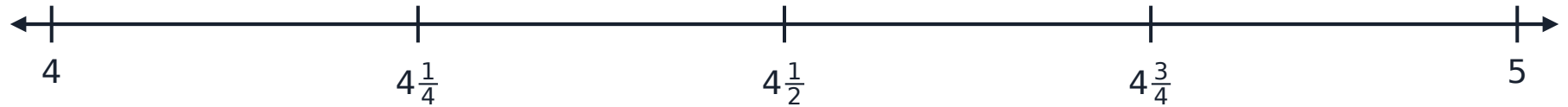
SET 47 • 3.MD.B.4

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.4

Make a line plot of the data.

Pencil lengths (inches): 4, $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $4\frac{1}{4}$, $4\frac{1}{2}$



Shell lengths (inches): 2 , $2\frac{1}{2}$, $2\frac{1}{2}$, 3 , $2\frac{1}{2}$

In a line plot of the data, how many X's go above $2\frac{1}{2}$?

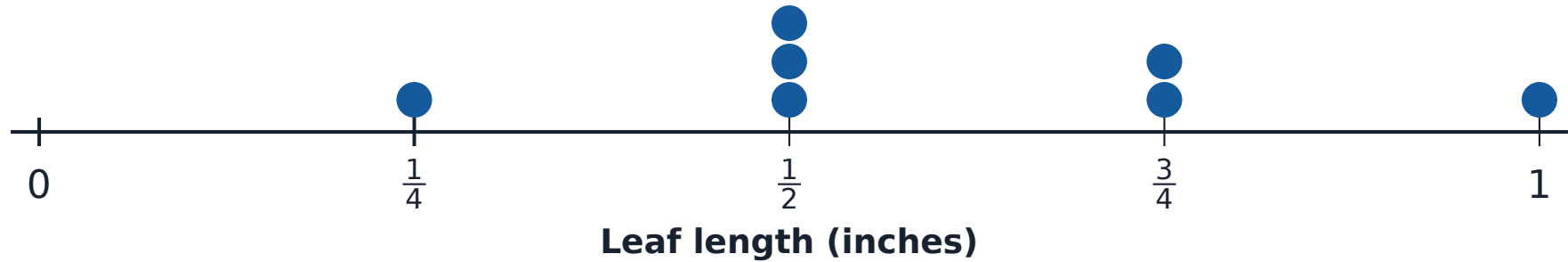
X's: _____

Which number line is best for a line plot of $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, and 2 inches?

- A.** From 1 to 2, marked in halves
- B.** From 0 to 10, marked in ones
- C.** From 1 to 2, marked in quarters
- D.** From 0 to 1, marked in halves

True or false?

The line plot shows that 2 leaves are $\frac{3}{4}$ inch long.

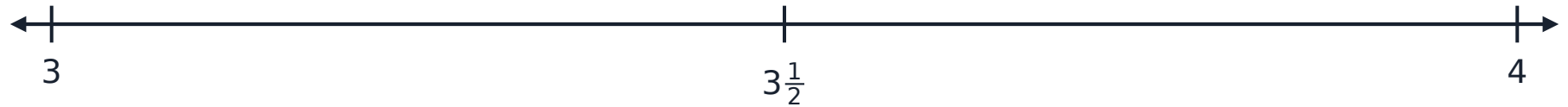


A. True

B. False

Make a line plot of the data.

Ribbon lengths (inches): 3, $3\frac{1}{2}$, 4, 3, $3\frac{1}{2}$, 3



SET 47 • 3.MD.B.4

BACKWARD 1

Make a line plot of whole-number lengths • Grade 2 • Measurement & Data •
2.MD.D.9

Make a line plot of the data.

Lengths (inches): 5, 6, 6, 8, 6



True or false?

A line plot shows each measurement as an X above a number line.

- A.** True
- B.** False

Which number line fits a line plot of 3, 4, 4, and 7?

- A.** A number line from 10 to 20
- B.** A number line from 5 to 9
- C.** A number line from 0 to 10
- D.** A number line from 0 to 4

Lengths (cm): 9, 10, 10, 12

In a line plot of the data, how many X's go above 10?

X's: _____

True or false?

A line plot of 2, 2, 2, and 3 has 2 X's above 2.

- A. True
- B. False

SET 47 • 3.MD.B.4

BACKWARD 2

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line, numbers get greater to the right.

- A.** True
- B.** False

Which number is between 4 and 6 on a number line?

- A.** 7
- B.** 3
- C.** 10
- D.** 5

What number is at point B?



B = _____

True or false?

On a number line, 9 is to the left of 6.

A. True

B. False

SET 47 • 3.MD.B.4

FORWARD 1

Make a line plot of measurements in halves, fourths, and eighths • Grade 4 •
Measurement & Data • 4.MD.B.4

Make a line plot of the data.

Screw lengths (inches): $\frac{3}{8}$, $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{4}$, $\frac{3}{8}$



True or false?

$\frac{1}{2}$ inch and $\frac{4}{8}$ inch go above the same mark on a line plot marked in eighths.

- A. True
- B. False

Lengths (inches): $\frac{1}{4}$, $\frac{2}{8}$, $\frac{3}{8}$, $\frac{1}{4}$

In a line plot marked in eighths, how many X's go above $\frac{2}{8}$?

A. 4

B. 1

C. 3

D. 2

Make a line plot of the data.

Bead sizes (inches): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$



True or false?

A line plot marked in eighths has a mark for $\frac{1}{3}$ inch.

- A.** True
- B.** False

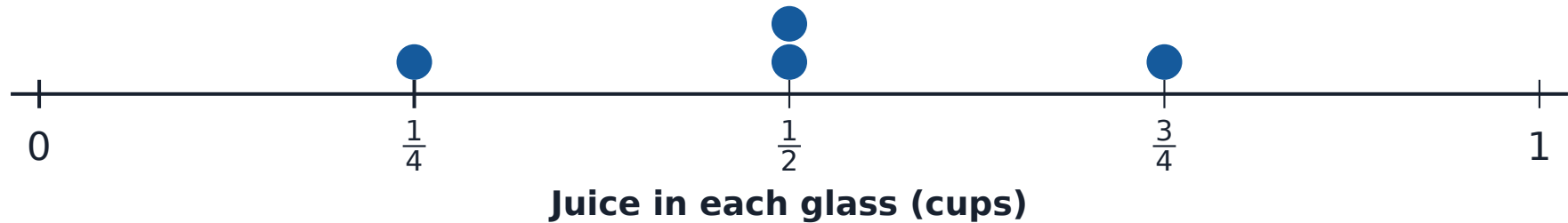
SET 47 • 3.MD.B.4

FORWARD 2

Use operations on fractions to solve problems with line plot data • Grade 5 •
Measurement & Data • 5.MD.B.2

The line plot shows the juice in 4 glasses.

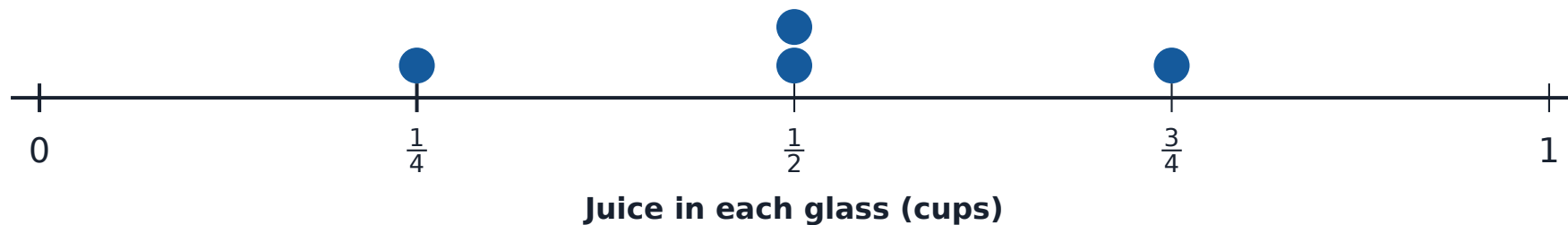
What is the total amount of juice?



Total: _____

True or false?

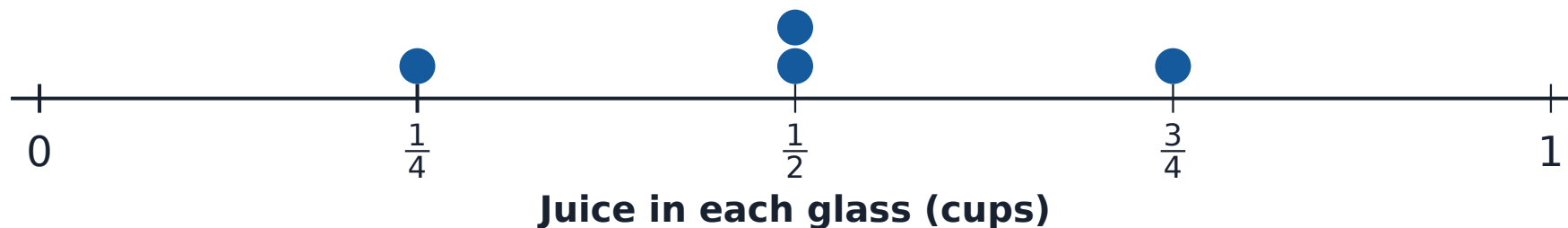
If the juice were shared equally among the 4 glasses, each glass would hold $\frac{1}{2}$ cup.



A. True

B. False

How much more juice is in the fullest glass than in the emptiest glass?



A. $\frac{3}{4}$ cup

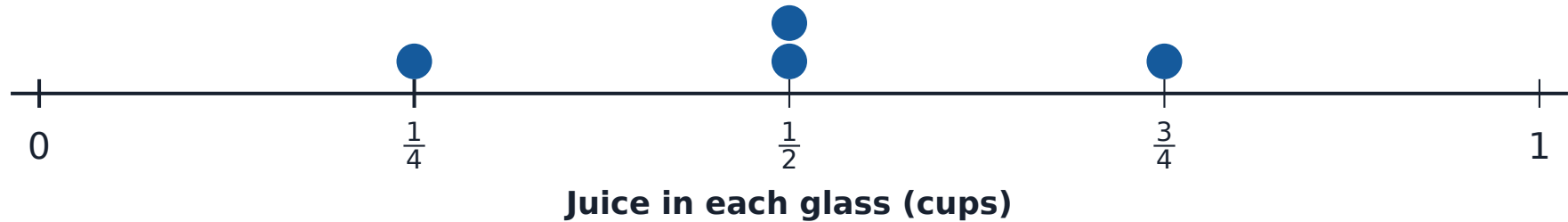
B. 1 cup

C. $\frac{1}{2}$ cup

D. $\frac{1}{4}$ cup

The juice in the two $\frac{1}{2}$ -cup glasses is poured together.

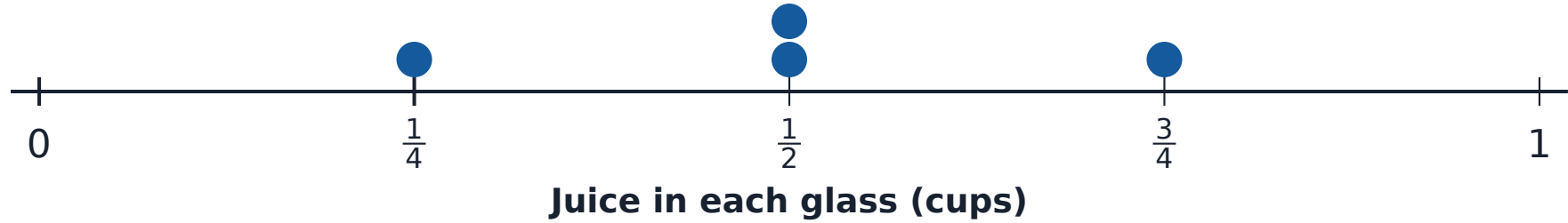
How much juice is that?



Juice: _____

True or false?

The glasses with less than $\frac{1}{2}$ cup hold $\frac{3}{4}$ cup in all.



A. True

B. False

Understand a unit square and that it is used to measure area

MAIN	Grade 3 main question	3.MD.C.5.a
BACKWARD 1	Count the same-size squares in a rectangle	2.G.A.2
BACKWARD 2	Defining attributes of a square	1.G.A.1
FORWARD 1	Find the area of a rectangle with the area formula	4.MD.A.3
FORWARD 2	Understand a unit cube and that it is used to measure volume —...	5.MD.C.3.a

SET 48 • 3.MD.C.5.a

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides 1 unit long is a unit square, and its area is 1 square unit.

A. True

B. False

Which shape is a unit square?

- A.** A rectangle 1 cm by 2 cm
- B.** A square with sides 1 cm long
- C.** A circle 1 cm across
- D.** A square with sides 2 cm long

What is the area of a square with sides 1 inch long?

Area: _____

Why can unit squares be used to measure area?

Explanation: _____

True or false?

A square with sides 1 meter long has an area of 1 square centimeter.

- A. True
- B. False

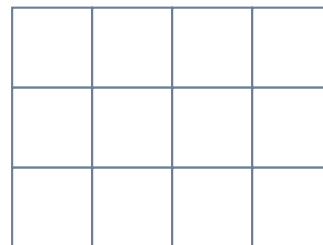
SET 48 • 3.MD.C.5.a

BACKWARD 1

Count the same-size squares in a rectangle • Grade 2 • Geometry • 2.G.A.2

How many same-size squares
are in the rectangle?

Squares: _____



True or false?

A rectangle split into 2 rows of 3 same-size squares has 6 squares.

- A. True
- B. False

A rectangle has 4 rows of 2 same-size squares.

How many squares are there?

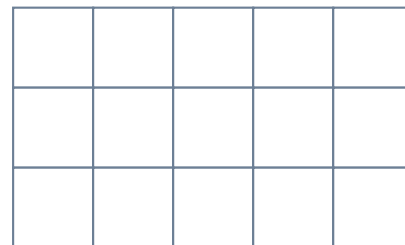
A. 2

B. 6

C. 8

D. 4

How many same-size squares
are in each column of the
rectangle?



Squares in each column: ____

True or false?

A rectangle split into 3 rows of 3 same-size squares has 6 squares.

- A. True
- B. False

SET 48 • 3.MD.C.5.a

BACKWARD 2

Defining attributes of a square • Grade 1 • Geometry • 1.G.A.1

True or false?

Every square has 4 sides of the same length.

A. True

B. False

Which is a defining attribute of a square?

- A.** It is big.
- B.** 4 equal sides and 4 square corners
- C.** It is turned on a corner.
- D.** It is blue.

How many sides does a square have?

Sides: _____

Is a very small square still a square?

Answer: _____

True or false?

A shape with 3 sides can be a square.

A. True

B. False

SET 48 • 3.MD.C.5.a

FORWARD 1

Find the area of a rectangle with the area formula • Grade 4 • Measurement & Data • 4.MD.A.3

Find the area of a rectangle that is 12 m by 7 m.

Area: _____

True or false?

A 9-by-11 rectangle has an area of 99 square units.

- A. True
- B. False

A rug is 15 feet by 6 feet.

What is its area?

- A.** 21 square feet
- B.** 42 square feet
- C.** 90 square feet
- D.** 96 square feet

A square has sides that are 13 cm long.

What is its area?

Area: _____

True or false?

A rectangle that is 20 in. by 5 in. has an area of 50 square inches.

- A.** True
- B.** False

FORWARD 2

Understand a unit cube and that it is used to measure volume • Grade 5 •
Measurement & Data • 5.MD.C.3.a

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

True or false?

A cube with edges 1 unit long is a unit cube, and its volume is 1 cubic unit.

- A. True
- B. False

Which can be used to measure volume?

- A.** A clock
- B.** A ruler
- C.** A unit square
- D.** A unit cube

What is the volume of a cube with edges 1 cm long?

Volume: _____

How is a unit cube like a unit square?

Answer: _____

True or false?

A unit cube has a volume of 6 cubic units because it has 6 faces.

A. True

B. False

A figure covered by n unit squares without gaps or overlaps has an area of n square units

MAIN	Grade 3 main question	3.MD.C.5.b
BACKWARD 1	Measure length with same-size units and no gaps or overlaps	1.MD.A.2
BACKWARD 2	Count the same-size squares that fill a rectangle	2.G.A.2
FORWARD 1	Find an unknown side length of a rectangle from its area	4.MD.A.3
FORWARD 2	A solid packed with n unit cubes has a volume of n cubic units —...	5.MD.C.3.b

SET 49 • 3.MD.C.5.b

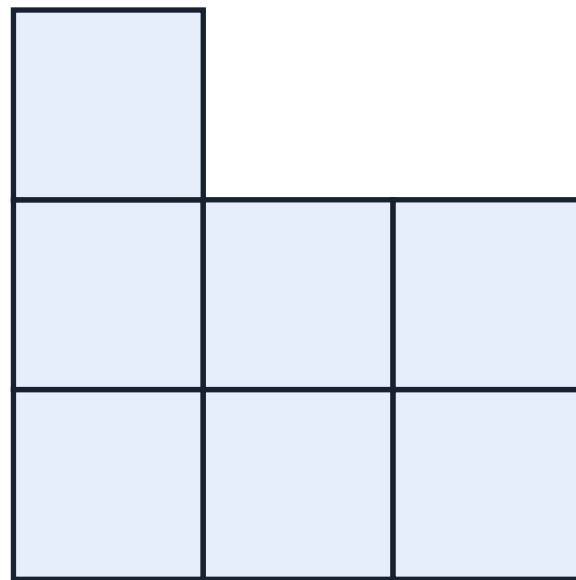
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.5.b

Each small square is 1 square unit.

What is the area of the shape?

Area: _____



True or false?

A shape covered by 15 unit squares with no gaps or overlaps has an area of 15 square units.

- A. True
- B. False

A rectangle is covered by 12 unit squares with no gaps or overlaps.

What is its area?

- A.** 12 square units
- B.** 6 square units
- C.** 24 square units
- D.** 12 units

Ben covers a shape with unit squares, but he leaves gaps between them. He counts 10 squares.

Is the area exactly 10 square units? Explain.

Answer: _____

Explanation: _____

True or false?

If unit squares overlap, counting them gives the exact area.

- A. True
- B. False

SET 49 • 3.MD.C.5.b

BACKWARD 1

Measure length with same-size units and no gaps or overlaps • Grade 1 •
Measurement & Data • 1.MD.A.2

A crayon is as long as 6 cubes laid end to end with no gaps.

How many cubes long is it?

Cubes: _____

True or false?

Length units laid end to end must have no gaps or overlaps.

A. True

B. False

Which shows the correct way to measure with paper clips?

- A.** Clips in a pile
- B.** Clips that overlap
- C.** Clips with spaces between them
- D.** Clips end to end with no gaps

A desk is 9 hands long. A table is 4 hands longer.
How many hands long is the table?

Hands: _____

True or false?

If the cubes overlap when you measure, the count is still correct.

A. True

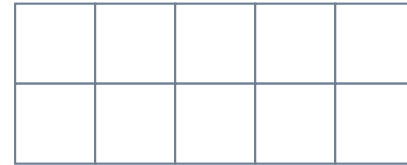
B. False

SET 49 • 3.MD.C.5.b

BACKWARD 2

Count the same-size squares that fill a rectangle • Grade 2 • Geometry •
2.G.A.2

How many same-size squares
fill the rectangle?



Squares: _____

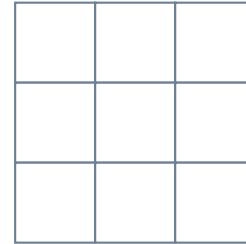
True or false?

A rectangle split into 3 rows of 4 same-size squares has 12 squares.

- A.** True
- B.** False

How many same-size squares
fill the rectangle?

- A.** 6
- B.** 9
- C.** 12
- D.** 3



A rectangle has 5 rows of 2 same-size squares.
How many squares fill it?

Squares: _____

True or false?

A rectangle with 2 rows of 4 same-size squares has 6 squares.

- A.** True
- B.** False

SET 49 • 3.MD.C.5.b

FORWARD 1

Find an unknown side length of a rectangle from its area • Grade 4 •
Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a length of 8 feet.

What is its width?

Width: _____

True or false?

A rectangle with an area of 72 square cm and a width of 6 cm has a length of 12 cm.

- A. True
- B. False

A garden has an area of 90 square meters. One side is 9 meters long.

How long is the other side?

A. 81 m

B. 18 m

C. 10 m

D. 99 m

A square has an area of 64 square inches.
How long is each side?

Side: _____

True or false?

A rectangle with an area of 56 square units and a length of 7 units has a width of 9 units.

- A. True
- B. False

FORWARD 2

A solid packed with n unit cubes has a volume of n cubic units • Grade 5 •
Measurement & Data • 5.MD.C.3.b

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

A solid built from 18 unit cubes has a volume of 18 cubic units.

- A. True
- B. False

A prism has 4 layers with 6 unit cubes in each layer.

What is its volume?

- A.** 46 cubic units
- B.** 10 cubic units
- C.** 24 square units
- D.** 24 cubic units

How many unit cubes fill a box with a volume of 36 cubic units?

Unit cubes: _____

True or false?

If the cubes in a box have gaps between them, counting them gives the exact volume.

A. True

B. False

Measure areas by counting unit squares in square centimeters, meters, inches, feet, and other units

MAIN	Grade 3 main question	3.MD.C.6
BACKWARD 1	Count objects to tell how many	K.CC.B.5
BACKWARD 2	Estimate lengths in inches, feet, centimeters, and meters	2.MD.A.3
FORWARD 1	Find areas of rectangles in real-world problems with the area for...	4.MD.A.3
FORWARD 2	Measure volume by counting unit cubes in cubic cm, cubic in., and...	5.MD.C.4

SET 50 • 3.MD.C.6

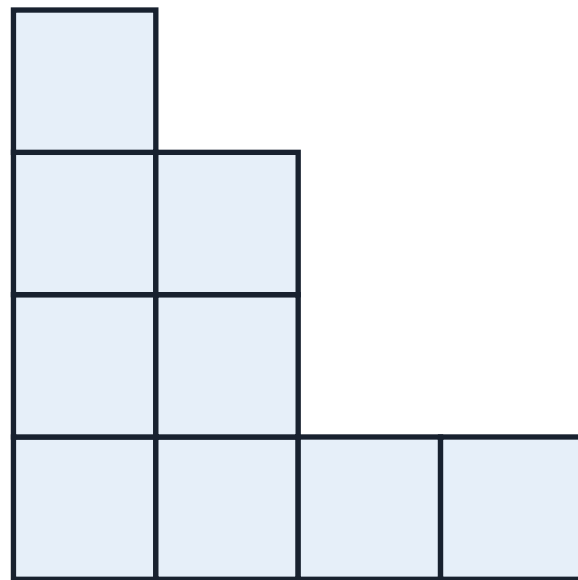
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.6

Each square is 1 square centimeter.

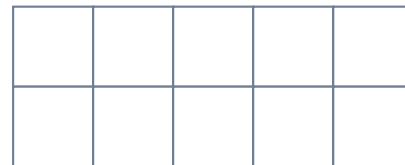
What is the area of the shape?

Area: _____



Each square is 1 square foot.

What is the area of the rug?

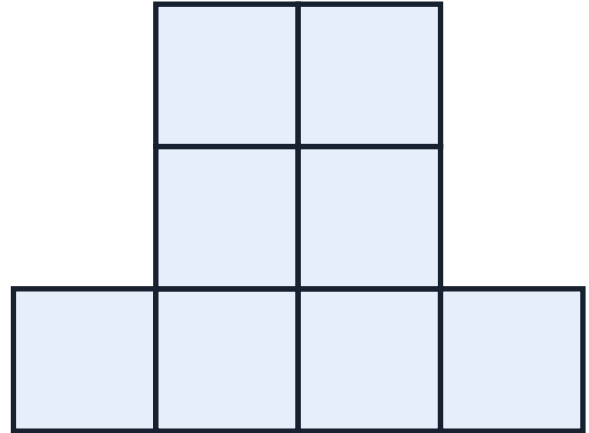


Area: _____

Each square is 1 square inch.

What is the area of the shape?

- A.** 6 square inches **B.** 8 inches
- C.** 12 square inches **D.** 8 square inches

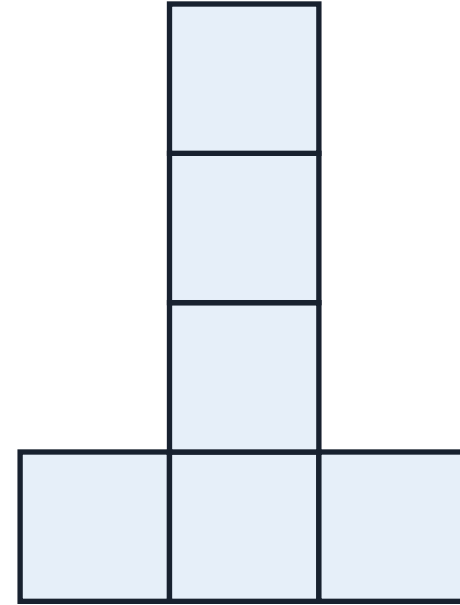


True or false?

Each square is 1 square meter. The shape has an area of 6 square meters.

A. True

B. False

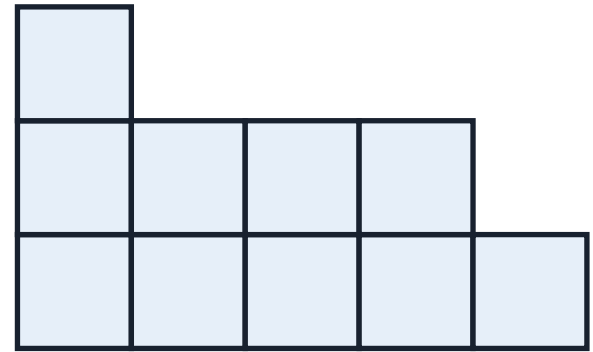


True or false?

Each square is 1 square unit. The shape has an area of 11 square units.

A. True

B. False

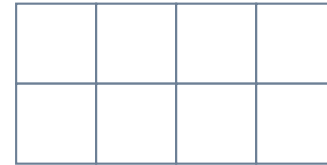


SET 50 • 3.MD.C.6

BACKWARD 1

Count objects to tell how many • Grade K • Counting & Cardinality • K.CC.B.5

How many squares are in the grid?



Squares: _____

True or false?

Counting each square once, a row of 6 squares has 6 squares.

- A.** True
- B.** False

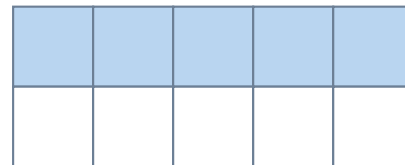
How many squares are shaded?

A. 4

B. 6

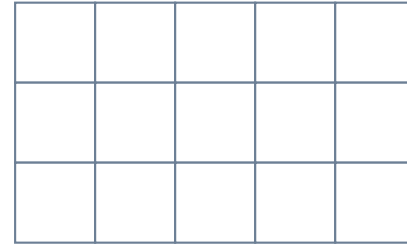
C. 10

D. 5



How many squares are in the grid?

Squares: _____



True or false?

When you count objects, you may count one object twice.

A. True

B. False

SET 50 • 3.MD.C.6

BACKWARD 2

Estimate lengths in inches, feet, centimeters, and meters • Grade 2 •
Measurement & Data • 2.MD.A.3

Would you measure the length of a classroom in meters or in centimeters?

Unit: _____

True or false?

A centimeter is shorter than a meter.

A. True

B. False

Which unit is best for the length of a bug?

- A.** Meters
- B.** Feet
- C.** Yards
- D.** Centimeters

Would you measure a book in inches or in feet?

Unit: _____

True or false?

A foot is shorter than an inch.

A. True

B. False

SET 50 • 3.MD.C.6

FORWARD 1

Find areas of rectangles in real-world problems with the area formula • Grade 4
• Measurement & Data • 4.MD.A.3

A classroom floor is 9 m by 8 m.

What is its area?

Area: _____

True or false?

A 14-ft by 10-ft deck has an area of 140 square feet.

A. True

B. False

A poster is 24 in. by 18 in.

What is its area?

- A.** 84 square inches
- B.** 422 square inches
- C.** 42 square inches
- D.** 432 square inches

A garden is 25 ft by 4 ft.

What is its area?

Area: _____

True or false?

A 16-cm by 5-cm card has an area of 42 square cm.

A. True

B. False

FORWARD 2

Measure volume by counting unit cubes in cubic cm, cubic in., and cubic ft •

Grade 5 • Measurement & Data • 5.MD.C.4

▮ No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box holds 2 layers of 6 cubes. Each cube is 1 cubic centimeter.

What is the volume of the box?

Volume: _____

True or false?

A stack of 3 layers of 4 one-inch cubes has a volume of 12 cubic inches.

- A. True
- B. False

A prism is made of 20 cubes that are each 1 cubic foot.
What is its volume?

- A.** 20 square feet
- B.** 20 cubic feet
- C.** 40 cubic feet
- D.** 20 feet

A figure is built from 15 unit cubes.

What is its volume?

Volume: _____

True or false?

A box that holds 2 layers of 9 one-centimeter cubes has a volume of 11 cubic centimeters.

- A. True
- B. False

Find the area of a rectangle by tiling, and show that it equals the product of the side lengths

MAIN	Grade 3 main question	3.MD.C.7.a
BACKWARD 1	Find the total of an array by adding equal addends	2.OA.C.4
BACKWARD 2	Split a rectangle into rows and columns of same-size squares	2.G.A.2
FORWARD 1	Write and apply the area formula $A = l \times w$	4.MD.A.3
FORWARD 2	Find the volume of a prism by packing it with unit cubes and by...	5.MD.C.5.a

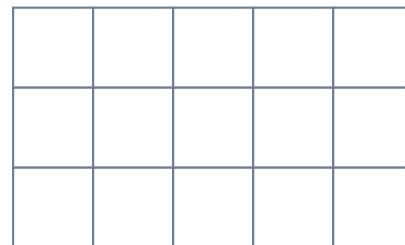
SET 51 • 3.MD.C.7.a

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.a

The rectangle is tiled with unit squares.

Count the squares. Then multiply the side lengths. Do you get the same area?



Count: _____

Multiply: _____

Same? _____

A rectangle is tiled with 4 rows of 6 unit squares.
Which equation gives its area?

A. $4 + 6 = 10$

B. $6 - 4 = 2$

C. $4 \times 6 = 24$

D. $4 + 4 + 6 + 6 = 20$

True or false?

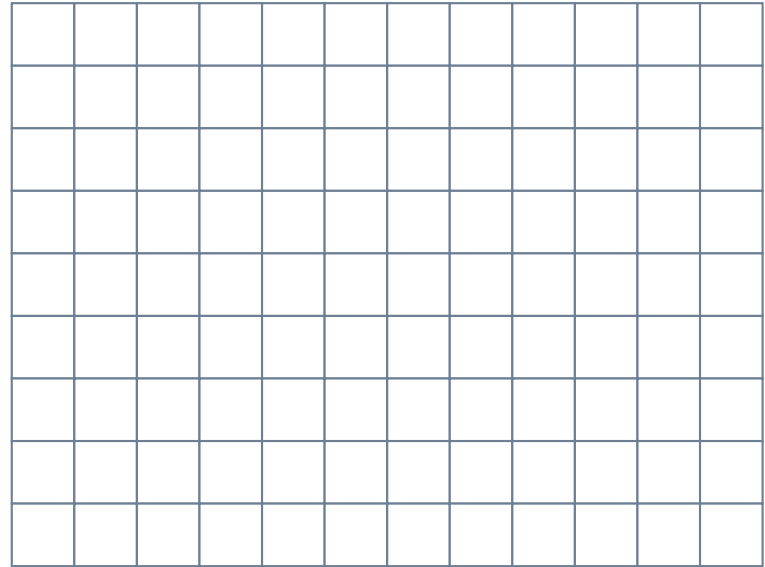
A rectangle 5 units long and 2 units wide can be tiled with 10 unit squares.

- A. True
- B. False

Draw a rectangle 4 units long
and 3 units wide on the grid,
and show its unit squares.

Write a multiplication
equation for its area.

Equation: _____



True or false?

Tiling a 6-by-3 rectangle takes 9 unit squares.

A. True

B. False

SET 51 • 3.MD.C.7.a

BACKWARD 1

Find the total of an array by adding equal addends • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many squares are in 3 rows of 5 squares? Write an addition equation.

Equation: _____

True or false?

An array of 4 rows of 4 has 16 objects.

A. True

B. False

Which addition matches 2 rows of 5 squares?

A. $2 + 2$

B. $5 + 2 + 5$

C. $5 + 5$

D. $2 + 5$

How many squares are in 5 rows of 3 squares?

Squares: _____

True or false?

$3 + 3 + 3 + 3$ matches 3 rows of 3 squares.

A. True

B. False

SET 51 • 3.MD.C.7.a

BACKWARD 2

Split a rectangle into rows and columns of same-size squares • Grade 2 •
Geometry • 2.G.A.2

Draw lines to split the rectangle into 2 rows of 4 same-size squares.



True or false?

A rectangle split into 3 rows and 2 columns has 6 same-size squares.

- A. True
- B. False

A rectangle is split into 2 rows and 3 columns of same-size squares.

How many squares are there?

A. 4

B. 9

C. 6

D. 5

A rectangle is split into 4 rows and 4 columns of same-size squares.

How many squares are there?

Squares: _____

True or false?

A rectangle split into 5 rows and 1 column has 6 squares.

A. True

B. False

SET 51 • 3.MD.C.7.a

FORWARD 1

Write and apply the area formula $A = l \times w$ • Grade 4 • Measurement & Data •
4.MD.A.3

Write a formula for the area A of a rectangle with length l and width w .

Formula: _____

True or false?

A rectangle with $l = 13$ and $w = 6$ has $A = 78$.

A. True

B. False

Use $A = l \times w$.

What is the area when $l = 25$ cm and $w = 4$ cm?

- A.** 104 square cm
- B.** 58 square cm
- C.** 29 square cm
- D.** 100 square cm

Find the area of a rectangle with $l = 18$ ft and $w = 5$ ft.

Area: _____

True or false?

A rectangle with $l = 11$ and $w = 9$ has $A = 40$.

A. True

B. False

FORWARD 2

Find the volume of a prism by packing it with unit cubes and by multiplying •
Grade 5 • Measurement & Data • 5.MD.C.5.a

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box is packed with unit cubes: 3 cubes long, 2 cubes wide, and 4 cubes tall.

Count the cubes and then multiply. Do you get the same volume?

Count: _____

Multiply: _____

Same? _____

True or false?

A prism 5 units by 2 units by 3 units can be packed with 30 unit cubes.

- A.** True
- B.** False

Which equation gives the volume of a prism 4 by 3 by 2?

A. $2 \times 4 + 3 = 11$

B. $4 + 3 + 2 = 9$

C. $4 \times 3 = 12$

D. $4 \times 3 \times 2 = 24$

A prism has a base area of 6 square units and a height of 5 units.

What is its volume?

Volume: _____

True or false?

A 2-by-2-by-2 cube can be packed with 6 unit cubes.

A. True

B. False

Multiply side lengths to find the area of a rectangle in real-world problems

MAIN	Grade 3 main question	3.MD.C.7.b
BACKWARD 1	Measure side lengths with a ruler	2.MD.A.1
BACKWARD 2	Find the total of equal rows by repeated addition	2.OA.C.4
FORWARD 1	Use the area formula to solve real-world problems with larger num...	4.MD.A.3
FORWARD 2	Find the area of a rectangle with fractional side lengths	5.NF.B.4.b

SET 52 • 3.MD.C.7.b

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rug is 6 feet long and 4 feet wide.

What is its area?

Area: _____

Find the area of the rectangle.

Area: _____



A garden is 8 yards long and 9 yards wide.

What is its area?

- A.** 34 square yards
- B.** 72 yards
- C.** 72 square yards
- D.** 17 square yards

True or false?

A poster 3 feet by 2 feet has an area of 5 square feet.

A. True

B. False

True or false?

A tile 8 inches by 8 inches has an area of 16 square inches.

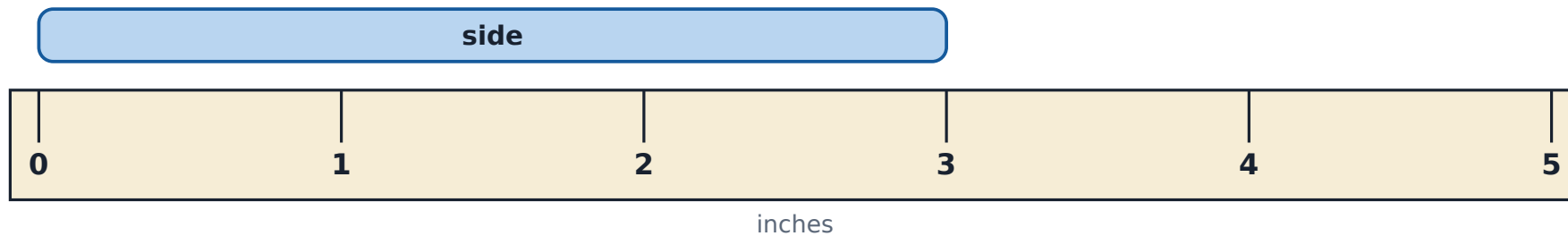
- A. True
- B. False

SET 52 • 3.MD.C.7.b

BACKWARD 1

Measure side lengths with a ruler • Grade 2 • Measurement & Data • 2.MD.A.1

How long is the side?



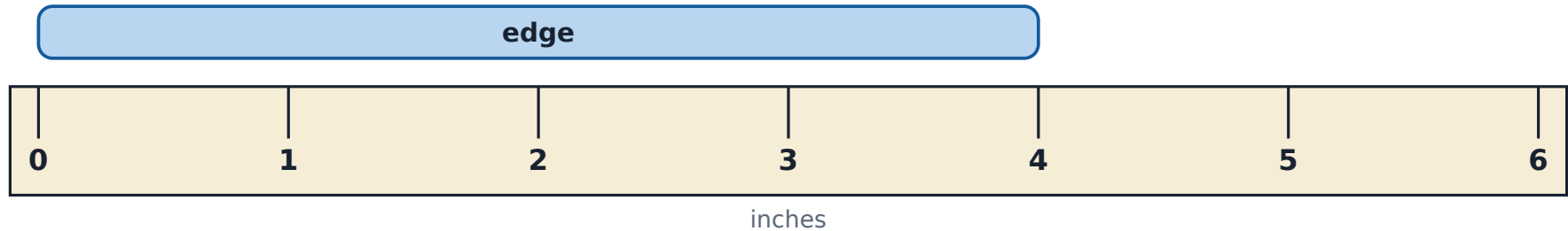
Length: _____

True or false?

To measure with a ruler, line up one end of the side with 0.

- A.** True
- B.** False

How long is the edge?



A. 4 inches

B. 3 inches

C. 6 inches

D. 5 inches

A side reaches from 0 to 7 on a centimeter ruler.

How long is it?

Length: _____

True or false?

A side that reaches from 0 to 2 on an inch ruler is 3 inches long.

- A. True
- B. False

SET 52 • 3.MD.C.7.b

BACKWARD 2

Find the total of equal rows by repeated addition • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

A tray has 4 rows of 5 cups.

Use repeated addition to find the number of cups.

Cups: _____

True or false?

3 rows of 5 tiles is $5 + 5 + 5 = 15$ tiles.

A. True

B. False

Which addition gives the number of squares in 2 rows of 4?

A. $2 + 2 + 4$

B. $4 + 4$

C. $2 + 4$

D. $4 + 2$

How many tiles are in 3 rows of 3?

Tiles: _____

True or false?

4 rows of 4 tiles is $4 + 4 = 16$ tiles.

A. True

B. False

SET 52 • 3.MD.C.7.b

FORWARD 1

Use the area formula to solve real-world problems with larger numbers • Grade 4 • Measurement & Data • 4.MD.A.3

A patio is 12 feet by 11 feet.

What is its area?

Area: _____

True or false?

A 15-m by 20-m field has an area of 300 square meters.

A. True

B. False

A wall is 9 ft tall and 14 ft wide.

What is its area?

- A.** 126 square feet
- B.** 46 square feet
- C.** 23 square feet
- D.** 116 square feet

A pool is 25 m long and 10 m wide.

What is its area?

Area: _____

True or false?

A 13-cm by 7-cm card has an area of 81 square cm.

A. True

B. False

SET 52 • 3.MD.C.7.b

FORWARD 2

Find the area of a rectangle with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.b

Find the area of a rectangle that is $\frac{2}{3}$ m by $\frac{3}{4}$ m.

Area: _____

True or false?

A rectangle $2\frac{1}{2}$ ft by 2 ft has an area of 5 square feet.

- A.** True
- B.** False

A square has sides that are $\frac{1}{3}$ yard long.

What is its area?

A. $\frac{1}{6}$ square yard

B. $\frac{4}{3}$ square yards

C. $\frac{2}{3}$ square yard

D. $\frac{1}{9}$ square yard

A rug is 4 feet by $1\frac{1}{2}$ feet.

What is its area?

Area: _____

True or false?

A rectangle $\frac{1}{2}$ inch by $\frac{1}{4}$ inch has an area of $\frac{3}{4}$ square inch.

- A. True
- B. False

Use tiling to show that the area of a rectangle with sides a and $b + c$ is $a \times b + a \times c$

MAIN	Grade 3 main question	3.MD.C.7.c
BACKWARD 1	Split an array into two parts and add	2.OA.C.4
BACKWARD 2	Add tens and ones within 100	2.NBT.B.5
FORWARD 1	Use area models and the distributive property to multiply a two-di...	4.NBT.B.5
FORWARD 2	Multiply two-digit numbers with partial products and the standard...	5.NBT.B.5

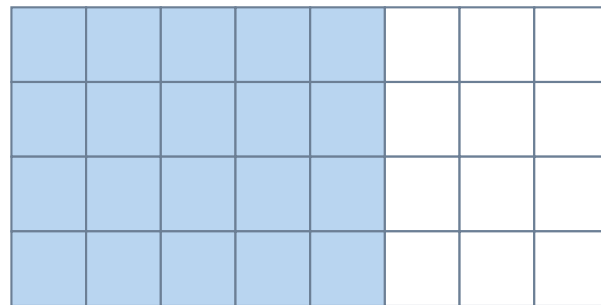
SET 53 • 3.MD.C.7.c

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.c

The 4-by-8 rectangle is split into a 4-by-5 part (shaded) and a 4-by-3 part.

Write the area as a sum of two products. Then find the total area.



Sum of products: _____

Total area: _____

A 6-by-9 rectangle is split into a 6-by-4 part and a 6-by-5 part.

Which expression gives its area?

A. $4 \times 5 + 6$

B. $6 \times 4 + 6 \times 5$

C. $6 \times 4 \times 5$

D. $6 + 4 + 5$

True or false?

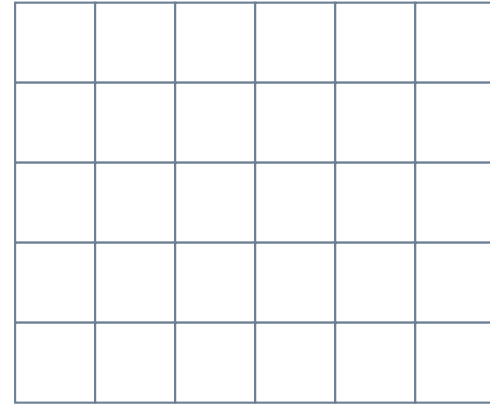
The area of a 3-by-7 rectangle equals $3 \times 5 + 3 \times 2$.

A. True

B. False

Draw one line along the grid to split the 5-by-6 rectangle into two smaller rectangles.

Write the total area as a sum of the two areas.



Sum: _____

True or false?

A 2-by-10 rectangle split into a 2-by-6 part and a 2-by-4 part has an area of $2 \times 6 \times 4$.

A. True

B. False

SET 53 • 3.MD.C.7.c

BACKWARD 1

Split an array into two parts and add • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

An array has 3 rows of 5 dots. A line splits it into 3 rows of 3 and 3 rows of 2.

How many dots are in each part? How many are there in all?

Parts: _____

Total: _____

True or false?

Splitting an array of 4 rows of 5 into 4 rows of 2 and 4 rows of 3 keeps 20 dots in all.

A. True

B. False

An array of 2 rows of 5 is split into 2 rows of 2 and 2 rows of 3.

How many dots are in the 2-rows-of-3 part?

A. 5

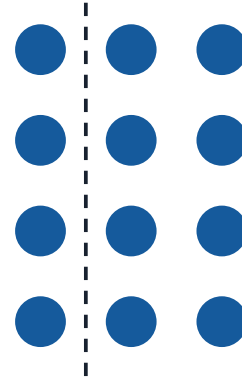
B. 4

C. 10

D. 6

The dashed line splits the array.

How many dots are there in all?



Dots: _____

True or false?

An array of 5 rows of 4 split into two parts has 10 dots in all.

A. True

B. False

SET 53 • 3.MD.C.7.c

BACKWARD 2

Add tens and ones within 100 • Grade 2 • Number & Operations in Base Ten •
2.NBT.B.5

Add.

$$20 + 15$$

Sum: _____

True or false?

$$30 + 24 = 54$$

A. True

B. False

$$40 + 32 = ?$$

A. 62

B. 8

C. 82

D. 72

Add.

$$18 + 12$$

Sum: _____

True or false?

$$35 + 25 = 50$$

A. True

B. False

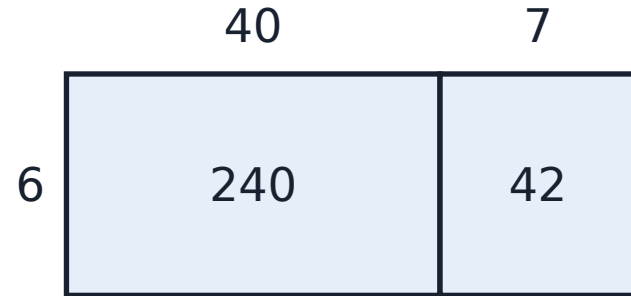
SET 53 • 3.MD.C.7.c

FORWARD 1

Use area models and the distributive property to multiply a two-digit number •
Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Use the area model to find
 6×47 .

Product: _____



True or false?

$$8 \times 53 = 8 \times 50 + 8 \times 3$$

A. True

B. False

Which two parts of an area model show 4×26 ?

- A.** 20×6 and 4
- B.** 4×2 and 4×6
- C.** 4×20 and 6
- D.** 4×20 and 4×6

Use an area model or the distributive property to find 7×38 . Show your work.

Product: _____

True or false?

$$5 \times 64 = 5 \times 60 + 4$$

A. True

B. False

SET 53 • 3.MD.C.7.c

FORWARD 2

Multiply two-digit numbers with partial products and the standard algorithm •

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

Use partial products to find 34×26 .

Product: _____

True or false?

$$25 \times 43 = 25 \times 40 + 25 \times 3$$

A. True

B. False

Which partial products give 12×35 ?

A. $12 \times 30 + 5$

B. $10 \times 35 + 2 \times 35$

C. $10 \times 35 + 2$

D. $10 \times 30 + 2 \times 5$

Use the standard algorithm to multiply. Show your work.

$$48 \times 27$$

Product: _____

True or false?

$$16 \times 25 = 16 \times 20 + 5$$

A. True

B. False

Find areas of rectilinear figures by splitting them into rectangles and adding the areas

MAIN	Grade 3 main question	3.MD.C.7.d
BACKWARD 1	Put shapes together to make a new shape	1.G.A.2
BACKWARD 2	Add two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Find areas of figures made of rectangles in real-world problems	4.MD.A.3
FORWARD 2	Find the volume of a solid made of two prisms by adding their volu...	5.MD.C.5.c

SET 54 • 3.MD.C.7.d

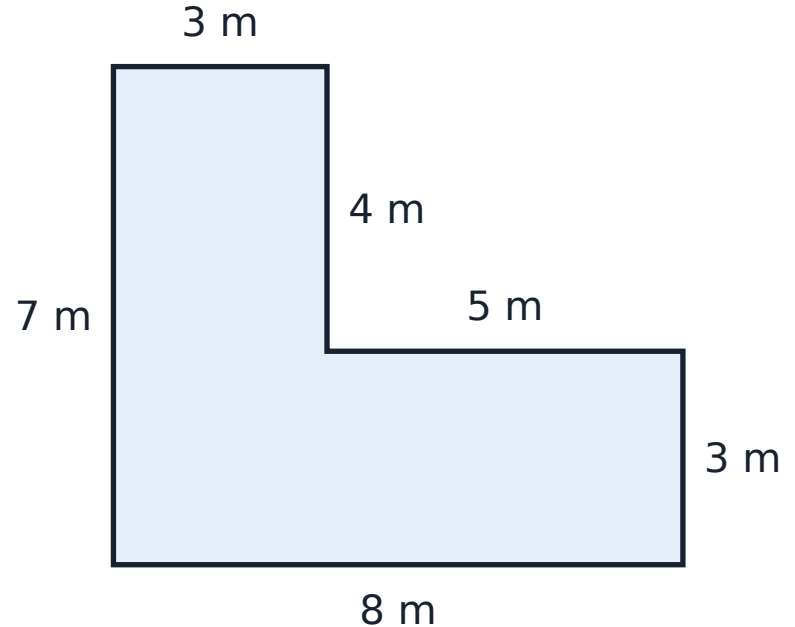
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.
Split it into two rectangles
and add their areas.

Areas of the parts: _____

Total area: _____



A figure is split into a 6-by-4 rectangle and a 2-by-3 rectangle that do not overlap.

What is the area of the figure?

- A.** 24 square units
- B.** 15 square units
- C.** 144 square units
- D.** 30 square units

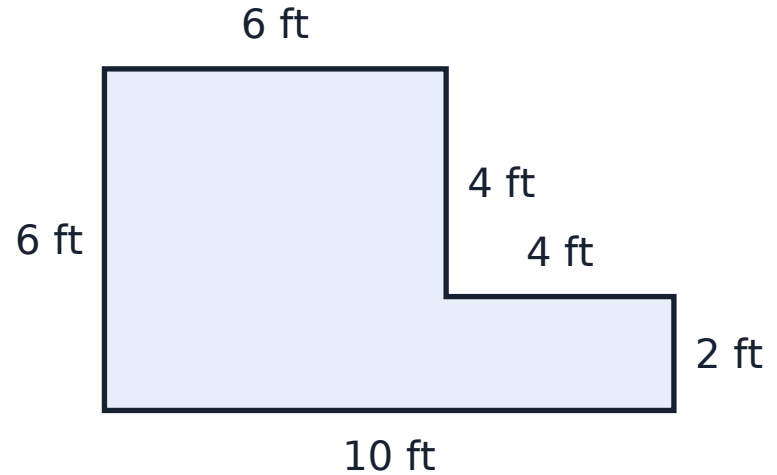
True or false?

The area of a figure made of two rectangles that do not overlap is the sum of the areas of the rectangles.

- A.** True
- B.** False

Find the area of the figure.

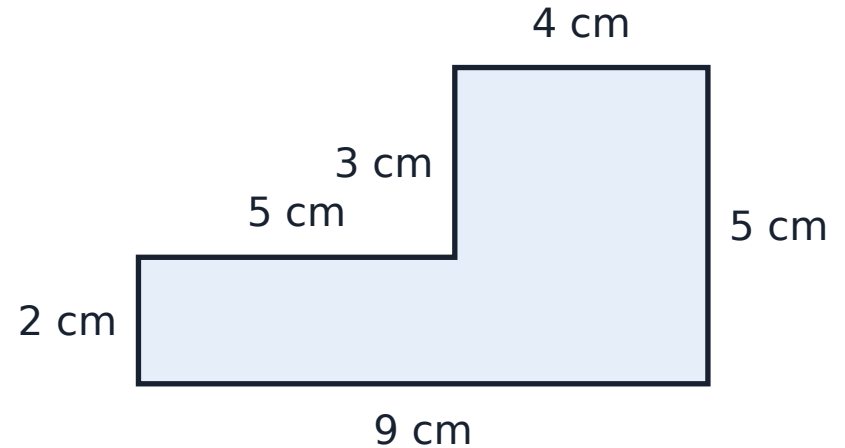
Area: _____



True or false?

The figure has an area of 50 square centimeters.

- A. True
- B. False



SET 54 • 3.MD.C.7.d

BACKWARD 1

Put shapes together to make a new shape • Grade 1 • Geometry • 1.G.A.2

True or false?

Two rectangles can be put together to make a new shape.

A. True

B. False

Two same-size squares are put side by side.
What shape do they make?

- A.** A triangle
- B.** A rectangle
- C.** A circle
- D.** A hexagon

How many same-size squares make a rectangle that is 3 squares long and 1 square wide?

Squares: _____

A shape is made of one square and one rectangle put together.

How many smaller shapes make it?

Shapes: _____

True or false?

Putting two rectangles together always makes a square.

- A.** True
- B.** False

SET 54 • 3.MD.C.7.d

BACKWARD 2

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$24 + 18$$

Sum: _____

True or false?

$$36 + 30 = 66$$

A. True

B. False

$$45 + 27 = ?$$

A. 62

B. 72

C. 18

D. 82

Add.

$$32 + 49$$

Sum: _____

True or false?

$$28 + 16 = 34$$

A. True

B. False

SET 54 • 3.MD.C.7.d

FORWARD 1

Find areas of figures made of rectangles in real-world problems • Grade 4 •
Measurement & Data • 4.MD.A.3

A room is made of a 12-ft by 10-ft rectangle and a 5-ft by 4-ft rectangle.

What is its total area?

Area: _____

True or false?

A garden made of a 15-m by 6-m bed and a 6-m by 6-m bed has an area of 126 square meters.

- A. True
- B. False

A deck is a 20-ft by 8-ft rectangle with a 6-ft by 5-ft hole cut out for a tree.

What is the area of the deck?

- A.** 160 square feet
- B.** 30 square feet
- C.** 130 square feet
- D.** 190 square feet

A parking lot is made of a 30-m by 15-m part and a 10-m by 15-m part.

What is its area?

Area: _____

True or false?

A figure made of a 9-by-7 rectangle and a 3-by-4 rectangle has an area of 23 square units.

- A. True
- B. False

FORWARD 2

Find the volume of a solid made of two prisms by adding their volumes • Grade 5 • Measurement & Data • 5.MD.C.5.c

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

Two prisms that do not overlap have volumes of 20 and 18 cubic units.

What is their total volume?

Volume: _____

True or false?

A 2-by-3-by-4 prism and a 2-by-3-by-1 prism together have a volume of 30 cubic units.

- A. True
- B. False

A solid is a 4-by-2-by-3 prism joined to a 2-by-2-by-3 prism.

What is its volume?

- A.** 24 cubic units
- B.** 36 cubic units
- C.** 288 cubic units
- D.** 12 cubic units

A building is a 10-by-6-by-3 prism with a 4-by-6-by-3 prism attached.

What is its total volume?

Volume: _____

True or false?

The volume of two joined prisms is the product of their volumes.

- A. True
- B. False

Find the perimeter of a polygon from its side lengths

MAIN	Grade 3 main question	3.MD.D.8
BACKWARD 1	Add lengths in word problems within 100	2.MD.B.5
BACKWARD 2	Add up to four two-digit numbers	2.NBT.B.6
FORWARD 1	Apply the perimeter formula for rectangles	4.MD.A.3
FORWARD 2	Find perimeters of shapes with fractional side lengths	5.NF.A.2

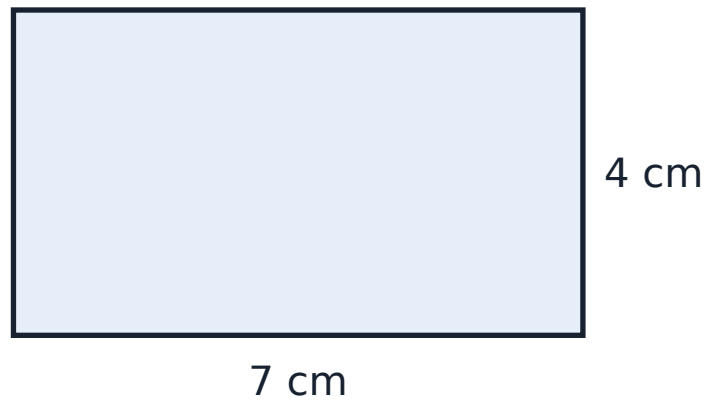
SET 55 • 3.MD.D.8

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



A triangle has sides of 6 in., 8 in., and 10 in.
What is its perimeter?

Perimeter: _____

A square has sides that are 9 m long.

What is its perimeter?

A. 81 m

B. 18 m

C. 27 m

D. 36 m

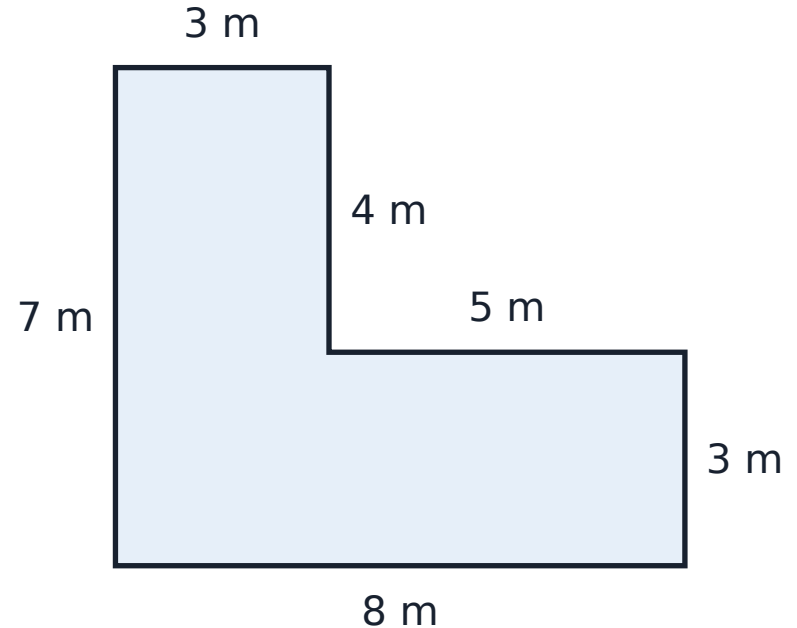
True or false?

A pentagon with five sides that are each 5 ft long has a perimeter of 25 ft.

- A.** True
- B.** False

Find the perimeter of the figure.

Perimeter: _____



SET 55 • 3.MD.D.8

BACKWARD 1

Add lengths in word problems within 100 • Grade 2 • Measurement & Data •
2.MD.B.5

A fence has parts that are 12 ft, 15 ft, and 20 ft long.
What is the total length?

Length: _____

True or false?

Ribbons of 18 cm and 25 cm have a total length of 43 cm.

A. True

B. False

Three sticks are 9 in., 14 in., and 21 in. long.

What is their total length?

- A.** 44 inches
- B.** 54 inches
- C.** 34 inches
- D.** 23 inches

A path has two parts, 35 m and 46 m long.

How long is the path?

Length: _____

True or false?

Boards of 30 in., 30 in., and 15 in. have a total length of 65 in.

A. True

B. False

SET 55 • 3.MD.D.8

BACKWARD 2

Add up to four two-digit numbers • Grade 2 • Number & Operations in Base Ten
• 2.NBT.B.6

Add.

$$12 + 15 + 12 + 15$$

Sum: _____

True or false?

$$20 + 8 + 20 + 8 = 56$$

A. True

B. False

$$9 + 16 + 9 + 16 = ?$$

A. 40

B. 41

C. 25

D. 50

Add.

$$30 + 25 + 18$$

Sum: _____

True or false?

$$11 + 11 + 11 + 11 = 40$$

A. True

B. False

SET 55 • 3.MD.D.8

FORWARD 1

Apply the perimeter formula for rectangles • Grade 4 • Measurement & Data •
4.MD.A.3

Use $P = 2 \times l + 2 \times w$ to find the perimeter of a 15-m by 9-m rectangle.

Perimeter: _____

True or false?

A 25-ft by 10-ft rectangle has a perimeter of 70 ft.

A. True

B. False

A square field has sides that are 45 yd long.

What is its perimeter?

- A.** 135 yd
- B.** 2,025 yd
- C.** 180 yd
- D.** 90 yd

A rectangle is 36 cm long and 14 cm wide.
What is its perimeter?

Perimeter: _____

True or false?

A 30-m by 20-m rectangle has a perimeter of 50 m.

A. True

B. False

SET 55 • 3.MD.D.8

FORWARD 2

Find perimeters of shapes with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A rectangle is $\frac{3}{4}$ ft by $\frac{1}{2}$ ft.

What is its perimeter?

Perimeter: _____

True or false?

A triangle with sides of $\frac{1}{2}$ m, $\frac{1}{3}$ m, and $\frac{1}{6}$ m has a perimeter of 1 m.

- A. True
- B. False

A square has sides that are $\frac{5}{8}$ inch long.

What is its perimeter?

A. $1\frac{1}{4}$ inches

B. $\frac{5}{32}$ inch

C. $2\frac{1}{2}$ inches

D. $\frac{20}{32}$ inch

A rectangle is $2\frac{1}{4}$ yd by $1\frac{1}{2}$ yd.

What is its perimeter?

Perimeter: _____

True or false?

A rectangle $\frac{2}{3}$ m by $\frac{1}{6}$ m has a perimeter of $\frac{5}{6}$ m.

A. True

B. False

Find an unknown side length of a polygon from its perimeter

MAIN	Grade 3 main question	3.MD.D.8
BACKWARD 1	Find an unknown part in a word problem within 100	2.OA.A.1
BACKWARD 2	Find the unknown number in an addition equation within 20	1.OA.D.8
FORWARD 1	Find an unknown side length from the perimeter of a rectangle wi...	4.MD.A.3
FORWARD 2	Find an unknown side length from a perimeter with fractions	5.NF.A.2

SET 56 • 3.MD.D.8

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.D.8

A rectangle has a perimeter of 20 cm. Its length is 6 cm.
What is its width?

Width: _____

A triangle has a perimeter of 30 in. Two of its sides are 9 in. and 12 in.

How long is the third side?

Length: _____

A square has a perimeter of 28 m.

How long is each side?

- A.** 24 m
- B.** 112 m
- C.** 7 m
- D.** 14 m

True or false?

A rectangle with a perimeter of 18 ft and a length of 5 ft has a width of 8 ft.

- A. True
- B. False

A quadrilateral has a perimeter of 40 cm. Three of its sides are 10 cm, 12 cm, and 8 cm.

How long is the fourth side?

Length: _____

SET 56 • 3.MD.D.8

BACKWARD 1

Find an unknown part in a word problem within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A fence will be 50 m long. 32 m are built.

How many meters are left to build?

Meters: _____

True or false?

Kim needs 45 beads. She has 27 beads. She needs 18 more.

A. True

B. False

A path is 80 ft long. Lu walked some of it and has 35 ft left.

How far did she walk?

A. 35 ft

B. 45 ft

C. 55 ft

D. 115 ft

A lot has some cars and 14 trucks. There are 40 vehicles in all.

How many cars are there?

Cars: _____

True or false?

A tape is 60 cm long. 25 cm are used, so 45 cm are left.

A. True

B. False

SET 56 • 3.MD.D.8

BACKWARD 2

Find the unknown number in an addition equation within 20 • Grade 1 •
Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

$$12 + ? = 20$$

Unknown: _____

True or false?

In $? + 9 = 15$, the unknown number is 6.

A. True

B. False

$$18 = 10 + ?$$

- A.** 9
- B.** 8
- C.** 10
- D.** 28

Find the unknown number.

$$? + 7 = 19$$

Unknown: _____

True or false?

In $11 + ? = 17$, the unknown number is 5.

A. True

B. False

SET 56 • 3.MD.D.8

FORWARD 1

Find an unknown side length from the perimeter of a rectangle with the formula

• Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has a perimeter of 64 m and a length of 20 m.

What is its width?

Width: _____

True or false?

A square with a perimeter of 96 in. has sides that are 24 in. long.

A. True

B. False

A rectangle has a perimeter of 50 ft and a width of 9 ft.
What is its length?

- A.** 32 ft
- B.** 25 ft
- C.** 41 ft
- D.** 16 ft

A rectangular garden has a perimeter of 100 yd. Its length is 30 yd.

Write an equation with a letter for the width. Then solve it.

Equation: _____

Width: _____

True or false?

A rectangle with a perimeter of 44 cm and a length of 15 cm has a width of 14 cm.

A. True

B. False

SET 56 • 3.MD.D.8

FORWARD 2

Find an unknown side length from a perimeter with fractions • Grade 5 •
Number & Operations–Fractions • 5.NF.A.2

A triangle has a perimeter of 3 ft. Two of its sides are $\frac{3}{4}$ ft and $1\frac{1}{4}$ ft.

How long is the third side?

Length: _____

True or false?

A square with a perimeter of 5 m has sides that are $1\frac{1}{4}$ m long.

- A. True
- B. False

A rectangle has a perimeter of 6 in. and a length of $1\frac{3}{4}$ in.

What is its width?

A. $4\frac{1}{4}$ in.

B. $3\frac{1}{4}$ in.

C. $1\frac{1}{4}$ in.

D. $2\frac{1}{2}$ in.

A rectangle has a perimeter of 4 yd and a width of $\frac{2}{3}$ yd.

What is its length?

Length: _____

True or false?

A square with a perimeter of 2 m has sides that are $\frac{1}{4}$ m long.

A. True

B. False

Show rectangles with the same perimeter and different areas, or the same area and different perimeters

MAIN	Grade 3 main question	3.MD.D.8
BACKWARD 1	Recognize and draw rectangles	2.G.A.1
BACKWARD 2	Arrays with the same total	2.OA.C.4
FORWARD 1	Compare the areas and perimeters of rectangles with the formulas	4.MD.A.3
FORWARD 2	Find different prisms with the same volume — nearest related (no...	5.MD.C.5.b

SET 57 • 3.MD.D.8

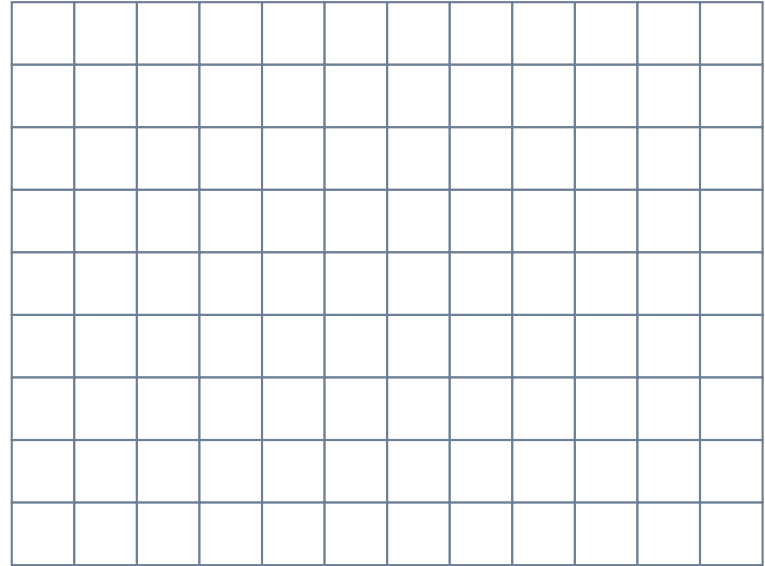
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.D.8

Draw two different rectangles on the grid that each have a perimeter of 12 units.

What is the area of each rectangle?

Areas: _____



Which rectangle has the same area as a 4-by-6 rectangle but a different perimeter?

A. 2 by 10

B. 3 by 8

C. 4 by 5

D. 5 by 5

True or false?

A 2-by-6 rectangle and a 3-by-4 rectangle have the same area but different perimeters.

A. True

B. False

A 1-by-8 rectangle and a 2-by-4 rectangle both have an area of 8 square units.

Find the perimeter of each.

1 by 8: _____

2 by 4: _____

True or false?

Two rectangles with the same perimeter always have the same area.

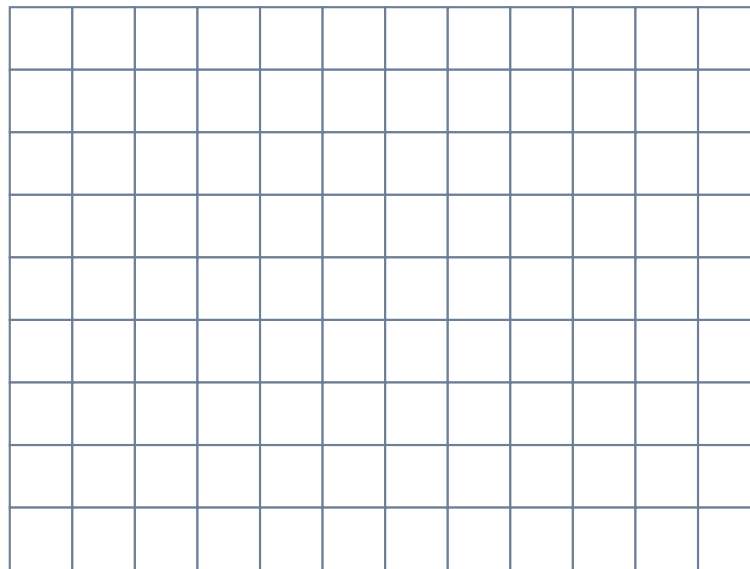
- A. True
- B. False

SET 57 • 3.MD.D.8

BACKWARD 1

Recognize and draw rectangles • Grade 2 • Geometry • 2.G.A.1

Draw a rectangle on the grid.



True or false?

A rectangle has 4 sides and 4 corners.

A. True

B. False

Which shape has 4 sides, with opposite sides the same length, and 4 square corners?

- A.** Triangle
- B.** Rectangle
- C.** Hexagon
- D.** Pentagon

How many sides does a rectangle have?

Sides: _____

True or false?

A rectangle has 3 corners.

A. True

B. False

SET 57 • 3.MD.D.8

BACKWARD 2

Arrays with the same total • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 2 rows of 6 dots. Another array has 3 rows of 4 dots.

How many dots does each array have?

2 rows of 6: _____

3 rows of 4: _____

True or false?

An array of 4 rows of 5 and an array of 5 rows of 4 have the same number of dots.

- A.** True
- B.** False

Which array has 10 dots?

- A.** 2 rows of 5
- B.** 3 rows of 3
- C.** 4 rows of 3
- D.** 2 rows of 4

How many dots are in 3 rows of 5?

Dots: _____

True or false?

An array of 2 rows of 3 and an array of 3 rows of 3 have the same number of dots.

- A.** True
- B.** False

SET 57 • 3.MD.D.8

FORWARD 1

Compare the areas and perimeters of rectangles with the formulas • Grade 4 •
Measurement & Data • 4.MD.A.3

A rectangle is 10 m by 6 m.

Find its area and its perimeter.

Area: _____

Perimeter: _____

True or false?

A 9-by-4 rectangle and a 12-by-3 rectangle have the same area.

- A. True
- B. False

Which rectangle has a perimeter of 30 ft and an area of 50 square feet?

- A.** 15 ft by 2 ft
- B.** 25 ft by 2 ft
- C.** 6 ft by 9 ft
- D.** 10 ft by 5 ft

A 2-by-10 rectangle and a 6-by-6 square both have a perimeter of 24 cm.

Which has the greater area?

Answer: _____

True or false?

A 7-by-7 square and an 8-by-6 rectangle have the same perimeter and the same area.

A. True

B. False

FORWARD 2

Find different prisms with the same volume • Grade 5 • Measurement & Data •
5.MD.C.5.b

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

Give the dimensions of two different prisms that each have a volume of 24 cubic units.

Prism 1: _____

Prism 2: _____

True or false?

A 2-by-2-by-6 prism and a 3-by-4-by-2 prism have the same volume.

- A. True
- B. False

Which prism has a volume of 36 cubic units?

A. 6 by 6 by 2

B. 2 by 2 by 8

C. 3 by 4 by 4

D. 3 by 3 by 4

A box is 5 by 2 by 3. Another box is 3 by 2 by ? and has the same volume.

What is the missing dimension?

Missing dimension: _____

True or false?

A 1-by-1-by-10 prism and a 2-by-2-by-2 prism have the same volume.

- A. True
- B. False

Understand that shapes in different categories can share attributes, and that quadrilaterals form a category

MAIN	Grade 3 main question	3.G.A.1
BACKWARD 1	Name triangles, quadrilaterals, pentagons, and hexagons	2.G.A.1
BACKWARD 2	Tell defining attributes from other attributes	1.G.A.1
FORWARD 1	Classify quadrilaterals by parallel sides and right angles	4.G.A.2
FORWARD 2	Attributes of a category belong to all of its subcategories	5.G.B.3

SET 58 • 3.G.A.1

MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.1

What do a rhombus, a rectangle, and a square always have in common?

- A.** They all have 4 sides.
- B.** They all have 4 square corners.
- C.** They all have 4 sides of equal length.
- D.** They all have 3 sides.

True or false?

A trapezoid and a square are both quadrilaterals.

A. True

B. False

A closed shape has 4 straight sides and 4 angles.
What category does it belong to?

Category: _____

Name an attribute that every rectangle and every square share, and an attribute that a square must have but a rectangle might not.

Shared: _____

Only the square must have: _____

True or false?

A triangle is a quadrilateral because it has straight sides.

A. True

B. False

SET 58 • 3.G.A.1

BACKWARD 1

Name triangles, quadrilaterals, pentagons, and hexagons • Grade 2 •
Geometry • 2.G.A.1

How many sides does a hexagon have?

Sides: _____

True or false?

A quadrilateral has 4 sides.

A. True

B. False

Which shape has 5 angles?

- A.** Triangle
- B.** Quadrilateral
- C.** Hexagon
- D.** Pentagon

What is a closed shape with 3 straight sides called?

Name: _____

True or false?

A pentagon has 6 sides.

A. True

B. False

SET 58 • 3.G.A.1

BACKWARD 2

Tell defining attributes from other attributes • Grade 1 • Geometry • 1.G.A.1

True or false?

A square can be any color and still be a square.

A. True

B. False

Which is a defining attribute of a triangle?

- A.** It has 3 sides.
- B.** It is red.
- C.** It is big.
- D.** It points up.

Is size a defining attribute of a rectangle?

Answer: _____

A closed shape has 4 straight sides.

Name one shape it could be.

Shape: _____

True or false?

A shape must be blue to be a hexagon.

A. True

B. False

SET 58 • 3.G.A.1

FORWARD 1

Classify quadrilaterals by parallel sides and right angles • Grade 4 • Geometry
• 4.G.A.2

A quadrilateral has exactly one pair of parallel sides.
What is it called?

Name: _____

True or false?

A rectangle has two pairs of parallel sides.

A. True

B. False

Which shape must have four right angles?

- A.** Rhombus
- B.** Parallelogram
- C.** Square
- D.** Trapezoid

Name a quadrilateral that has two pairs of parallel sides and no right angles.

Shape: _____

True or false?

Every quadrilateral has at least one pair of parallel sides.

- A. True
- B. False

SET 58 • 3.G.A.1

FORWARD 2

Attributes of a category belong to all of its subcategories • Grade 5 • Geometry
• 5.G.B.3

True or false?

All rectangles have 4 right angles, and every square is a rectangle, so every square has 4 right angles.

- A.** True
- B.** False

A shape is a square.

Which statement must be true?

- A.** It has 5 sides.
- B.** It is a rectangle.
- C.** It has exactly one pair of parallel sides.
- D.** It is a triangle.

Every parallelogram has opposite sides of equal length.
A rhombus is a parallelogram.

What must be true about the opposite sides of a rhombus?

Answer: _____

Does every rectangle have 4 sides? Explain using categories.

Answer: _____

Explanation: _____

True or false?

Every rhombus is a square.

A. True

B. False

Recognize rhombuses, rectangles, and squares as quadrilaterals, and draw quadrilaterals that are not in these subcategories

MAIN	Grade 3 main question	3.G.A.1
BACKWARD 1	Draw shapes with a given number of sides or angles	2.G.A.1
BACKWARD 2	Square corners and equal sides as defining attributes	1.G.A.1
FORWARD 1	Draw and identify parallel and perpendicular sides in quadrilatera...	4.G.A.1
FORWARD 2	Classify quadrilaterals in a hierarchy	5.G.B.4

SET 59 • 3.G.A.1

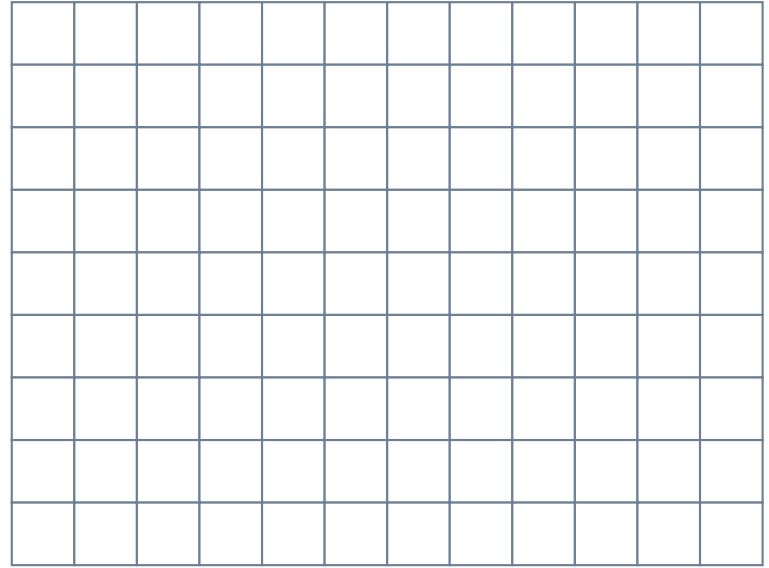
MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.1

Draw a quadrilateral that is not a rhombus, not a rectangle, and not a square.

What makes your shape a quadrilateral?

Reason: _____



Which shape is a quadrilateral but is NOT a rhombus, a rectangle, or a square?

- A.** A square
- B.** A rhombus
- C.** A rectangle whose length and width are different
- D.** A trapezoid with exactly one pair of parallel sides

True or false?

A square is a rectangle, and it is also a rhombus.

A. True

B. False

A shape has 4 sides of equal length and no square corners.

What is it?

A. A triangle

B. A rectangle whose length and width are different

C. A square

D. A rhombus that is not a square

True or false?

Every quadrilateral is a rectangle.

A. True

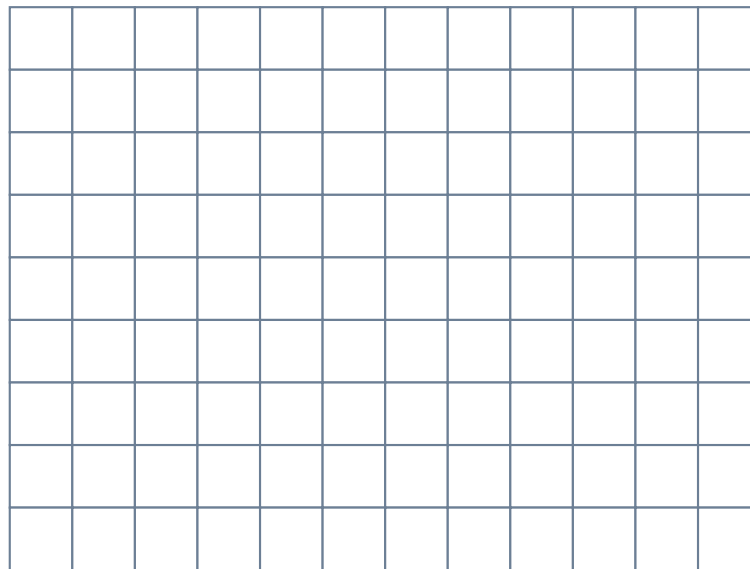
B. False

SET 59 • 3.G.A.1

BACKWARD 1

Draw shapes with a given number of sides or angles • Grade 2 • Geometry •
2.G.A.1

Draw a shape with 4 sides.



True or false?

A closed shape with 4 angles has 4 sides.

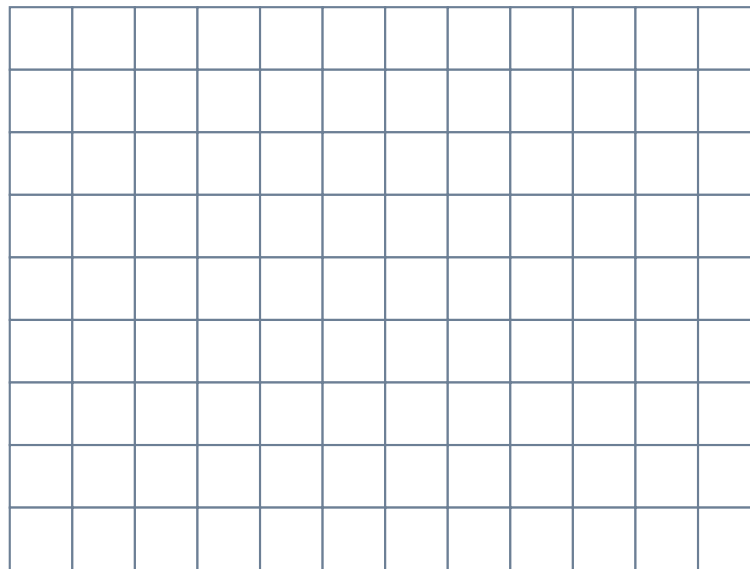
A. True

B. False

Which shape has 4 sides?

- A.** Triangle
- B.** Quadrilateral
- C.** Pentagon
- D.** Hexagon

Draw a shape with 3 angles.



True or false?

A closed shape with 5 sides has 4 angles.

A. True

B. False

SET 59 • 3.G.A.1

BACKWARD 2

Square corners and equal sides as defining attributes • Grade 1 • Geometry •
1.G.A.1

True or false?

A square has 4 sides of equal length.

A. True

B. False

How many square corners does a rectangle have?

Square corners: _____

Which shape always has 4 square corners?

- A.** Rectangle
- B.** Triangle
- C.** Hexagon
- D.** Circle

Is a rectangle that is turned on its corner still a rectangle?

Answer: _____

True or false?

A rectangle must have 4 sides of equal length.

A. True

B. False

SET 59 • 3.G.A.1

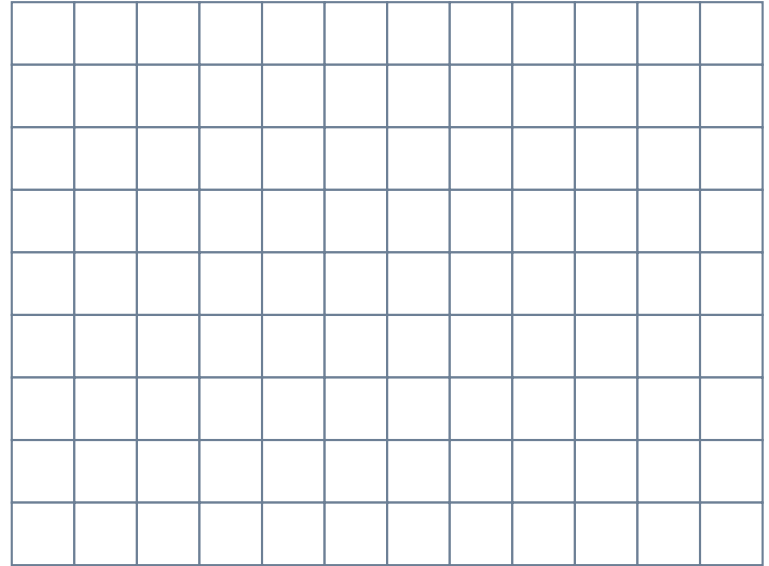
FORWARD 1

Draw and identify parallel and perpendicular sides in quadrilaterals • Grade 4 •
Geometry • 4.G.A.1

Draw a quadrilateral with exactly one pair of parallel sides.

What is it called?

Name: _____



True or false?

A rectangle has 4 right angles, so the sides that meet at each corner are perpendicular.

A. True

B. False

Which quadrilateral always has perpendicular sides?

- A.** A trapezoid with no right angles
- B.** A parallelogram with no right angles
- C.** A rectangle
- D.** A rhombus that is not a square

How many pairs of parallel sides does a square have?

Pairs: _____

True or false?

A rhombus that is not a square has perpendicular sides.

- A.** True
- B.** False

SET 59 • 3.G.A.1

FORWARD 2

Classify quadrilaterals in a hierarchy • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Square, rectangle, parallelogram, quadrilateral
- B.** Parallelogram, square, quadrilateral, rectangle
- C.** Quadrilateral, parallelogram, rectangle, square
- D.** Rectangle, quadrilateral, square, parallelogram

True or false?

Every square is a rhombus.

A. True

B. False

Which of these categories does every square belong to?
quadrilateral, parallelogram, rectangle, rhombus

Categories: _____

True or false?

Every rectangle is a square.

A. True

B. False

A parallelogram has 4 right angles and 4 sides of equal length.

What is its most specific name?

A. Rhombus

B. Quadrilateral

C. Rectangle

D. Square

Split shapes into parts with equal areas

MAIN	Grade 3 main question	3.G.A.2
BACKWARD 1	Split circles and rectangles into 2, 3, or 4 equal shares	2.G.A.3
BACKWARD 2	Draw rows and columns of same-size squares in a rectangle	2.G.A.2
FORWARD 1	Draw lines of symmetry that split a figure into matching parts —...	4.G.A.3
FORWARD 2	Tile a rectangle with unit-fraction squares or rectangles to find its...	5.NF.B.4.b

SET 60 • 3.G.A.2

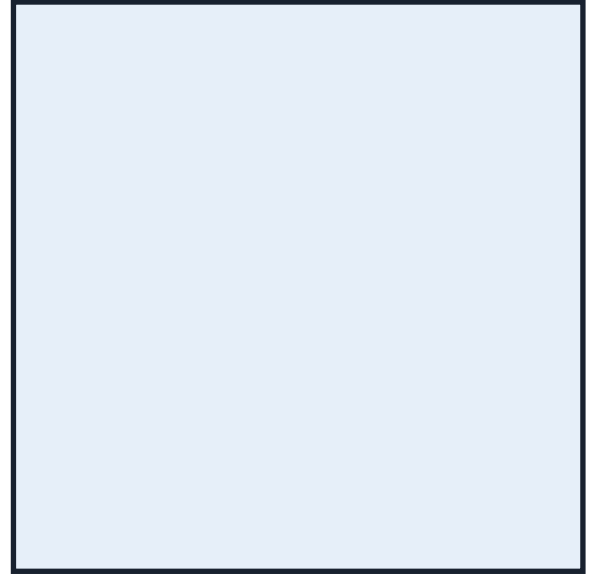
MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.2

Draw lines to split the rectangle into 4 parts with equal areas.



Draw lines to split the square into 3 parts with equal areas.



Which line always splits a rectangle into 2 parts with equal areas?

- A.** A line from a corner to the middle of the opposite side
- B.** Any line that crosses the rectangle
- C.** A line through the middle from one long side to the opposite long side, parallel to the short sides
- D.** A line close to one end

True or false?

One diagonal splits a square into 2 triangles with equal areas.

- A. True
- B. False

True or false?

If a rectangle is split into 4 parts, the parts always have equal areas.

A. True

B. False

SET 60 • 3.G.A.2

BACKWARD 1

Split circles and rectangles into 2, 3, or 4 equal shares • Grade 2 • Geometry •
2.G.A.3

Draw a line to split the rectangle into 2 equal shares.



True or false?

A rectangle can be split into 3 equal shares called thirds.

A. True

B. False

How many equal shares are there when a shape is split into halves?

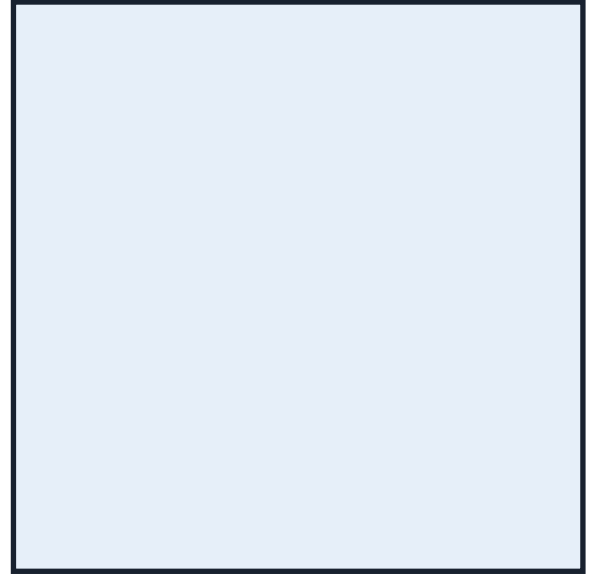
A. 2

B. 3

C. 1

D. 4

Draw lines to split the square into 4 equal shares.



True or false?

Two shares of different sizes are halves.

A. True

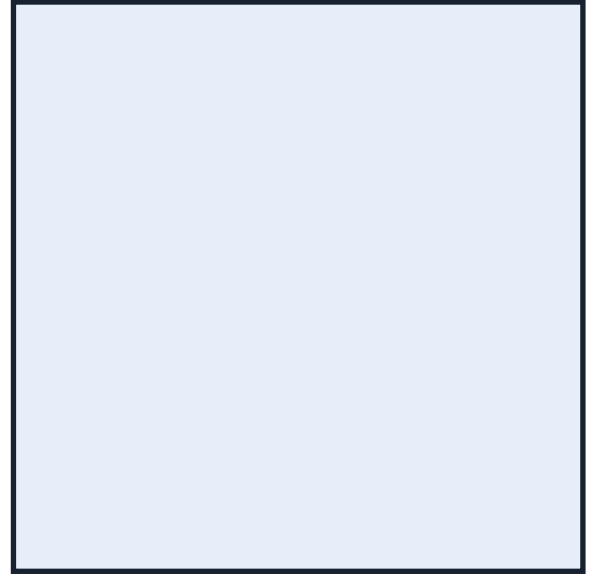
B. False

SET 60 • 3.G.A.2

BACKWARD 2

Draw rows and columns of same-size squares in a rectangle • Grade 2 •
Geometry • 2.G.A.2

Draw lines to split the square into 3 rows of 3 same-size squares.



True or false?

Splitting a square into 2 rows of 2 same-size squares makes 4 squares.

- A. True
- B. False

A rectangle is split into 1 row of 4 same-size squares.

How many squares are there?

A. 4

B. 8

C. 1

D. 5

A rectangle is split into 2 rows of 3 same-size squares.
How many squares are there?

Squares: _____

True or false?

A rectangle split into 3 rows of 2 same-size squares has 5 squares.

- A. True
- B. False

FORWARD 1

Draw lines of symmetry that split a figure into matching parts • Grade 4 •
Geometry • 4.G.A.3

► No Grade 4 standard directly continues this skill. This branch uses the nearest related Grade 4 standard.

Draw all the lines of symmetry of the rectangle.



True or false?

A line of symmetry splits a figure into two matching parts.

A. True

B. False

How many lines of symmetry does a square have?

A. 2

B. 1

C. 4

D. 0

How many lines of symmetry does a triangle with three sides of equal length have?

Lines: _____

True or false?

Every line that splits a rectangle into two parts with equal areas is a line of symmetry.

A. True

B. False

SET 60 • 3.G.A.2

FORWARD 2

Tile a rectangle with unit-fraction squares or rectangles to find its area • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.b

A square with an area of 1 square unit is split into 4 equal small squares.

What is the area of each small square?

Area: _____

True or false?

A rectangle $\frac{1}{2}$ unit by $\frac{1}{3}$ unit has an area of $\frac{1}{6}$ square unit.

- A. True
- B. False

A unit square is split into 3 equal columns and 2 equal rows.

What is the area of each small rectangle?

A. $\frac{1}{5}$ square unit

B. $\frac{1}{6}$ square unit

C. $\frac{1}{3}$ square unit

D. 6 square units

A rectangle $\frac{3}{4}$ unit by $\frac{1}{2}$ unit is tiled with rectangles that are $\frac{1}{4}$ unit by $\frac{1}{2}$ unit.

How many tiles fit? What is the area of the rectangle?

Tiles: _____

Area: _____

True or false?

A unit square split into 8 equal parts has parts with an area of $\frac{1}{4}$ square unit each.

- A. True
- B. False

Express the area of each equal part as a unit fraction of the whole

MAIN	Grade 3 main question	3.G.A.2
BACKWARD 1	Name one of 2, 3, or 4 equal shares	2.G.A.3
BACKWARD 2	Halves and quarters	1.G.A.3
FORWARD 1	Add fractions as joining parts of the same whole	4.NF.B.3.a
FORWARD 2	Add fractions with unlike denominators in word problems about pa...	5.NF.A.2

SET 61 • 3.G.A.2

MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.2

The rectangle is split into parts with equal areas.

What fraction of the whole area is each part?



Fraction: _____

A circle is split into 8 parts with equal areas.
What fraction of the area is one part?

A. $\frac{7}{8}$

B. $\frac{1}{7}$

C. $\frac{1}{8}$

D. $\frac{8}{1}$

True or false?

If a hexagon is split into 6 parts with equal areas, each part is $\frac{1}{6}$ of the area of the hexagon.

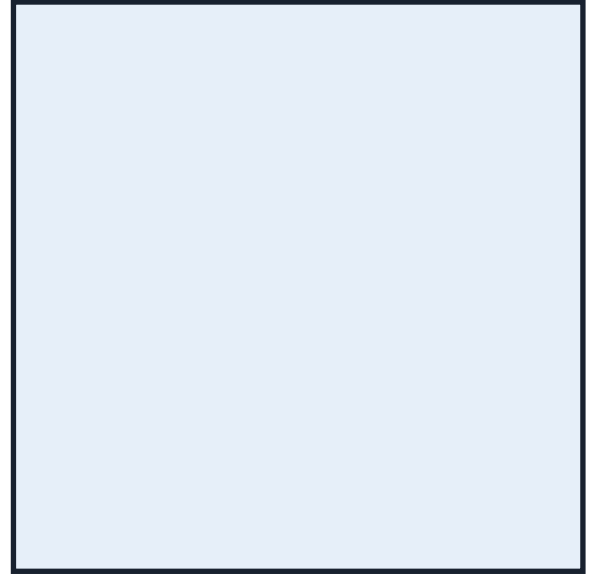
A. True

B. False

Split the square into 2 parts
with equal areas.

What fraction of the area of
the square is each part?

Fraction: _____



True or false?

A rectangle split into 3 parts with equal areas has parts that are each $\frac{1}{4}$ of the area.

- A. True
- B. False

SET 61 • 3.G.A.2

BACKWARD 1

Name one of 2, 3, or 4 equal shares • Grade 2 • Geometry • 2.G.A.3

A circle is split into 4 equal shares.

What is one share called?

Name: _____

True or false?

One of 2 equal shares is called a half.

A. True

B. False

A rectangle is split into 3 equal shares.

What is one share called?

- A.** A fourth
- B.** A third
- C.** A whole
- D.** A half

A shape is split into 2 equal shares.
What do the 2 shares make together?

Answer: _____

True or false?

One of 4 equal shares is called a third.

A. True

B. False

SET 61 • 3.G.A.2

BACKWARD 2

Halves and quarters • Grade 1 • Geometry • 1.G.A.3

A square is cut into 4 equal parts.

What is one part called?

Name: _____

True or false?

Two halves make a whole.

A. True

B. False

A pizza is cut into 2 equal parts.

What is each part called?

- A.** A quarter
- B.** A third
- C.** A half
- D.** A whole

How many halves are in one whole?

Halves: _____

True or false?

One of 4 equal parts is called a half.

A. True

B. False

SET 61 • 3.G.A.2

FORWARD 1

Add fractions as joining parts of the same whole • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

A rectangle is split into 6 equal parts. 2 parts are red and 3 parts are blue.

What fraction of the rectangle is red or blue?

Fraction: _____

True or false?

$\frac{2}{8} + \frac{3}{8}$ means joining 2 eighths and 3 eighths of the same whole.

- A.** True
- B.** False

A square is split into 4 equal parts. 1 part is shaded, and then 2 more parts are shaded.

Which equation shows the shaded fraction?

A. $\frac{1}{4} + \frac{2}{4} = \frac{2}{4}$

B. $\frac{1}{4} + \frac{2}{4} = \frac{3}{8}$

C. $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$

D. $\frac{1}{2} + \frac{2}{4} = \frac{3}{4}$

A pie is cut into 8 equal pieces. 3 pieces are eaten.
What fraction of the pie is left?

Fraction: _____

True or false?

$$\frac{1}{3} + \frac{1}{3} = \frac{2}{6}$$

A. True

B. False

SET 61 • 3.G.A.2

FORWARD 2

Add fractions with unlike denominators in word problems about parts of a whole • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

$\frac{1}{2}$ of a garden is corn and $\frac{1}{3}$ of it is beans.

What fraction of the garden is corn or beans?

Fraction: _____

True or false?

$\frac{1}{4}$ of a field is grass and $\frac{1}{2}$ is dirt. Together they are $\frac{3}{4}$ of the field.

A. True

B. False

A flag is $\frac{1}{3}$ red and $\frac{1}{6}$ white.

What fraction of the flag is red or white?

A. $\frac{1}{18}$

B. $\frac{1}{2}$

C. $\frac{2}{6}$

D. $\frac{2}{9}$

A cake is $\frac{3}{8}$ chocolate and $\frac{1}{4}$ vanilla.

What fraction of the cake is chocolate or vanilla?

Fraction: _____

True or false?

$\frac{1}{2}$ of a wall is blue and $\frac{1}{3}$ is green. The rest of the wall is $\frac{1}{5}$ of it.

A. True

B. False